

CHAPTER 12

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CHAPTER OBJECTIVES

- 1 Be able to calculate the Laplace transform of a function using the definition of Laplace transform, the Laplace transform table, and/or a table of operational transforms.
- 2 Be able to calculate the inverse Laplace transform using partial fraction expansion and the Laplace transform table.
- 3 Be able to find and plot the poles and zeros for a rational function in the s domain.
- 4 Understand and know how to use the initial-value theorem and the final-value theorem.

Introduction to the Laplace Transform

We now introduce the **Laplace transform**, which forms the basis of a powerful technique that is widely used to analyze linear, lumped-parameter circuits. We need the Laplace transform analysis technique because we can use it to solve the following types of problems.

- Finding the transient behavior of circuits whose describing equations consist of more than a single node-voltage or mesh-current differential equation. In other words, analyzing multiple-node and multiple-mesh circuits that are described by sets of linear differential equations.
- Determining the transient response of circuits whose signal sources vary in ways more complicated than the simple dc level jumps considered in Chapters 7 and 8.
- Calculating the transfer function for a circuit and using it to find the steady-state sinusoidal response of that circuit when the frequency of the sinusoidal source is varied. We discuss the transfer function in Chapter 13.
- Relating the time-domain behavior of a circuit to its frequency-domain behavior, in a systematic fashion.

In this chapter, we define the Laplace transform, discuss its pertinent characteristics, and present a systematic method for transforming expressions from the frequency domain to the time domain.

■ Practical Perspective

Transient Effects

As we learned in Chapter 9, power delivered from electrical wall outlets in the United States can be modeled as a sinusoidal voltage or current source, where the frequency of the sinusoid is 60 Hz. We used the phasor transform, introduced in Chapter 9, to find the steady-state response of a circuit to a sinusoidal source.

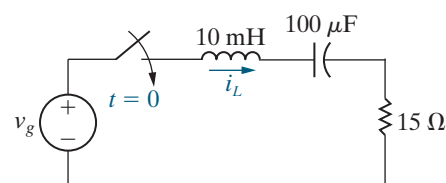
But in many cases, we need to consider the complete response of a circuit to a sinusoidal source. Remember that the complete response has two parts—the steady-state response that takes the same form as the input to the circuit, and the transient response that decays to zero as time progresses. When a circuit's source is a 60 Hz sinusoid, the steady-state response is also a 60 Hz sinusoid whose magnitude and phase angle can be calculated using phasor circuit analysis. The transient response depends on the components that make up the circuit, the values of those components, and the way the components are interconnected. Once the source is switched into the circuit, the voltage and current for every circuit component are the sum of a transient expression and a steady-state expression.

The transient part of the voltage and current eventually decays to zero. But, initially, the sum of the transient part and the steady-state part might exceed the voltage or current rating of the circuit component. This is why it is important to determine the complete response of a circuit. The Laplace transform techniques introduced in this chapter can be used to find the complete response of a circuit to a sinusoidal source.

Consider the *RLC* circuit shown, comprised of components from Appendix H and powered by a 60 Hz sinusoidal source. As detailed in Appendix H, the 10 mH inductor has a current rating of 40 mA. The amplitude of the sinusoidal source has been chosen so that this rating is not exceeded in the steady state. Once we have presented the Laplace transform method, we will be able to determine whether this current rating is exceeded when the source is first switched on and both the transient and steady-state components of the inductor current exist.



Arctic ice/Shutterstock



12.1 Definition of the Laplace Transform

The **Laplace transform** of a function is given by the expression

LAPLACE TRANSFORM

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t)e^{-st} dt, \quad (12.1)$$

where the symbol $\mathcal{L}\{f(t)\}$ is read “the Laplace transform of $f(t)$.”

The Laplace transform of $f(t)$ is also denoted $F(s)$; that is,

$$F(s) = \mathcal{L}\{f(t)\}. \quad (12.2)$$

This notation emphasizes that when the integral in Eq. 12.1 has been evaluated, the resulting expression is a function of s . In our applications, t represents the time domain and s represents the frequency domain. Note that the dimension of s must be reciprocal time, or frequency, because the exponent of e in the integral of Eq. 12.1 must be dimensionless. The Laplace transform transforms the problem from the time domain to the frequency domain. Once we solve the problem in the frequency domain, we inverse-transform the solution back to the time domain.

Recall that the phasor is also a transform. As we know from Chapter 9, it converts a sinusoidal signal into a complex number for easier, algebraic computation of circuit values. After determining the phasor value of a signal, we transform it back to its time-domain expression. Both the Laplace transform and the phasor transform exhibit an essential feature of mathematical transforms: They create a new domain to make the mathematical manipulations easier. After finding the unknown in the new domain, we inverse-transform it back to the original domain.

In circuit analysis, we use the Laplace transform to transform a set of integrodifferential equations in the time domain to a set of algebraic equations in the frequency domain. We can therefore find the solution for an unknown quantity by solving a set of algebraic equations.

Before we illustrate some of the important properties of the Laplace transform, some general comments are in order. First, note that the integral in Eq. 12.1 is improper because the upper limit is infinite. Thus, we are confronted immediately with the question of whether the integral converges. In other words, does a given $f(t)$ have a Laplace transform? Obviously, the functions of primary interest in engineering analysis have Laplace transforms; otherwise we would not be interested in the transform. In linear circuit analysis, we excite circuits with sources that have Laplace transforms. Excitation functions such as t^t or e^{t^2} , which do not have Laplace transforms, are of no interest here.

Second, because the lower limit on the integral is zero, the Laplace transform ignores $f(t)$ for negative values of t . That is, $F(s)$ is determined by the behavior of $f(t)$ only for positive values of t . To emphasize that the lower limit is zero, Eq. 12.1 is frequently referred to as the **one-sided**, or **unilateral**, Laplace transform. In the two-sided, or bilateral, Laplace transform, the lower limit is $-\infty$. We do not use the bilateral form here; hence, $F(s)$ is understood to be the one-sided transform.

The Laplace transform’s lower limit creates a concern: what happens when $f(t)$ has a discontinuity at the origin? If $f(t)$ is continuous at the

origin, like the function in Fig. 12.1(a), $f(0)$ is not ambiguous. However, if $f(t)$ has a finite discontinuity at the origin, like the function in Fig. 12.1(b), should the Laplace transform integral include or exclude the discontinuity? That is, should we make the lower limit 0^- and include the discontinuity, or choose 0^+ as the lower limit and exclude the discontinuity? (We use the notation 0^- and 0^+ to denote values of t just to the left and right of the origin, respectively.) Actually, we may choose either as long as we are consistent. For reasons we explain later, we choose 0^- as the lower limit.

Because we are using 0^- as the lower limit, we note that the integral from 0^- to 0^+ is zero, except when the discontinuity at the origin is an impulse function. We discuss this situation in Section 12.3. The two functions shown in Fig. 12.1 have the same unilateral Laplace transform because there is no impulse function at the origin for the function in Fig. 12.1(b).

The one-sided Laplace transform ignores $f(t)$ for $t < 0^-$. We use initial conditions to account for what happens prior to 0^- . Thus, the Laplace transform predicts the response to a function that begins after initial conditions have been established.

In the discussion that follows, we divide the Laplace transforms into two types: functional transforms and operational transforms. A **functional transform** is the Laplace transform of a specific function, such as $\sin \omega t$, t , e^{-at} , and so on. An **operational transform** defines a general mathematical property of the Laplace transform, such as finding the transform of the derivative of $f(t)$. Before considering functional and operational transforms, we introduce the step and impulse functions.

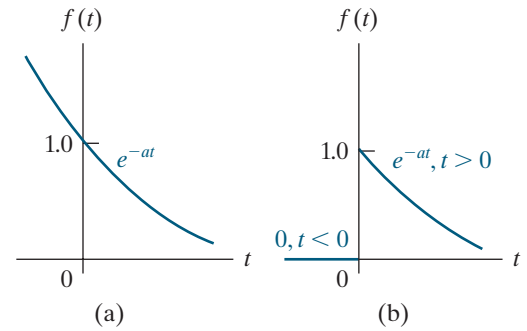


Figure 12.1 ▲ A continuous and discontinuous function at the origin. (a) $f(t)$ is continuous at the origin. (b) $f(t)$ is discontinuous at the origin.

12.2 The Step Function

When a circuit contains a switch, a change in the switch position creates abrupt changes in currents and voltages, as we have seen in previous chapters. An abrupt change is represented mathematically as a discontinuity, which can occur at any instant in time. We represent such discontinuities in the functions that describe the currents and voltages using step and impulse functions.

Figure 12.2 illustrates the step function. The symbol for the step function is $Ku(t)$. The mathematical definition of the **step function** is

$$\begin{aligned} Ku(t) &= 0, & t < 0, \\ Ku(t) &= K, & t > 0. \end{aligned} \quad (12.3)$$

If K is 1, the function in Eq. 12.3 is the **unit step**, $u(t)$.

The step function is not defined at $t = 0$. If we need to define the transition between 0^- and 0^+ , we assume that it is linear and that

$$Ku(0) = 0.5K.$$

As before, 0^- and 0^+ represent symmetric points arbitrarily close to the left and right of the origin. Figure 12.3 illustrates the linear transition from 0^- to 0^+ .

A discontinuity may occur at some time other than $t = 0$; for example, it may occur in sequential switching. A step that occurs at $t = a$ is represented by the function $Ku(t - a)$. Thus

$$\begin{aligned} Ku(t - a) &= 0, & t < a, \\ Ku(t - a) &= K, & t > a. \end{aligned}$$

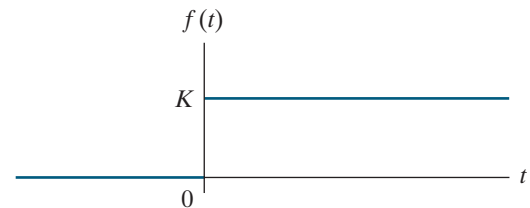


Figure 12.2 ▲ The step function.

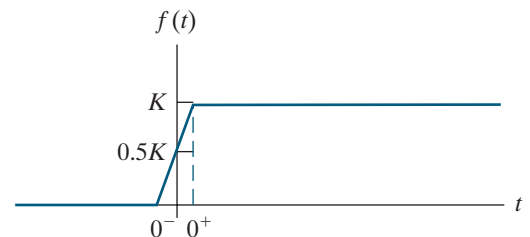


Figure 12.3 ▲ The linear approximation to the step function.

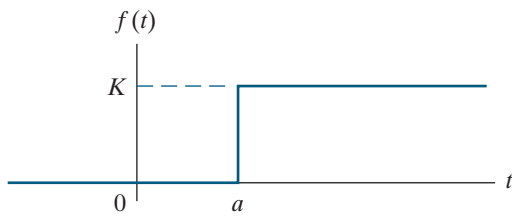


Figure 12.4 ▲ A step function occurring at $t = a$ when $a > 0$.

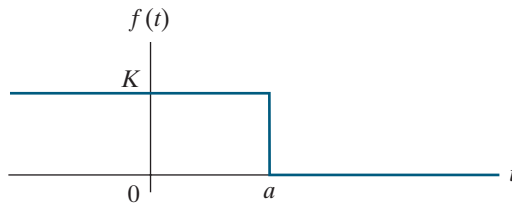


Figure 12.5 ▲ A step function $Ku(a - t)$ for $a > 0$.

If $a > 0$, the step occurs to the right of the origin, and if $a < 0$, the step occurs to the left of the origin. The step function's value is 0 when the argument $t - a$ is negative, and it is K when the argument is positive. Figure 12.4 illustrates a step that occurs at $t = a$.

A step function equal to K for $t < a$ is written as $Ku(a - t)$. Thus

$$Ku(a - t) = K, \quad t < a,$$

$$Ku(a - t) = 0, \quad t > a.$$

The discontinuity is to the left of the origin when $a < 0$. This type of step is shown in Fig. 12.5.

We can add two step functions to create a function that describes a finite-width pulse. For example, the function $K[u(t - 1) - u(t - 3)]$ has the value K for $1 < t < 3$ and the value 0 everywhere else, so it is a finite-width pulse of height K initiated at $t = 1$ and terminated at $t = 3$. In defining this pulse, think of the step function $u(t - 1)$ as “turning on” the constant value K at $t = 1$, and the step function $-u(t - 3)$ as “turning off” the constant value K at $t = 3$. We use step functions to turn on and turn off linear functions at desired times in Example 12.1.

EXAMPLE 12.1

Using Step Functions to Represent a Function of Finite Duration

Use step functions to write an expression for the function illustrated in Fig. 12.6.

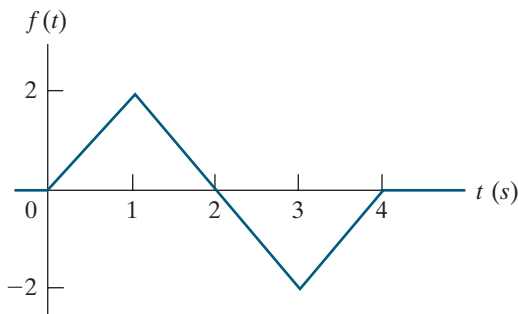


Figure 12.6 ▲ The function for Example 12.1.

Solution

The function shown in Fig. 12.6 is made up of linear segments, so it is a piecewise linear function. You have probably seen such functions defined using a different functional form for each of its time segments. For this function, the piecewise-linear definition is

$$\begin{aligned} &= 0, & t &\leq 0; \\ &= 2t, & 0 &\leq t \leq 1 \text{ s}; \\ f(t) &= -2t + 4, & 1 \text{ s} &\leq t \leq 3 \text{ s}; \\ &= 2t - 8, & 3 \text{ s} &\leq t \leq 4 \text{ s}; \\ &= 0, & t &\geq 4 \text{ s}. \end{aligned}$$

We can construct a single continuous definition for this function using step functions to initiate and

terminate each linear segment at the proper times. In other words, we use the step function to turn on and turn off the three nonzero pieces of the function:

$$\begin{aligned} &+2t, \text{ on at } t = 0, \text{ off at } t = 1; \\ &-2t + 4, \text{ on at } t = 1, \text{ off at } t = 3; \\ &+2t - 8, \text{ on at } t = 3, \text{ off at } t = 4. \end{aligned}$$

These straight line segments and their equations are shown in Fig. 12.7. The expression for $f(t)$, valid for all values of t , is

$$\begin{aligned} f(t) &= 2t[u(t) - u(t - 1)] + (-2t + 4)[u(t - 1) \\ &\quad - u(t - 3)] + (2t - 8)[u(t - 3) - u(t - 4)]. \end{aligned}$$

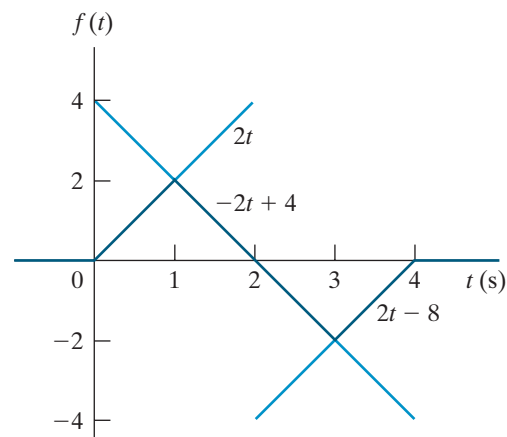


Figure 12.7 ▲ Definition of the three line segments turned on and off with step functions to form the function shown in Fig. 12.6.

SELF-CHECK: Assess your understanding of step functions by trying Chapter Problems 12.1 and 12.2.

12.3 The Impulse Function

An **impulse** is a signal of infinite amplitude and zero duration. Such signals don't exist in nature, but some circuit signals come very close to approximating this definition, so a mathematical model of an impulse is useful. For example, the impulse function¹ enables us to define the derivative at a discontinuity, such as the one in the function of Fig. 12.1(b), and thus to define the Laplace transform of that derivative. Also, voltage and current impulses occur in circuit analysis either because of a switching operation or because the circuit is excited by an impulsive source. We will analyze these situations in Chapter 13, but here we define the impulse function generally.

Describing the Impulse Function

We describe the impulse function by considering how we would define the derivative of the function in Fig. 12.1(b) at its discontinuity. First, we assume that the function varies linearly across the discontinuity, as shown in Fig. 12.8. In this figure, note that as $\epsilon \rightarrow 0$ an abrupt discontinuity occurs at the origin. When we differentiate the function, the derivative between $-\epsilon$ and $+\epsilon$ is constant, with a value of $1/2\epsilon$. For $t > \epsilon$, the derivative is $-ae^{-a(t-\epsilon)}$. Figure 12.9 shows these observations graphically. As ϵ approaches zero, the value of $f'(t)$ between $\pm\epsilon$ approaches infinity. At the same time, the duration of this large value is approaching zero. Furthermore, the area under $f'(t)$ between $\pm\epsilon$ remains constant as $\epsilon \rightarrow 0$. In this example, the area is unity. As ϵ approaches zero, we say that the function between $\pm\epsilon$ approaches a **unit impulse function**, denoted $\delta(t)$. Thus, the derivative of $f(t)$ at the origin approaches a unit impulse function as ϵ approaches zero, or

$$f'(0) \rightarrow \delta(t) \quad \text{as } \epsilon \rightarrow 0.$$

If the area under the impulse function curve is other than unity, the impulse function is denoted $K\delta(t)$, where K is the area. K is often referred to as the **strength** of the impulse function.

To summarize, an impulse function is created from a variable-parameter function whose parameter approaches zero. The variable-parameter function must exhibit the following three characteristics as the parameter approaches zero:

1. The amplitude of the function approaches infinity.
2. The duration of the function approaches zero.
3. The area under the variable-parameter function is constant as the parameter value changes.

Many different variable-parameter functions have these three characteristics. In Fig. 12.8, we used a linear function $f(t) = 0.5t/\epsilon + 0.5$. Another example of a variable-parameter function with the three characteristics is the exponential function:

$$f(t) = \frac{K}{2\epsilon} e^{-|t|/\epsilon}.$$

As ϵ approaches zero, the function becomes infinite at the origin and at the same time decays to zero in an infinitesimal length of time.

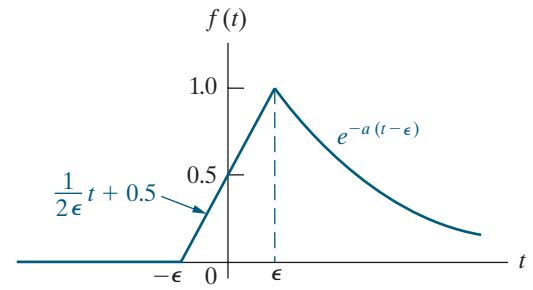


Figure 12.8 ▲ A magnified view of the discontinuity in Fig. 12.1(b), assuming a linear transition between $-\epsilon$ and $+\epsilon$.

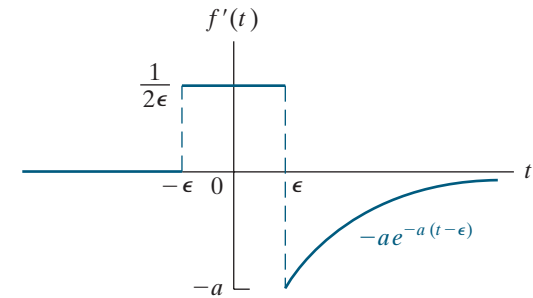


Figure 12.9 ▲ The derivative of the function shown in Fig. 12.8.

¹The impulse function is also known as the Dirac delta function.

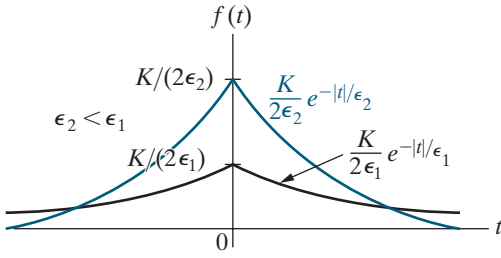


Figure 12.10 ▲ A variable-parameter function used to generate an impulse function.

Figure 12.10 illustrates $f(t)$ as $\epsilon \rightarrow 0$. An impulse function is created as $\epsilon \rightarrow 0$ if the area under the function is independent of ϵ . Thus,

$$\begin{aligned} \text{Area} &= \int_{-\infty}^0 \frac{K}{2\epsilon} e^{t/\epsilon} dt + \int_0^{\infty} \frac{K}{2\epsilon} e^{-t/\epsilon} dt \\ &= \frac{K}{2\epsilon} \cdot \frac{e^{t/\epsilon}}{1/\epsilon} \Big|_{-\infty}^0 + \frac{K}{2\epsilon} \cdot \frac{e^{-t/\epsilon}}{-1/\epsilon} \Big|_0^{\infty} \\ &= \frac{K}{2} + \frac{K}{2} = K, \end{aligned}$$

so the area under the curve is constant and equal to K units. Therefore, as $\epsilon \rightarrow 0$, $f(t) \rightarrow K\delta(t)$.

Defining the Impulse Function and Its Sifting Property

Mathematically, the **impulse function** is defined as

$$\int_{-\infty}^{\infty} K\delta(t) dt = K; \quad (12.4)$$

$$\delta(t) = 0, \quad t \neq 0. \quad (12.5)$$

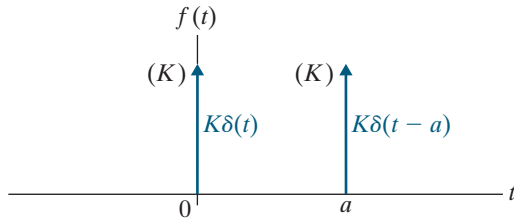


Figure 12.11 ▲ A graphic representation of the impulse $K\delta(t)$ and $K\delta(t-a)$.

Equation 12.4 states that the area under the impulse function is constant. This area represents the strength of the impulse. Equation 12.5 states that the impulse is zero everywhere except at $t = 0$. An impulse that occurs at $t = a$ is denoted $K\delta(t-a)$. The graphic symbol for the impulse function is an arrow. The strength of the impulse is given parenthetically next to the head of the arrow. Figure 12.11 shows the impulses $K\delta(t)$ and $K\delta(t-a)$.

An important property of the impulse function is the **sifting property**, which is expressed as

$$\int_{-\infty}^{\infty} f(t)\delta(t-a) dt = f(a), \quad (12.6)$$

where we assume the function $f(t)$ is continuous at $t = a$, the location of the impulse. Equation 12.6 shows that the impulse function sifts out all values of $f(t)$ except the one at $t = a$. Equation 12.6 follows from Eqs. 12.4 and 12.5, noting that $\delta(t-a)$ is zero everywhere except at $t = a$, and hence the integral can be written

$$\int_{-\infty}^{\infty} f(t)\delta(t-a) dt = \int_{a-\epsilon}^{a+\epsilon} f(t)\delta(t-a) dt.$$

But because $f(t)$ is continuous at a , it takes on the value $f(a)$ as $t \rightarrow a$, so

$$\begin{aligned} \int_{a-\epsilon}^{a+\epsilon} f(a)\delta(t-a) dt &= f(a) \int_{a-\epsilon}^{a+\epsilon} \delta(t-a) dt \\ &= f(a). \end{aligned}$$

Laplace Transform and Derivatives of the Impulse Function

We use the sifting property of the impulse function to find its Laplace transform:

$$\mathcal{L}\{\delta(t)\} = \int_{0^-}^{\infty} \delta(t)e^{-st} dt = e^{-s(0)} = 1,$$

which is an important Laplace transform. We can also define the derivatives of the impulse function and the Laplace transform of these derivatives. We discuss the first derivative, along with its transform, and then state the result for the higher-order derivatives.

The function illustrated in Fig. 12.12(a) generates an impulse function as $\epsilon \rightarrow 0$. Figure 12.12(b) shows the derivative of this impulse-generating function, which is defined as the derivative of the impulse $[\delta'(t)]$ as $\epsilon \rightarrow 0$. The derivative of the impulse function sometimes is referred to as a moment function, or unit doublet.

To find the Laplace transform of $\delta'(t)$, we apply the defining integral to the function shown in Fig. 12.12(b) and, after integrating, let $\epsilon \rightarrow 0$. Then

$$\begin{aligned}\mathcal{L}\{\delta'(t)\} &= \lim_{\epsilon \rightarrow 0} \left[\int_{-\epsilon}^{0^-} \frac{1}{\epsilon^2} e^{-st} dt + \int_{0^+}^{\epsilon} \left(-\frac{1}{\epsilon^2}\right) e^{-st} dt \right] \\ &= \lim_{\epsilon \rightarrow 0} \frac{e^{s\epsilon} + e^{-s\epsilon} - 2}{s\epsilon^2} \\ &= \lim_{\epsilon \rightarrow 0} \frac{se^{s\epsilon} - se^{-s\epsilon}}{2\epsilon s} \\ &= \lim_{\epsilon \rightarrow 0} \frac{s^2 e^{s\epsilon} + s^2 e^{-s\epsilon}}{2s} \\ &= s.\end{aligned}$$

We used l'Hôpital's rule twice in this derivation, to evaluate the indeterminate form $0/0$.

Higher-order derivatives can be generated in a similar manner (see Problem 12.7), and the defining integral can then be used to find the Laplace transforms. The Laplace transform of the n th derivative of the impulse function is

$$\mathcal{L}\{\delta^{(n)}(t)\} = s^n.$$

Finally, note that the derivative of a step function is an impulse function; that is,

$$\delta(t) = \frac{du(t)}{dt}.$$

Figure 12.13 depicts the relationship between the impulse function and the step function. The function shown in Fig. 12.13(a) approaches a unit step function as $\epsilon \rightarrow 0$. The function shown in Fig. 12.13(b)—the derivative of the function in Fig. 12.13(a)—approaches a unit impulse as $\epsilon \rightarrow 0$.

The impulse function is an extremely useful concept in circuit analysis, and we say more about it in the following chapters. We introduced the concept here so that we can include discontinuities at the origin in our definition of the Laplace transform.

SELF-CHECK: Assess your understanding of the impulse function by trying Chapter Problems 12.9 and 12.10.

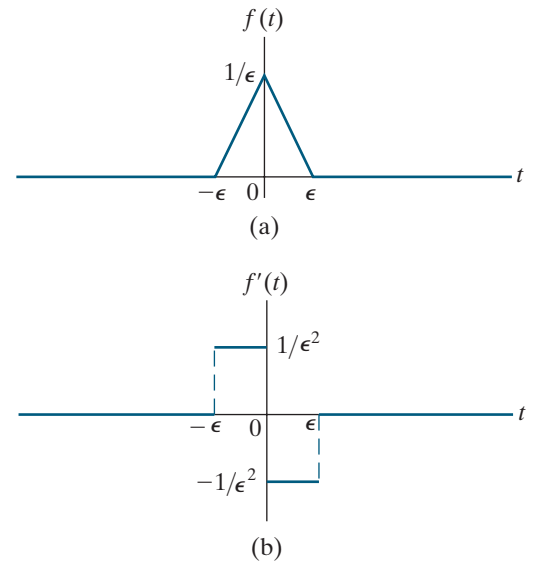


Figure 12.12 ▲ The first derivative of the impulse function. (a) The impulse-generating function used to define the first derivative of the impulse. (b) The first derivative of the impulse-generating function that approaches $\delta'(t)$ as $\epsilon \rightarrow 0$.

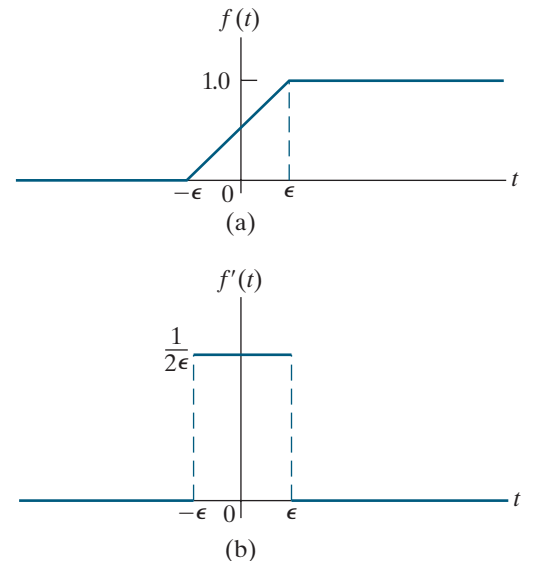


Figure 12.13 ▲ The impulse function as the derivative of the step function: (a) $f(t) \rightarrow u(t)$ as $\epsilon \rightarrow 0$; and (b) $f'(t) \rightarrow \delta(t)$ as $\epsilon \rightarrow 0$.

12.4 Functional Transforms

The Laplace transform of a specified function of t is called a functional transform. Because we are using the unilateral, or one-sided, Laplace transform, we define all functions to be zero for $t < 0^-$.

We derived one functional transform pair in Section 12.3, where we showed that the Laplace transform of the unit impulse function equals 1. Next, we find the Laplace transform of the unit step function at the origin, where

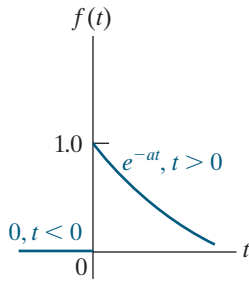


Figure 12.14 ▲ A decaying exponential function.

$$\begin{aligned}\mathcal{L}\{u(t)\} &= \int_0^\infty f(t)e^{-st} dt = \int_{0^+}^\infty 1e^{-st} dt \\ &= \left. \frac{e^{-st}}{-s} \right|_{0^+}^\infty = \frac{1}{s}.\end{aligned}$$

Thus, the Laplace transform of the unit step function is $1/s$.

The Laplace transform of the decaying exponential function shown in Fig. 12.14 is

$$\mathcal{L}\{e^{-at}\} = \int_{0^+}^\infty e^{-at}e^{-st} dt = \int_{0^+}^\infty e^{-(a+s)t} dt = \frac{1}{s+a}.$$

In deriving the Laplace transforms of the unit step function and the decaying exponential function, we used the fact that integration across the discontinuity at the origin is zero.

Let's find the Laplace transform of the sinusoidal function shown in Fig. 12.15. The expression for $f(t)$ for $t > 0^-$ is $\sin \omega t$; hence, the Laplace transform is

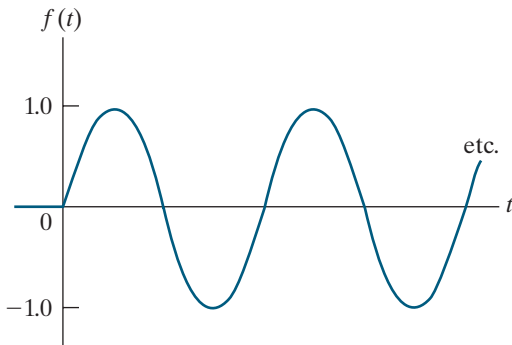


Figure 12.15 ▲ A sinusoidal function for $t > 0$.

$$\begin{aligned}\mathcal{L}\{\sin \omega t\} &= \int_{0^-}^\infty (\sin \omega t)e^{-st} dt \\ &= \int_{0^-}^\infty \left(\frac{e^{j\omega t} - e^{-j\omega t}}{2j} \right) e^{-st} dt \\ &= \int_{0^-}^\infty \frac{e^{-(s-j\omega)t} - e^{-(s+j\omega)t}}{2j} dt \\ &= \frac{1}{2j} \left(\frac{1}{s-j\omega} - \frac{1}{s+j\omega} \right) \\ &= \frac{\omega}{s^2 + \omega^2}.\end{aligned}$$

Table 12.1 gives an abbreviated list of functional Laplace transform pairs. It includes the functions of most interest in an introductory course on circuit applications.

TABLE 12.1 An Abbreviated List of Laplace Transform Pairs

Type	$f(t)$ ($t > 0^-$)	$F(s)$
(impulse)	$\delta(t)$	1
(step)	$u(t)$	$\frac{1}{s}$
(ramp)	t	$\frac{1}{s^2}$
(exponential)	e^{-at}	$\frac{1}{s + a}$
(sine)	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
(cosine)	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
(damped ramp)	te^{-at}	$\frac{1}{(s + a)^2}$
(damped sine)	$e^{-at} \sin \omega t$	$\frac{\omega}{(s + a)^2 + \omega^2}$
(damped cosine)	$e^{-at} \cos \omega t$	$\frac{s + a}{(s + a)^2 + \omega^2}$

ASSESSMENT PROBLEM

Objective 1 — Be able to calculate the Laplace transform of a function using the definition of Laplace transform

12.1 Use the defining integral to

- find the Laplace transform of $\sin(\omega t + \theta)$;
- find the Laplace transform of $\cos(\omega t + \theta)$.

Answer: a) $\frac{\omega \cos \theta + s \sin \theta}{s^2 + \omega^2}$;
b) $\frac{s \cos \theta - \omega \sin \theta}{s^2 + \omega^2}$.

SELF-CHECK: Also try Chapter Problem 12.13.

12.5 Operational Transforms

Operational Laplace transforms define how mathematical operations performed on $f(t)$ affect its Laplace transform, $F(s)$. Operational transforms also define how mathematical operations performed on $F(s)$ affect its corresponding time-domain function $f(t)$. The operations we consider include (1) multiplication by a constant; (2) addition and subtraction; (3) differentiation; (4) integration; (5) translation in the time domain; (6) translation in the frequency domain; and (7) scale changing.

Multiplication by a Constant

From the defining integral, if

$$\mathcal{L}\{f(t)\} = F(s),$$

then

$$\mathcal{L}\{Kf(t)\} = KF(s).$$

Thus, multiplication of $f(t)$ by a constant corresponds to multiplying $F(s)$ by the same constant.

Addition and Subtraction

Addition and subtraction in the time domain translate into addition and subtraction in the frequency domain. Thus if

$$\mathcal{L}\{f_1(t)\} = F_1(s),$$

$$\mathcal{L}\{f_2(t)\} = F_2(s),$$

$$\mathcal{L}\{f_3(t)\} = F_3(s),$$

then

$$\mathcal{L}\{f_1(t) + f_2(t) - f_3(t)\} = F_1(s) + F_2(s) - F_3(s).$$

Use the defining integral to derive this operation transform by recognizing that the integral of a sum of functions equals the sum of each function's integral.

Differentiation

We use the definition of the Laplace transform to find the operational transform for differentiation in the time domain:

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = \int_{0^-}^{\infty} \left[\frac{df(t)}{dt}\right] e^{-st} dt.$$

We integrate by parts to evaluate this integral. Let $u = e^{-st}$ and $dv = [df(t)/dt] dt$ to give

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = e^{-st} f(t) \Big|_{0^-}^{\infty} - \int_{0^-}^{\infty} f(t) (-se^{-st} dt).$$

Because we are assuming that $f(t)$ has a Laplace transform, evaluating $e^{-st} f(t)$ at $t = \infty$ gives zero. We complete the evaluation of the integral to get

$$-f(0^-) + s \int_{0^-}^{\infty} f(t) e^{-st} dt = sF(s) - f(0^-).$$

Thus, the Laplace transform of the derivative of $f(t)$ is

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = sF(s) - f(0^-).$$

This important result shows that differentiation in the time domain transforms to an algebraic operation in the s domain.

Use the Laplace transform of the first derivative of $f(t)$ to find the Laplace transform of higher-order derivatives. For example, to find the Laplace transform of the second derivative of $f(t)$, we first let

$$g(t) = \frac{df(t)}{dt}.$$

Now we find the Laplace transform of $g(t)$:

$$G(s) = sF(s) - f(0^-).$$

But because

$$\frac{dg(t)}{dt} = \frac{d^2f(t)}{dt^2},$$

we know

$$\mathcal{L}\left\{\frac{dg(t)}{dt}\right\} = \mathcal{L}\left\{\frac{d^2f(t)}{dt^2}\right\} = sG(s) - g(0^-).$$

Therefore,

$$\mathcal{L}\left\{\frac{d^2f(t)}{dt^2}\right\} = s^2F(s) - sf(0^-) - \frac{df(0^-)}{dt}.$$

We find the Laplace transform of the n th derivative by successively applying the preceding process, which leads to the general result

$$\begin{aligned} \mathcal{L}\left\{\frac{d^n f(t)}{dt^n}\right\} &= s^n F(s) - s^{n-1} f(0^-) - s^{n-2} \frac{df(0^-)}{dt} \\ &\quad - s^{n-3} \frac{d^2 f(0^-)}{dt^2} - \dots - \frac{d^{n-1} f(0^-)}{dt^{n-1}}. \end{aligned}$$

Integration

We find the Laplace transform of the integral of $f(t)$ by again applying the defining integral:

$$\mathcal{L}\left\{\int_{0^-}^t f(x) dx\right\} = \int_{0^-}^{\infty} \left[\int_{0^-}^t f(x) dx\right] e^{-st} dt.$$

Integrate by parts to evaluate the integral on the right-hand side of this expression. Let

$$u = \int_{0^-}^t f(x) dx,$$

$$dv = e^{-st} dt.$$

Then

$$du = f(t) dt,$$

$$v = -\frac{e^{-st}}{s}.$$

The integration-by-parts formula yields

$$\mathcal{L}\left\{\int_{0^-}^t f(x) dx\right\} = -\frac{e^{-st}}{s} \int_{0^-}^t f(x) dx \Big|_{0^-}^{\infty} + \int_{0^-}^{\infty} \frac{e^{-st}}{s} f(t) dt.$$

The first term on the right-hand side is zero at both the upper and lower limits. The value at the lower limit is zero because both limits on the integral are the same. The value at the upper limit is zero because $e^{-st} \rightarrow 0$ as $t \rightarrow \infty$. The second term on the right-hand side is $F(s)/s$; therefore

$$\mathcal{L}\left\{\int_{0^-}^t f(x) dx\right\} = \frac{F(s)}{s},$$

which reveals that integration in the time domain transforms to multiplication by $1/s$ in the s domain. We have therefore demonstrated that the Laplace transform translates a set of integrodifferential equations into a set of algebraic equations.

Translation in the Time Domain

If we start with any function $f(t)u(t)$, we can represent the same function, translated in time by the constant a , as $f(t-a)u(t-a)$.² To find the Laplace transform of $f(t-a)u(t-a)$, start with the defining integral:

$$\begin{aligned} \mathcal{L}\{f(t-a)u(t-a)\} &= \int_{0^-}^{\infty} u(t-a)f(t-a)e^{-st} dt \\ &= \int_a^{\infty} f(t-a)e^{-st} dt. \end{aligned}$$

In writing this equation, we took advantage of $u(t-a) = 0$ for $t < a$. Now change the variable of integration by letting $x = t-a$. Then $x = 0$ when $t = a$, $x = \infty$ when $t = \infty$ and $dx = dt$. Rewrite the integral as

$$\begin{aligned} \mathcal{L}\{f(t-a)u(t-a)\} &= \int_0^{\infty} f(x)e^{-s(x+a)} dx \\ &= e^{-sa} \int_0^{\infty} f(x)e^{-sx} dx \\ &= e^{-as} F(s). \end{aligned}$$

Thus

$$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as} F(s), \quad a > 0.$$

Translation in the time domain corresponds to multiplication by an exponential in the frequency domain.

²Note that throughout we multiply any arbitrary function $f(t)$ by the unit step function $u(t)$ to ensure that the resulting function is defined for all positive time.

For example, knowing that

$$\mathcal{L}\{tu(t)\} = \frac{1}{s^2},$$

we can use this operational transform to find the Laplace transform of $(t - a)u(t - a)$:

$$\mathcal{L}\{(t - a)u(t - a)\} = \frac{e^{-as}}{s^2}.$$

Translation in the Frequency Domain

Translation in the frequency domain corresponds to multiplication by an exponential in the time domain:

$$\mathcal{L}\{e^{-at} f(t)\} = F(s + a),$$

which follows from the defining integral. Problem 12.15 asks you to derive this result.

We can use this operational transform to derive new transform pairs. Thus, knowing that

$$\mathcal{L}\{\cos \omega t\} = \frac{s}{s^2 + \omega^2},$$

we use the effect of translation in the frequency domain to deduce that

$$\mathcal{L}\{e^{-at} \cos \omega t\} = \frac{s + a}{(s + a)^2 + \omega^2}.$$

Scale Changing

The scale-change property gives the relationship between $f(t)$ and $F(s)$ when the time variable is multiplied by a positive constant:

$$\mathcal{L}\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right), \quad a > 0.$$

The derivation is left to Problem 12.16. The scale-change property is particularly useful in experimental work, when time-scale changes are made to assist in building a model of a system.

We use this operational transform to formulate new transform pairs. Thus, knowing that

$$\mathcal{L}\{\cos t\} = \frac{s}{s^2 + 1},$$

we use the effect of scale changing to show that

$$\mathcal{L}\{\cos \omega t\} = \frac{1}{\omega} \frac{s/\omega}{(s/\omega)^2 + 1} = \frac{s}{s^2 + \omega^2}.$$

Table 12.2 lists these operational transforms.

TABLE 12.2 An Abbreviated List of Operational Transforms

Operation	$f(t)$	$F(s)$
Multiplication by a constant	$Kf(t)$	$KF(s)$
Addition/subtraction	$f_1(t) + f_2(t) - f_3(t) + \dots$	$F_1(s) + F_2(s) - F_3(s) + \dots$
First derivative (time)	$\frac{df(t)}{dt}$	$sF(s) - f(0^-)$
Second derivative (time)	$\frac{d^2f(t)}{dt^2}$	$s^2F(s) - sf(0^-) - \frac{df(0^-)}{dt}$
n th derivative (time)	$\frac{d^nf(t)}{dt^n}$	$s^nF(s) - s^{n-1}f(0^-) - s^{n-2}\frac{df(0^-)}{dt} - s^{n-3}\frac{d^2f(0^-)}{dt^2} - \dots - \frac{d^{n-1}f(0^-)}{dt^{n-1}}$
Time integral	$\int_0^t f(x) \, dx$	$\frac{F(s)}{s}$
Translation in time	$f(t - a)u(t - a), \, a > 0$	$e^{-as}F(s)$
Translation in frequency	$e^{-at}f(t)$	$F(s + a)$
Scale changing	$f(at), \, a > 0$	$\frac{1}{a}F\left(\frac{s}{a}\right)$
First derivative (s)	$tf(t)$	$-\frac{dF(s)}{ds}$
n th derivative (s)	$t^n f(t)$	$(-1)^n \frac{d^n F(s)}{ds^n}$
s integral	$\frac{f(t)}{t}$	$\int_s^\infty F(u) \, du$

ASSESSMENT PROBLEM

Objective 1 — Be able to calculate the Laplace transform of a function using the Laplace transform table or a table of operational transforms

12.2 Use the appropriate operational transform from Table 12.2 to find the Laplace transform of each function:

- a) $t^3 e^{-at}$;
- b) $\frac{d}{dt} (e^{-at} \cos \omega t)$;
- c) $\int_{0^-}^t e^{-ax} \sin \omega x \, dx$.

- Answer:
- a) $\frac{6}{(s + a)^4}$;
 - b) $\frac{-a^2 - sa - \omega^2}{(s + a)^2 + \omega^2}$;
 - c) $\frac{\omega}{s[(s + a)^2 + \omega^2]}$.

SELF-CHECK: Also try Chapter Problems 12.17 and 12.18.

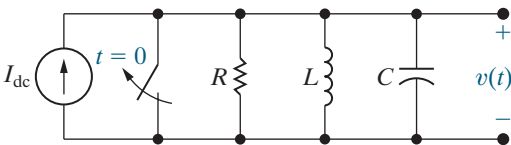


Figure 12.16 A parallel RLC circuit.

12.6 Applying the Laplace Transform

We now use the Laplace transform to solve the ordinary integrodifferential equations that describe the behavior of a lumped-parameter circuit, such as the one shown in Fig. 12.16. We assume that no initial energy is stored in the circuit at the instant the switch is opened. The problem is to find the time-domain expression for $v(t)$ when $t \geq 0$.

We begin by writing the integrodifferential equation that $v(t)$ must satisfy, using a single KCL equation to describe the circuit. Summing the currents away from the top node in the circuit gives:

$$\frac{v(t)}{R} + \frac{1}{L} \int_0^t v(x) dx + C \frac{dv(t)}{dt} = I_{\text{dc}} u(t).$$

Note that we represented the switch opening at $t = 0$ with the product of the source current and the unit step function, $I_{\text{dc}} u(t)$.

We transform the KCL equation into the s -domain using four operational transforms (multiplication by a constant, addition, integration, and differentiation) and one functional transform (unit step) to get

$$\frac{V(s)}{R} + \frac{1}{L} \frac{V(s)}{s} + C[sV(s) - v(0^-)] = I_{\text{dc}} \left(\frac{1}{s} \right).$$

The result is an algebraic equation with one unknown variable, $V(s)$. We are assuming that the circuit parameters R , L , and C , as well as the source current I_{dc} , are known; the initial voltage on the capacitor $v(0^-)$ is zero because the initial energy stored in the circuit is zero. Thus, we have reduced the problem to solving an algebraic equation.

Solving for $V(s)$ gives

$$V(s) \left(\frac{1}{R} + \frac{1}{sL} + sC \right) = \frac{I_{\text{dc}}}{s},$$

$$V(s) = \frac{I_{\text{dc}}/C}{s^2 + (1/RC)s + (1/LC)}.$$

To find $v(t)$, we must inverse-transform the expression for $V(s)$. We denote this inverse operation as

$$v(t) = \mathcal{L}^{-1}\{V(s)\}.$$

The inverse transform, which takes the solution from the s -domain to the time domain, is the subject of Section 12.7. In that section, we also present a final, critical step: checking the validity of the resulting time-domain expression. This final step is not unique to the Laplace transform; it is always a good idea to test any derived solution to be sure it makes sense in terms of known system behavior.

Before continuing, we simplify the notation by dropping the parenthetical t in time-domain expressions and the parenthetical s in frequency-domain expressions. We use lowercase letters for all time-domain variables, and we represent the corresponding s -domain variables with uppercase letters. Thus

$$\mathcal{L}\{v\} = V \quad \text{or} \quad v = \mathcal{L}^{-1}\{V\},$$

$$\mathcal{L}\{i\} = I \quad \text{or} \quad i = \mathcal{L}^{-1}\{I\},$$

$$\mathcal{L}\{f\} = F \quad \text{or} \quad f = \mathcal{L}^{-1}\{F\},$$

and so on.

Example 12.2 supplies component values for the circuit in Fig. 12.16 and uses Laplace transforms to predict the circuit's output voltage.

EXAMPLE 12.2 Using Laplace Transforms to Predict a Circuit's Response

Suppose for the circuit in Fig. 12.16, $I_{dc} = 24 \text{ mA}$, $R = 400 \Omega$, $L = 25 \text{ mH}$, and $C = 25 \text{ nF}$. There is no energy stored in the circuit when the switch opens at $t = 0$. Find the Laplace transform of $v(t)$ and use it to predict the functional form of $v(t)$.

Solution

Using the expression for $v(s)$ found for the circuit in Fig. 12.16,

$$\begin{aligned} V(s) &= \frac{I_{dc}/C}{s^2 + (1/RC)s + (1/LC)} \\ &= \frac{0.024/(25 \times 10^{-9})}{s^2 + [1/(400)(25 \times 10^{-9})]s + [1/(0.025)(25 \times 10^{-9})]} \\ &= \frac{96 \times 10^4}{s^2 + 10^5 s + 16 \times 10^8}. \end{aligned}$$

The expression for $V(s)$ is not a familiar functional Laplace transform, so we cannot use Tables 12.1 and 12.2 to find its inverse transform. But note that we can rewrite the denominator of $V(s)$ as the product of two factors:

$$V(s) = \frac{96 \times 10^4}{(s + 20,000)(s + 80,000)}.$$

Now, recognize that we can write $V(s)$ as the sum of two terms, with each factor appearing in the denominator of one term:

$$V(s) = \frac{K_1}{(s + 20,000)} + \frac{K_2}{(s + 80,000)}.$$

Each of these terms looks like a familiar Laplace transform. Using the transform for the exponential function (Table 12.1) and the operational transforms for multiplication by a constant and addition (Table 12.2) we predict that $v(t)$ is the sum of two exponential terms in the form $e^{-20,000t}$ and $e^{-80,000t}$.

SELF-CHECK: Assess your understanding of this material by trying Chapter Problem 12.32.

12.7 Inverse Transforms

The expression for $V(s)$ in Example 12.2 is a **rational** function of s . This means we can write $V(s)$ as a ratio of two polynomials in s where only integer powers of s appear in the polynomials. For linear, lumped-parameter circuits with constant component values, the s -domain expressions for the unknown voltages and currents are always rational functions of s . (For verification, work Problems 12.27–12.31.) If we can inverse-transform rational functions of s , we can find the time-domain expressions for the voltages and currents. This section presents a straightforward and systematic technique for finding the inverse transform of a rational function.

In general, we need to find the inverse transform of a function that has the form

$$F(s) = \frac{N(s)}{D(s)} = \frac{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}. \quad (12.7)$$

The coefficients a and b are real constants, and the exponents m and n are positive integers. The ratio $N(s)/D(s)$ is called a **proper rational function** if $m > n$, and an **improper rational function** if $m \leq n$. Only a proper rational function can be expanded as a sum of partial fractions. This restriction poses no problem, as we show at the end of this section.

Partial Fraction Expansion: Proper Rational Functions

A proper rational function is expanded into a sum of partial fractions by writing a term or a series of terms for each root of $D(s)$. Thus, $D(s)$ must be factored before we construct a partial fraction expansion. For each distinct root of $D(s)$, a single term appears in the sum of partial fractions. For each multiple root of $D(s)$ of multiplicity r , the sum of partial fractions contains r terms. For example, in the rational function

$$\frac{s + 6}{s(s + 3)(s + 1)^2},$$

the denominator has four roots. Two of these roots are distinct—namely, at $s = 0$ and $s = -3$. A multiple root of multiplicity 2 occurs at $s = -1$. Thus, the partial fraction expansion takes the form

$$\frac{s + 6}{s(s + 3)(s + 1)^2} \equiv \frac{K_1}{s} + \frac{K_2}{s + 3} + \frac{K_3}{(s + 1)^2} + \frac{K_4}{s + 1}.$$

To find the inverse transform from the sum of partial fractions, we identify the $f(t)$ corresponding to each term in the sum using the functional and operational transform tables. Use Tables 12.1 and 12.2 to verify that

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{K_1}{s} + \frac{K_2}{s + 3} + \frac{K_3}{(s + 1)^2} + \frac{K_4}{s + 1} \right\} \\ = (K_1 + K_2 e^{-3t} + K_3 t e^{-t} + K_4 e^{-t}) u(t). \end{aligned}$$

Now we need to find the numerator coefficients ($K_1, K_2, K_3, \dots$) that appear in each partial fraction term. There are only four different types of partial fraction terms because the roots of $D(s)$ can be (1) real and distinct; (2) complex and distinct; (3) real and repeated; or (4) complex and repeated. We develop a technique to determine the numerator coefficient for each type of partial fraction term. Before doing so, a few general comments are in order.

We used the identity sign $\equiv$ in the partial fraction expansion to emphasize that expanding a rational function into a sum of partial fractions establishes an identical equation. This means that both sides of the equation must be the same for all values of the variable s . Also, the identity relationship must hold when the same mathematical operation is applied to both sides. These observations will help us calculate the coefficient values.

Before creating a partial fraction expansion, you should verify that the rational function is proper. This check is important because the procedure for finding the partial fraction coefficients will not prevent you from generating invalid results if the rational function is improper. We present a procedure for checking the coefficients, but you can avoid wasted effort by always asking, “Is $F(s)$ a proper rational function?”

Partial Fraction Expansion: Distinct Real Roots of $D(s)$

Let’s find the partial fraction expansion of a proper rational function whose denominator has distinct real roots. For example,

$$F(s) = \frac{N(s)}{(s + p_1)(s + p_2)(s + p_3)} \equiv \frac{K_1}{(s + p_1)} + \frac{K_2}{(s + p_2)} + \frac{K_3}{(s + p_3)}.$$

To find K_1 for the first partial fraction term, we multiply both sides of the identity by the denominator beneath K_1 to get

$$\frac{N(s)}{(s + p_2)(s + p_3)} \equiv K_1 + \frac{K_2(s + p_1)}{(s + p_2)} + \frac{K_3(s + p_1)}{(s + p_3)}.$$

Then when we evaluate both sides of the identity for $s = -p_1$, which is the root of the partial fraction term whose coefficient is K_1 :

$$\begin{aligned} \left. \frac{N(s)}{(s + p_2)(s + p_3)} \right|_{s=-p_1} &\equiv K_1 + \left. \frac{K_2(s + p_1)}{(s + p_2)} \right|_{s=-p_1} + \left. \frac{K_3(s + p_1)}{(s + p_3)} \right|_{s=-p_1} = K_1. \end{aligned}$$

The right-hand side is always the desired K , and the left-hand side is always its numerical value. To find K_2 and K_3 , repeat the steps used to find K_1 .

Example 12.3 illustrates this process.

EXAMPLE 12.3

Finding the Inverse Laplace Transform When $F(s)$ Has Distinct Real Roots

Use partial fraction expansion to find the inverse Laplace transform of

$$F(s) = \frac{96(s + 5)(s + 12)}{s(s + 8)(s + 6)}.$$

Solution

We begin by noting that $F(s)$ is a proper rational function. The partial fraction expansion of $F(s)$ is

$$F(s) = \frac{96(s + 5)(s + 12)}{s(s + 8)(s + 6)} \equiv \frac{K_1}{s} + \frac{K_2}{(s + 8)} + \frac{K_3}{(s + 6)}.$$

To find the value of K_1 , we multiply both sides by s and then evaluate both sides at $s = 0$:

$$\left. \frac{96(s + 5)(s + 12)}{(s + 8)(s + 6)} \right|_{s=0} \equiv K_1 + \left. \frac{K_2 s}{s + 8} \right|_{s=0} + \left. \frac{K_3 s}{s + 6} \right|_{s=0},$$

or

$$\frac{96(5)(12)}{8(6)} = K_1 = 120.$$

To find the value of K_2 , we multiply both sides by $s + 8$ and then evaluate both sides at $s = -8$:

$$\begin{aligned} \left. \frac{96(s + 5)(s + 12)}{s(s + 6)} \right|_{s=-8} &\equiv \left. \frac{K_1(s + 8)}{s} \right|_{s=-8} + K_2 + \left. \frac{K_3(s + 8)}{(s + 6)} \right|_{s=-8}, \end{aligned}$$

or

$$\frac{96(-3)(4)}{(-8)(-2)} = K_2 = -72.$$

Then K_3 is

$$\left. \frac{96(s + 5)(s + 12)}{s(s + 8)} \right|_{s=-6} = K_3 = 48.$$

Therefore,

$$\frac{96(s + 5)(s + 12)}{s(s + 8)(s + 6)} \equiv \frac{120}{s} - \frac{72}{(s + 8)} + \frac{48}{(s + 6)}.$$

It is a good idea to test this result. While the choice of test values is completely open, we choose values that are easy to verify. For example, testing at either -5 or -12 is convenient because in both cases the left-hand side reduces to zero. Choosing -5 yields

$$\frac{120}{-5} - \frac{72}{3} + \frac{48}{1} = -24 - 24 + 48 = 0,$$

whereas testing -12 gives

$$\frac{120}{-12} - \frac{72}{-4} + \frac{48}{-6} = -10 + 18 - 8 = 0.$$

Now find the inverse transform of $F(s)$ using the tables of functional and operational transforms (Tables 12.1 and 12.2):

$$\mathcal{L}^{-1} \left\{ \frac{96(s + 5)(s + 12)}{s(s + 8)(s + 6)} \right\} = (120 - 78e^{-8t} + 48e^{-6t})u(t).$$

ASSESSMENT PROBLEMS

Objective 2—Be able to calculate the inverse Laplace transform using partial fraction expansion and the Laplace transform table

12.3 Find $f(t)$ if

$$F(s) = \frac{8s^2 + 37s + 32}{(s+1)(s+2)(s+4)}.$$

Answer: $f(t) = (e^{-t} + 5e^{-2t} + 2e^{-4t})u(t)$.

12.4 Find $f(t)$ if

$$F(s) = \frac{10s^2 + 76s + 160}{(s+2)(s+4)(s+6)}.$$

Answer: $f(t) = (6e^{-2t} - 4e^{-4t} + 8e^{-6t})u(t)$.

12.5 Suppose for the circuit in Fig. 12.16,

$I_{dc} = 15 \text{ mA}$, $R = 800 \Omega$, $C = 1 \mu\text{F}$, and $L = 4 \text{ H}$. There is no energy stored in the circuit when the switch opens at $t = 0$.

a) Find $V(s)$, the Laplace transform of $v(t)$.

b) Find $v(t)$ by finding the inverse transform of the partial fraction expansion of $V(s)$.

Answer: a) $V(s) = \frac{15,000}{s^2 + 800s + 25 \times 10^4}$;

b) $v(t) = [20e^{-250t} - 20e^{-1000t}]u(t) \text{ V}$.

SELF-CHECK: Also try Chapter Problems 12.42(a) and (b).

Partial Fraction Expansion: Distinct Complex Roots of $D(s)$

We begin by noting that if $F(s)$ describes the Laplace transform of a voltage or current in a physically realizable circuit, the factors of $D(s)$ that have complex roots will always come in conjugate pairs. Let's assume that

$$F(s) = \frac{A(s + z_1)}{(s + \alpha - j\beta)(s + \alpha + j\beta)} \equiv \frac{K_1}{(s + \alpha - j\beta)} + \frac{K_2}{(s + \alpha + j\beta)}.$$

Here we assume that z_1 is a real number.

We find the unknown coefficients, K_1 and K_2 , using the same technique we employed when the factors of $D(s)$ were real and distinct. As you will see, the only difference is that the algebra involves complex numbers. To find K_1 , we multiply both sides of the identity by the denominator beneath K_1 to get

$$\frac{A(s + z_1)}{(s + \alpha + j\beta)} \equiv K_1 + \frac{K_2(s + \alpha - j\beta)}{(s + \alpha + j\beta)}.$$

Then when we evaluate both sides of the identity for $s = -\alpha + j\beta$, which is the root of the partial fraction term whose coefficient is K_1 :

$$\left. \frac{A(s + z_1)}{(s + \alpha + j\beta)} \right|_{s=-\alpha+j\beta} = K_1 + \left. \frac{K_2(s + \alpha - j\beta)}{(s + \alpha + j\beta)} \right|_{s=-\alpha+j\beta} = K_1.$$

We evaluate the left-hand side of the identity to get an expression for K_1 :

$$K_1 = \frac{A(-\alpha + z_1 + j\beta)}{(j2\beta)} = \frac{A}{2} + j\left(\frac{\alpha - z_1}{2\beta}\right).$$

As expected, K_1 is a complex number. We repeat the steps used to find K_1 when finding the expression for K_2 :

$$\begin{aligned}\frac{A(s + z_1)}{(s + \alpha - j\beta)} &= \frac{K_1(s + \alpha + j\beta)}{(s + \alpha - j\beta)} + K_2; \\ \left. \frac{A(s + z_1)}{(s + \alpha - j\beta)} \right|_{s=-\alpha-j\beta} &= \left. \frac{K_1(s + \alpha + j\beta)}{(s + \alpha - j\beta)} \right|_{s=-\alpha-j\beta} + K_2; \\ K_2 &= \frac{A(-\alpha + z_1 - j\beta)}{(-j2\beta)} = \frac{A}{2} - j\left(\frac{\alpha - z_1}{2\beta}\right).\end{aligned}$$

Compare the expression for K_1 with the expression for K_2 ; K_1 and K_2 are conjugates. This will always be the case when $D(s)$ has complex roots, so we need only calculate one coefficient, since the other is its conjugate. We can use the following general polar form for K_1 and K_2 :

$$K_1 = |K|e^{j\theta} \quad \text{and} \quad K_2 = |K|e^{-j\theta}.$$

Using the polar form for the general coefficients K_1 and K_2 , we can find the inverse Laplace transform of $F(s)$, with the help of the functional and operational transform tables (Tables 12.1 and 12.2) and Euler's identity:

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{|K|e^{j\theta}}{s + \alpha - j\beta} + \frac{|K|e^{-j\theta}}{s + \alpha + j\beta}\right\} &= |K|e^{j\theta}e^{-(\alpha - j\beta)t} + |K|e^{-j\theta}e^{-(\alpha + j\beta)t} \\ &= |K|e^{-\alpha t}e^{j(\beta t + \theta)} + |K|e^{-\alpha t}e^{-j(\beta t + \theta)} \\ &= |K|e^{-\alpha t}[\cos(\beta t + \theta) + j\sin(\beta t + \theta)] \\ &\quad + |K|e^{-\alpha t}[\cos(\beta t + \theta) - j\sin(\beta t + \theta)] \\ &= 2|K|e^{-\alpha t}\cos(\beta t + \theta).\end{aligned}$$

Because distinct complex roots appear frequently in lumped-parameter linear circuit analysis, we summarize these results with a new transform pair. Whenever $D(s)$ contains distinct complex roots—that is, factors of the form $(s + \alpha - j\beta)(s + \alpha + j\beta)$ —a pair of terms of the form

$$\frac{K}{s + \alpha - j\beta} + \frac{K^*}{s + \alpha + j\beta}$$

appears in the partial fraction expansion, where the partial fraction coefficient is, in general, a complex number. In polar form,

$$K = |K|e^{j\theta} = |K|\angle\theta, \quad (12.8)$$

where $|K|$ denotes the magnitude of the complex coefficient. Then

$$K^* = |K|e^{-j\theta} = |K|\angle-\theta. \quad (12.9)$$

The pair of complex conjugate partial fraction terms always inverse-transforms as

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{K}{s + \alpha - j\beta} + \frac{K^*}{s + \alpha + j\beta}\right\} \\ = 2|K|e^{-\alpha t}\cos(\beta t + \theta).\end{aligned} \quad (12.10)$$

In applying Eq. 12.10, it is important to note that K is defined as the coefficient associated with the denominator $s + \alpha - j\beta$, and K^* is defined as the coefficient associated with the denominator $s + \alpha + j\beta$.

Example 12.4 finds the inverse Laplace transform of an s -domain function with distinct complex roots.

EXAMPLE 12.4

Finding the Inverse Laplace Transform When $F(s)$ Has Distinct Complex Roots

Use partial fraction expansion to find the inverse Laplace transform of

$$F(s) = \frac{100(s+3)}{(s+6)(s^2+6s+25)}.$$

Solution

We begin by noting that $F(s)$ is a proper rational function. Next we find the roots of the quadratic term $s^2 + 6s + 25$:

$$s^2 + 6s + 25 = (s + 3 - j4)(s + 3 + j4).$$

With the denominator in factored form, we create the partial fraction expansion:

$$\begin{aligned} \frac{100(s+3)}{(s+6)(s^2+6s+25)} &\equiv \\ \frac{K_1}{s+6} + \frac{K_2}{s+3-j4} + \frac{K_2^*}{s+3+j4}. \end{aligned}$$

Find K_1 and K_2 using the same process employed in Example 12.3:

$$K_1 = \left. \frac{100(s+3)}{s^2+6s+25} \right|_{s=-6} = \frac{100(-3)}{25} = -12,$$

$$\begin{aligned} K_2 &= \left. \frac{100(s+3)}{(s+6)(s+3+j4)} \right|_{s=-3-j4} = \frac{100(j4)}{(3+j4)(j8)} \\ &= 6 - j8 = 10e^{-j53.13^\circ}. \end{aligned}$$

Thus

$$\begin{aligned} \frac{100(s+3)}{(s+6)(s^2+6s+25)} &\equiv \\ &\equiv \frac{-12}{s+6} + \frac{10 \angle -53.13^\circ}{s+3-j4} + \frac{10 \angle 53.13^\circ}{s+3+j4}. \end{aligned}$$

Before inverse-transforming the terms in the partial fraction expansion, we check the expansion numerically. We test using $s = -3$ because the left-hand side reduces to zero at this value:

$$\begin{aligned} F(s) &= \frac{-12}{3} + \frac{10 \angle -53.13^\circ}{-j4} + \frac{10 \angle 53.13^\circ}{j4} \\ &= -4 + 2.5 \angle 36.87^\circ + 2.5 \angle -36.87^\circ \\ &= -4 + 2.0 + j1.5 + 2.0 - j1.5 = 0. \end{aligned}$$

Finally, perform the inverse-transform, using the functional and operational transform tables (Tables 12.1 and 12.2) and the new transform pair in Eq. 12.10:

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{100(s+3)}{(s+6)(s^2+6s+25)} \right\} &= \\ &= [-12e^{-6t} + 20e^{-3t} \cos(4t - 53.13^\circ)]u(t). \end{aligned}$$

ASSESSMENT PROBLEMS

Objective 2—Be able to calculate the inverse Laplace transform using partial fraction expansion and the Laplace transform table

12.6 Find $f(t)$ if

$$F(s) = \frac{22s^2 + 60s + 58}{(s+1)(s^2+4s+5)}.$$

Answer: $f(t) = [10e^{-t} + 20e^{-2t} \cos(t + 53.13^\circ)]u(t).$

- 12.7** Suppose for the circuit in Fig. 12.16, $I_{dc} = 15 \text{ mA}$, $R = 1250 \, \Omega$, $L = 4 \text{ H}$, and $C = 1 \, \mu\text{F}$. These are the same values used in Assessment Problem 12.5 except for the value of R . There is no energy stored in the circuit when the switch opens at $t = 0$.
- a) Find $V(s)$, the Laplace transform of $v(t)$.

- b) Find $v(t)$ by finding the inverse transform of the partial fraction expansion of $V(s)$.

Answer: a) $V(s) = \frac{15,000}{s^2 + 800s + 25 \times 10^4}$;
 b) $v(t) = (50e^{-400t} \sin 300t) u(t)$.

SELF-CHECK: Also try Chapter Problems 12.43(c) and (d).

Partial Fraction Expansion: Repeated Real Roots of $D(s)$

We describe two methods for finding the partial fraction coefficients for terms generated by a multiple root with multiplicity r . Method A requires you to solve several simultaneous algebraic equations. Method B is a modification of the process we have been using to find the coefficients for the partial fraction terms associated with distinct roots. Both methods begin with the identity relating the original s -domain function, which must be a proper rational function, and its partial fraction expansion:

$$\frac{N(s)}{D(s)} \equiv \frac{K_1}{(s + p)^r} + \frac{K_2}{(s + p)^{r-1}} + \cdots + \frac{K_r}{(s + p)}.$$

Method A

1. Combine all terms in the partial fraction expansion over the common denominator, $D(s)$. Call the numerator of this new function $N_1(s)$.
2. Collect all of the terms in the numerator $N_1(s)$ according to their power of s . The coefficient of each power of s will include some of the unknown partial fraction coefficients.
3. Equate the coefficient of each power of s in $N_1(s)$ with the coefficient of the corresponding power of s in $N(s)$. The result is a collection of simultaneous equations whose unknowns are the partial fraction coefficients.
4. Solve the simultaneous equations to find the partial fraction coefficients.

Method B

1. Multiply both sides of the identity defining the partial fraction expansion by the multiple root raised to its r th power. Call the resulting identity I_r .
2. Find K in the numerator of the factor raised to the r th power by evaluating both sides of I_r at the multiple root.
3. To find K in the numerator of the factor raised to the $(r - 1)$ power, differentiate both sides of I_r with respect to s . Call the resulting identity $I_{(r-1)}$. Evaluate both sides of $I_{(r-1)}$ at the multiple root. The right-hand side is always the desired K , and the left-hand side is always its numerical value.
4. Repeat Step 3 to find the remaining partial fraction coefficients by differentiating $I_{(r-1)}$ to get $I_{(r-2)}$ and so on. In total, you will have differentiated $I_r(r - 1)$ times.

Example 12.5 uses Method A to find the inverse Laplace transform for an s -domain function with repeated real roots.

EXAMPLE 12.5**Finding the Inverse Laplace Transform When $F(s)$ Has Repeated Real Roots**

Use partial fraction expansion to find the inverse Laplace transform of

$$F(s) = \frac{100(s + 25)}{s(s + 5)^3}.$$

Solution

We begin by noting that $F(s)$ is a proper rational function. Next we find the partial fraction expansion of $F(s)$:

$$\frac{100(s + 25)}{s(s + 5)^3} \equiv \frac{K_1}{s} + \frac{K_2}{(s + 5)^3} + \frac{K_3}{(s + 5)^2} + \frac{K_4}{s + 5}.$$

We will use Method A to find the partial fraction coefficients. Begin by multiplying the numerator and denominator of each term in the partial fraction expansion with an expression that creates a denominator of $s(s + 5)^3$ in each term:

$$\begin{aligned} \frac{100(s + 25)}{s(s + 5)^3} &\equiv \frac{K_1(s + 5)^3}{s(s + 5)^3} + \frac{K_2s}{s(s + 5)^3} \\ &\quad + \frac{K_3s(s + 5)}{s(s + 5)^3} + \frac{K_4s(s + 5)^2}{s(s + 5)^3}. \end{aligned}$$

Combine the terms on the right-hand side over their common denominator, and expand the resulting numerator by collecting the coefficients for each power of s . The numerator on the right-hand side is

$$\begin{aligned} (K_1 + K_4)s^3 + (15K_1 + K_3 + 10K_4)s^2 \\ + (75K_1 + K_2 + 5K_3 + 25K_4)s \\ + (125K_1). \end{aligned}$$

Equate the coefficients of each power of s in the numerators on the right-hand side and left-hand side to create four simultaneous equations:

$$K_1 + K_4 = 0;$$

$$15K_1 + K_3 + 10K_4 = 0;$$

$$75K_1 + K_2 + 5K_3 + 25K_4 = 100;$$

$$125K_1 = 2500.$$

Solve the simultaneous equations to find

$$K_1 = 20; \quad K_2 = -400; \quad K_3 = -100; \quad K_4 = -20.$$

Therefore, the partial fraction expansion is

$$\frac{100(s + 25)}{s(s + 5)^3} \equiv \frac{20}{s} - \frac{400}{(s + 5)^3} - \frac{100}{(s + 5)^2} - \frac{20}{s + 5}.$$

At this point, we can check our expansion by testing both sides at $s = -25$; for this value of s , both sides should equal zero. The result of evaluating the partial fraction expansion at $s = -25$ is

$$\frac{20}{-25} - \frac{400}{(-20)^3} - \frac{100}{(-20)^2} - \frac{20}{(-20)} = 0.$$

Use the functional and operational transform tables (Tables 12.1 and 12.2) to transform each term in the partial fraction expansion. Thus, the inverse transform of $F(s)$ is

$$\mathcal{L}^{-1} \left\{ \frac{100(s + 25)}{s(s + 5)^3} \right\} = [20 - 200t^2e^{-5t} - 100te^{-5t} - 20e^{-5t}]u(t).$$

ASSESSMENT PROBLEMS

Objective 2—Be able to calculate the inverse Laplace transform using partial fraction expansion and the Laplace transform table

12.8 Find $f(t)$ if

$$F(s) = \frac{80(s + 3)}{s(s + 2)^2}.$$

Answer: $f(t) = (60 - 40te^{-2t} - 60e^{-2t})u(t).$

12.9 Suppose for the circuit in Fig. 12.16, $I_{dc} = 15$ mA, $R = 1$ k Ω , $L = 4$ H, and $C = 1$ μ F. These are the same values used in

Assessment Problems 12.5 and 12.7 except for the value of R . There is no energy stored in the circuit when the switch opens at $t = 0$.

- Find $V(s)$, the Laplace transform of $v(t)$.
- Find $v(t)$ by finding the inverse transform of the partial fraction expansion of $V(s)$.

Answer: a) $V(s) = \frac{15,000}{s^2 + 1000s + 25 \times 10^4};$
b) $v(t) = (15,000te^{-500t})u(t)$ V.

SELF-CHECK: Also try Chapter Problems 12.44(b) and (c).

Partial Fraction Expansion: Repeated Complex Roots of $D(s)$

We can find the coefficients of the partial fraction terms corresponding to repeated complex roots using either Method A or Method B. The algebra involves complex numbers. Recall that complex roots always appear in conjugate pairs and that the coefficients associated with a conjugate pair are also conjugates, so that only half the K s need to be evaluated. We illustrate the process using Method B in Example 12.6.

EXAMPLE 12.6

Finding the Inverse Laplace Transform When $F(s)$ Has Repeated Complex Roots

Use partial fraction expansion to find the inverse Laplace transform of

$$F(s) = \frac{768}{(s^2 + 6s + 25)^2}.$$

Solution

After factoring the denominator polynomial, we write

$$\begin{aligned} F(s) &= \frac{768}{(s + 3 - j4)^2 (s + 3 + j4)^2} \\ &\equiv \frac{K_1}{(s + 3 - j4)^2} + \frac{K_2}{s + 3 - j4} \\ &\quad + \frac{K_1^*}{(s + 3 + j4)^2} + \frac{K_2^*}{s + 3 + j4}. \end{aligned}$$

Now we need to evaluate only K_1 and K_2 , because K_1^* and K_2^* are conjugate values. We use Method B to find these two partial fraction coefficients.

The value of K_1 is

$$K_1 = \left. \frac{768}{(s + 3 + j4)^2} \right|_{s=-3+j4} = \frac{768}{(j8)^2} = -12.$$

To find the value of K_2 , multiply $F(s)$ by $(s + 3 - j4)^2$, find the first derivative of the result with respect to s , and evaluate for $s = -3 + j4$:

$$\begin{aligned} K_2 &= \frac{d}{ds} \left[\frac{768}{(s + 3 + j4)^2} \right]_{s=-3+j4} \\ &= \left. \frac{-2(768)}{(s + 3 + j4)^3} \right|_{s=-3+j4} \\ &= \frac{-2(768)}{(j8)^3} = -j3 = 3 \angle -90^\circ. \end{aligned}$$

From the values for K_1 and K_2 , $K_1^* = -12$ and $K_2^* = j3 = 3 \angle 90^\circ$. Group the partial fraction expansion by conjugate terms to obtain

$$\begin{aligned} F(s) &= \left[\frac{-12}{(s + 3 - j4)^2} + \frac{-12}{(s + 3 + j4)^2} \right] \\ &\quad + \left[\frac{3 \angle -90^\circ}{(s + 3 - j4)} + \frac{3 \angle 90^\circ}{(s + 3 + j4)} \right]. \end{aligned}$$

Inverse-transform $F(s)$ by applying the functional and operational transforms (Tables 12.1 and 12.2) to the terms in the partial fraction expansion. The result is

$$f(t) = [-24te^{-3t} \cos 4t + 6e^{-3t} \cos(4t - 90^\circ)]u(t).$$

ASSESSMENT PROBLEM

Objective 2—Be able to calculate the inverse Laplace transform using partial fraction expansion and the Laplace transform table

12.10 Find $f(t)$ if

$$F(s) = \frac{14,400}{(s^2 + 60s + 2500)^2}.$$

Answer:

$$f(t) = (-4.5te^{-30t} \cos 40t + 0.1125e^{-30t} \sin 40t)u(t).$$

SELF-CHECK: Also try Chapter Problem 12.45(b).

Note that if $F(s)$ has a real root a of multiplicity r in its denominator, the partial fraction expansion has a term of the form

$$\frac{K}{(s + a)^r}.$$

The inverse transform of this term is

$$\mathcal{L}^{-1}\left\{\frac{K}{(s + a)^r}\right\} = \frac{Kt^{r-1}e^{-at}}{(r-1)!}u(t) \quad (12.11)$$

If $F(s)$ has a complex root of $\alpha + j\beta$ of multiplicity r in its denominator, the partial fraction expansion has a conjugate pair of terms in the form

$$\frac{K}{(s + \alpha - j\beta)^r} + \frac{K^*}{(s + \alpha + j\beta)^r}.$$

The inverse transform of this pair is

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{|K|\angle\theta}{(s + \alpha - j\beta)^r} + \frac{|K|\angle-\theta}{(s + \alpha + j\beta)^r}\right\} \\ = \left[\frac{2|K|t^{r-1}}{(r-1)!}e^{-\alpha t}\cos(\beta t + \theta)\right]u(t). \end{aligned} \quad (12.12)$$

Equations 12.11 and 12.12 are the key to finding the inverse transform for any partial fraction expansion with repeated roots. One further note regarding these two equations: In circuit analysis problems, r is seldom greater than 2. Therefore, the inverse transform of a rational function can be handled with four transform pairs. Table 12.3 lists these pairs.

Partial Fraction Expansion: Improper Rational Functions

An improper rational function can always be written as the sum of a polynomial and a proper rational function. The polynomial is then inverse-transformed into impulse functions and derivatives of impulse functions, while the proper rational function is inverse-transformed by the techniques outlined in this section.

We illustrate the procedure in Example 12.7.

TABLE 12.3 Four Useful Transform Pairs

Pair Number	Nature of Roots	$F(s)$	$f(t)$
1	Distinct real	$\frac{K}{s + a}$	$Ke^{-at}u(t)$
2	Repeated real	$\frac{K}{(s + a)^2}$	$Kte^{-at}u(t)$
3	Distinct complex	$\frac{K}{s + \alpha - j\beta} + \frac{K^*}{s + \alpha + j\beta}$	$2 K e^{-\alpha t}\cos(\beta t + \theta)u(t)$
4	Repeated complex	$\frac{K}{(s + \alpha - j\beta)^2} + \frac{K^*}{(s + \alpha + j\beta)^2}$	$2t K e^{-\alpha t}\cos(\beta t + \theta)u(t)$

Note: In pairs 1 and 2, K is a real quantity, whereas in pairs 3 and 4, K is the complex quantity $|K|\angle\theta$.

EXAMPLE 12.7**Finding the Inverse Laplace Transform of an Improper Rational Function**

Use partial fraction expansion to find the inverse Laplace transform of

$$F(s) = \frac{s^4 + 13s^3 + 66s^2 + 200s + 300}{s^2 + 9s + 20}.$$

Solution

The order of the numerator is 4, while the order of the denominator is 2, so $F(s)$ is an improper rational function. To write it as the sum of a polynomial and a proper rational function, divide the denominator into the numerator until the remainder is a proper rational function. The result is

$$\begin{aligned} F(s) &= (s^2 + 9s + 20) \overline{s^4 + 13s^3 + 66s^2 + 200s + 300} \\ &= s^2 + 4s + 10 + \frac{30s + 10}{s^2 + 9s + 20}. \end{aligned}$$

Next, expand the proper rational function into a sum of partial fractions:

$$\frac{30s + 10}{s^2 + 9s + 20} = \frac{30s + 10}{(s + 4)(s + 5)} \equiv \frac{-20}{s + 4} + \frac{50}{s + 5}.$$

Replace the proper rational function in $F(s)$ with the partial fraction expansion to get

$$F(s) = s^2 + 4s + 10 - \frac{20}{s + 4} + \frac{50}{s + 5}.$$

Using the tables of functional and operational transforms (Tables 12.1 and 12.2) we can now inverse-transform $F(s)$. Hence

$$f(t) = \frac{d^2\delta(t)}{dt^2} + 4\frac{d\delta(t)}{dt} + 10\delta(t) - (20e^{-4t} - 50e^{-5t})u(t).$$

ASSESSMENT PROBLEMS

Objective 2—Be able to calculate the inverse Laplace transform using partial fraction expansion and the Laplace transform table

12.11 Find $f(t)$ if

$$F(s) = \frac{25s^2 + 395s + 1494}{s^2 + 15s + 54}.$$

Answer: $25\delta(t) + [8e^{-6t} + 12e^{-9t}]u(t).$

SELF-CHECK: Also try Chapter Problem 12.45(d).

12.12 Find $f(t)$ if

$$F(s) = \frac{5s^3 + 20s^2 - 49s - 108}{s^2 + 7s + 10}.$$

Answer:

$$f(t) = 5\frac{d\delta(t)}{dt} - 15\delta(t) + [10e^{-2t} - 4e^{-5t}]u(t).$$

12.8 Poles and Zeros of $F(s)$

The rational function of Eq. 12.7 can also be expressed as the ratio of two factored polynomials. In other words, we can write $F(s)$ as

$$F(s) = \frac{K(s + z_1)(s + z_2)\cdots(s + z_n)}{(s + p_1)(s + p_2)\cdots(s + p_m)}, \quad (12.13)$$

where K is the constant a_n/b_m .

The roots of the denominator polynomial, that is, $-p_1, -p_2, -p_3, \dots, -p_m$, are called the **poles of $F(s)$** ; they are the values of s at which $F(s)$ becomes infinitely large. The roots of the numerator polynomial, that is, $-z_1, -z_2, -z_3, \dots, -z_n$ are called the **zeros of $F(s)$** ; they are the values of s at which $F(s)$ becomes zero. We can visualize the poles and zeros of $F(s)$ as

points on a complex s plane. In the complex s plane, we use the horizontal axis to plot the real values of s and the vertical axis to plot the imaginary values of s . Example 12.8 finds the poles and zeros for two different functions of s and plots the locations of the poles and zeros on the complex plane.

EXAMPLE 12.8

Finding and Plotting the Poles and Zeros of an s -Domain Function

a) Suppose $F_1(s)$ is given by

$$F_1(s) = \frac{40s^3 + 440s^2 + 2200s + 5000}{4s^4 + 88s^3 + 880s^2 + 4000s}.$$

Find the poles and zeros of $F_1(s)$ and plot them on the complex s plane.

b) Suppose $F_2(s)$ is given by

$$F_2(s) = \frac{8s^2 + 120s + 400}{2s^4 + 20s^3 + 70s^2 + 100s + 48}.$$

Find the poles and zeros of $F_2(s)$ and plot them on the complex s plane.

Solution

a) Begin by factoring out a constant in the numerator and denominator of $F_1(s)$. Then factor the numerator and denominator polynomials. The result is

$$\begin{aligned} F_1(s) &= \frac{40(s^3 + 11s^2 + 55s + 125)}{4(s^4 + 22s^3 + 220s^2 + 1000s)} \\ &= \frac{10(s + 5)(s + 3 - j4)(s + 3 + j4)}{s(s + 10)(s + 6 - j8)(s + 6 + j8)}. \end{aligned}$$

The poles of $F_1(s)$ are at 0, -10 , $-6 + j8$, and $-6 - j8$. The zeros are at -5 , $-3 + j4$, and $-3 - j4$. Figure 12.17 shows the poles and zeros plotted on the s plane, where X's represent poles and O's represent zeros.

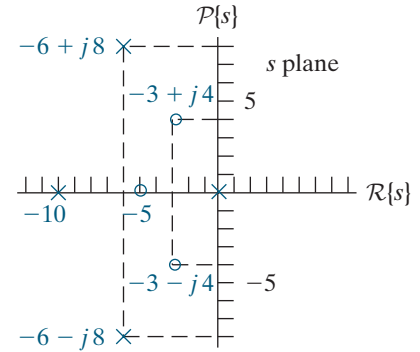


Figure 12.17 ▲ Plotting poles and zeros on the s plane for $F_1(s)$ in Example 12.8.

b) Factor out a constant in the numerator and denominator of $F_2(s)$. Then factor the numerator and denominator polynomials to give

$$\begin{aligned} F_2(s) &= \frac{8(s^2 + 15s + 50)}{2(s^4 + 10s^3 + 35s^2 + 50s + 24)} \\ &= \frac{4(s + 5)(s + 10)}{(s + 1)(s + 2)(s + 3)(s + 4)}. \end{aligned}$$

The zeros of $F_2(s)$ are -5 and -10 . The poles of $F_2(s)$ are -1 , -2 , -3 , and -4 . They are plotted in the complex s plane in Fig. 12.18. Note that $F_2(s)$ also has a second-order zero at infinity because for large values of s the function reduces to $4/s^2$, and $F_2(s) = 0$ when $s = \infty$. In general, $F(s)$ can have either an r th-order pole or an r th-order zero at infinity. In this text, we are interested in the poles and zeros located in the finite s plane. Therefore, when we refer to the poles and zeros of a rational function of s , we are referring to the finite poles and zeros.

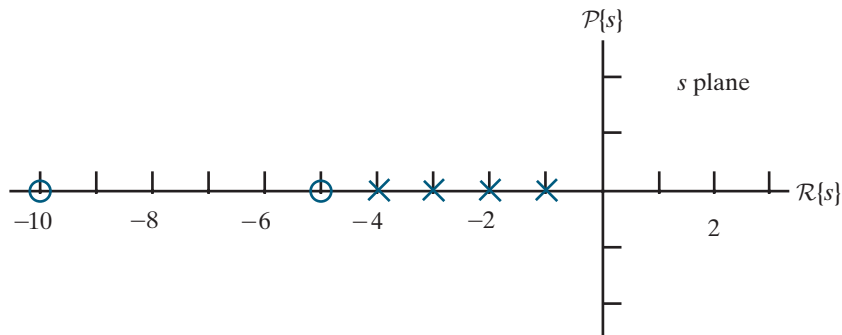


Figure 12.18 ▲ Plotting poles and zeros on the s plane for $F_2(s)$ in Example 12.8.

ASSESSMENT PROBLEM

Objective 3—Be able to find and plot the poles and zeros for a rational function in the s domain

12.13 Find the poles and zeros for the following rational functions of s .

a) $F(s) = \frac{10s^2 + 210s + 980}{s^3 + 14s^2 + 50s};$

b) $F(s) = \frac{5s^2 + 50s + 445}{4s^3 + 40s^2 + 176s + 480};$

c) $F(s) = \frac{25s^2 + 200s + 400}{s^4 + 8s^3 + 38s^2 + 56s + 25}.$

Answer: a) Zeros at -7 and -14 ; poles at 0 and $-7 \pm j$;
 b) Zeros at $-5 \pm j8$; poles at -6 and $-2 \pm j4$;
 c) Two zeros at -4 ; poles at $-3 \pm j4$ and two poles at -1 .

SELF-CHECK: Also try Chapter Problem 12.46.

12.9 Initial- and Final-Value Theorems

The initial- and final-value theorems enable us to determine the behavior of $f(t)$ at 0 and ∞ from $F(s)$. Hence, we can check the initial and final values of $f(t)$ to see if they conform to known circuit behavior, before actually finding the inverse transform of $F(s)$.

The initial-value theorem states that

INITIAL-VALUE THEOREM

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} sF(s), \quad (12.14)$$

and the final-value theorem states that

FINAL-VALUE THEOREM

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s). \quad (12.15)$$

The initial-value theorem assumes that $f(t)$ does not have an impulse function at the origin. The final-value theorem is valid only if the poles of $F(s)$, except for a first-order pole at the origin, lie in the left half of the s plane.

To prove Eq. 12.14, we start with the operational transform of the first derivative:

$$\mathcal{L}\left\{\frac{df}{dt}\right\} = sF(s) - f(0^-) = \int_{0^-}^{\infty} \frac{df}{dt} e^{-st} dt.$$

Now we take the limit as $s \rightarrow \infty$:

$$\lim_{s \rightarrow \infty} [sF(s) - f(0^-)] = \lim_{s \rightarrow \infty} \int_{0^-}^{\infty} \frac{df}{dt} e^{-st} dt. \quad (12.16)$$

Observe that the right-hand side of Eq. 12.16 can be written as

$$\lim_{s \rightarrow \infty} \left(\int_{0^-}^{0^+} \frac{df}{dt} e^0 dt + \int_{0^+}^{\infty} \frac{df}{dt} e^{-st} dt \right).$$

As $s \rightarrow \infty$, $(df/dt)e^{-st} \rightarrow 0$; hence, the second integral vanishes in the limit. The first integral reduces to $f(0^+) - f(0^-)$, which is independent of s . Thus, the right-hand side of Eq. 12.16 becomes

$$\lim_{s \rightarrow \infty} \int_{0^-}^{\infty} \frac{df}{dt} e^{-st} dt = f(0^+) - f(0^-).$$

Because $f(0^-)$ is independent of s , the left-hand side of Eq. 12.16 can be written as

$$\lim_{s \rightarrow \infty} [sF(s) - f(0^-)] = \lim_{s \rightarrow \infty} [sF(s)] - f(0^-).$$

Therefore,

$$\lim_{s \rightarrow \infty} sF(s) = f(0^+) = \lim_{t \rightarrow 0^+} f(t),$$

which completes the proof of the initial-value theorem.

The proof of the final-value theorem also starts with the operational transform of the first derivative. Here we take the limit as $s \rightarrow 0$:

$$\lim_{s \rightarrow 0} [sF(s) - f(0^-)] = \lim_{s \rightarrow 0} \left(\int_{0^-}^{\infty} \frac{df}{dt} e^{-st} dt \right) \quad (12.17)$$

The integration is with respect to t and the limit operation is with respect to s , so the right-hand side of Eq. 12.17 reduces to

$$\lim_{s \rightarrow 0} \left(\int_{0^-}^{\infty} \frac{df}{dt} e^{-st} dt \right) = \int_{0^-}^{\infty} \frac{df}{dt} dt.$$

Because the upper limit on the integral is infinite, this integral may also be written as a limit process:

$$\int_{0^-}^{\infty} \frac{df}{dt} dt = \lim_{t \rightarrow \infty} \int_{0^-}^t \frac{df}{dy} dy,$$

where we use y as the symbol of integration to avoid confusion with the upper limit on the integral. Carrying out the integration on the right-hand side gives

$$\lim_{t \rightarrow \infty} [f(t) - f(0^-)] = \lim_{t \rightarrow \infty} [f(t)] - f(0^-).$$

Substituting this expression into the right-hand side of Eq. 12.17 gives

$$\lim_{s \rightarrow 0} [sF(s)] - f(0^-) = \lim_{t \rightarrow \infty} [f(t)] - f(0^-).$$

Since $f(0^-)$ cancels, we get

$$\lim_{s \rightarrow 0} sF(s) = \lim_{t \rightarrow \infty} f(t),$$

which completes the proof of the final value theorem.

The final-value theorem is useful only if $f(\infty)$ exists. This condition is true only if all the poles of $F(s)$, except for a single pole at the origin, lie in the left half of the s plane.

Example 12.9 applies the initial- and final-value theorems to the s -domain function from Example 12.2.

EXAMPLE 12.9 Applying the Initial- and Final-Value Theorems

Suppose for the circuit in Fig. 12.16, $I_{dc} = 24$ mA, $R = 400\ \Omega$, $L = 25$ mH, and $C = 25$ nF. There is no energy stored in the circuit when the switch opens at $t = 0$. In Example 12.2, we found the Laplace transform of $v(t)$ is

$$V(s) = \frac{96 \times 10^4}{s^2 + 10^5 s + 16 \times 10^8}.$$

Use the initial- and final-value theorems to predict the initial and final values of $v(t)$ and verify that $V(s)$ correctly predicts the values of $v(0^+)$ and $v(\infty)$ from the circuit.

Solution

From the initial-value theorem,

$$\lim_{s \rightarrow \infty} sV(s) = \lim_{s \rightarrow \infty} \frac{96 \times 10^4 s}{s^2 + 10^5 s + 16 \times 10^8}.$$

To evaluate the limit on the right-hand side, divide numerator and denominator by the highest power of s in the denominator, in this case s^2 , and find the limit as $1/s \rightarrow 0$:

$$\begin{aligned} \lim_{s \rightarrow \infty} \frac{96 \times 10^4 s}{s^2 + 10^5 s + 16 \times 10^8} &= \lim_{1/s \rightarrow 0} \frac{96 \times 10^4 (1/s)}{1 + 10^5 (1/s) + 16 \times 10^8 (1/s)^2} \\ &= \frac{0}{1 + 0 + 0} = 0 = \lim_{t \rightarrow 0} v(t) = v(0^+). \end{aligned}$$

Since the problem states that there is no energy stored in the circuit prior to the switch opening at $t = 0$, we have confirmed that the initial voltage is zero.

Before applying the final-value theorem, find the poles of $V(s)$. They are $-20,000$ and $-80,000$, so both lie in the left-half complex plane, and we can use the final-value theorem to get

$$\lim_{s \rightarrow 0} sV(s) = \lim_{s \rightarrow 0} \frac{96 \times 10^4 s}{s^2 + 10^5 s + 16 \times 10^8} = 0.$$

As we expected from the circuit, as $t \rightarrow \infty$, the final-value theorem gives $v(\infty) = 0$. Thus, $V(s)$ correctly predicts the initial and final values of $V(t)$.

ASSESSMENT PROBLEM

Objective 4—Understand and know how to use the initial-value theorem and the final-value theorem

12.14 Use the initial- and final-value theorems to find the initial and final values of $f(t)$ in Assessment Problems 12.4, 12.8, and 12.10.

Answer: 10, 0; 0, 60; and 0, 0.

SELF-CHECK: Also try Chapter Problem 12.52.

Practical Perspective**Transient Effects**

The circuit introduced in the Practical Perspective at the beginning of the chapter is repeated in Fig. 12.19 with the switch closed and the chosen sinusoidal source.

We use Laplace methods to determine the complete response of the inductor current, i_L . To begin, use KVL to sum the voltage drops around the circuit, in the clockwise direction:

$$15i_L + 0.01 \frac{di_L}{dt} + \frac{1}{100 \times 10^{-6}} \int_0^t i_L(x) dx = \cos 120\pi t.$$

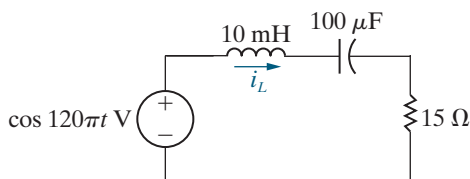


Figure 12.19 ▲ A series RLC circuit with a 60 Hz sinusoidal source.

Using Table 12.1 and 12.2, we find the Laplace transform of the KVL equation:

$$15I_L + 0.01sI_L + 10^4 \frac{I_L}{s} = \frac{s}{s^2 + (120\pi)^2}.$$

Solve this equation for I_L :

$$I_L = \frac{100s^2}{[s^2 + 1500s + 10^6][s^2 + (120\pi)^2]}.$$

Note that the expression for I_L has two complex conjugate pairs of poles, so the partial fraction expansion of I_L will have four terms:

$$I_L = \frac{K_1}{(s + 750 - j661.44)} + \frac{K_1^*}{(s + 750 + j661.44)} + \frac{K_2}{(s - j120\pi)} + \frac{K_2^*}{(s + j120\pi)}.$$

Determine the values of K_1 and K_2 :

$$\begin{aligned} K_1 &= \left. \frac{100s^2}{[s + 750s + j661.44][s^2 + (120\pi)^2]} \right|_{s=-750+j661.44} \\ &= 0.07357 \angle -97.89^\circ, \\ K_2 &= \left. \frac{100s^2}{[s^2 + 1500s + 10^6][s + j120\pi]} \right|_{s=j120\pi} = 0.018345 \angle 56.61^\circ. \end{aligned}$$

Therefore, the s-domain expression for the inductor current is

$$I_L = \frac{0.07357 \angle -97.89^\circ}{(s + 750 - j661.44)} + \frac{0.07357 \angle 97.89^\circ}{(s + 750 + j661.44)} + \frac{0.018345 \angle 56.61^\circ}{(s - j120\pi)} + \frac{0.018345 \angle -56.61^\circ}{(s + j120\pi)}.$$

Finally, we use Table 12.3 to calculate the inverse Laplace transform and find i_L :

$$i_L = 147.14e^{-750t} \cos(661.44t - 97.89^\circ) + 36.69 \cos(120\pi t + 56.61^\circ) \text{ mA}.$$

The first term in the inductor current is the transient response, which will decay to zero in about 7 ms. The second term in the inductor current is the steady-state response, which has the same frequency as the 60 Hz sinusoidal source and will persist as long as this source is connected in the circuit. Note that the amplitude of the steady-state response is 36.69 mA, which is less than the 40 mA current rating of the inductor. But the transient response has an initial amplitude of 147.14 mA, far greater than the 40 mA current rating. Calculate the value of the inductor current at $t = 0$:

$$i_L(0) = 147.14(1)\cos(-97.89^\circ) + 36.69\cos(56.61^\circ) = -6.21 \mu\text{A}.$$

Clearly, the transient part of the response does not cause the inductor current to exceed its rating initially. But we need a plot of the complete response to determine whether or not the current rating is ever exceeded, as shown in Fig. 12.20.

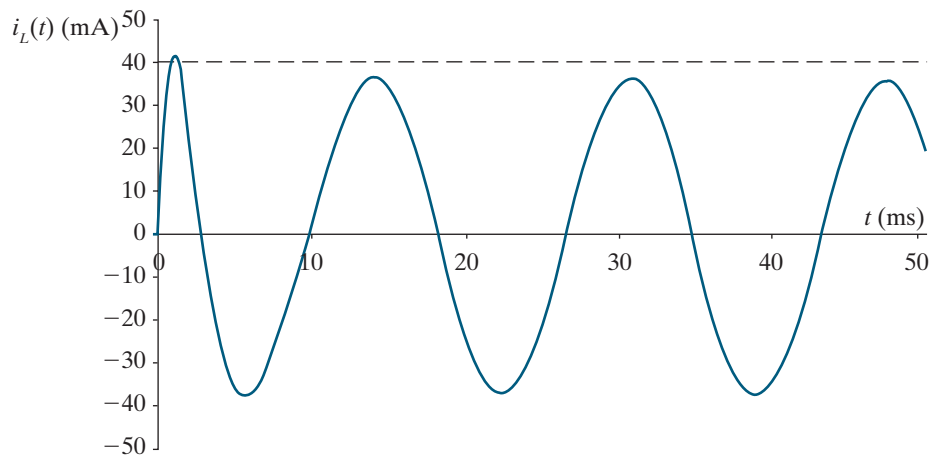


Figure 12.20 ▲ Plot of the inductor current for the circuit in Fig. 12.19.

The plot suggests we check the value of the inductor current at 1 ms:

$$i_L(0.001) = 147.14e^{-0.75} \cos(-59.99^\circ) + 36.69 \cos(78.21^\circ) = 42.4 \text{ mA.}$$

Thus, the current rating is exceeded in the inductor, at least momentarily. If we determine that we never want to exceed the current rating, we should reduce the magnitude of the sinusoidal source. This example illustrates the importance of considering the complete response of a circuit to a sinusoidal input, even if we are satisfied with the steady-state response.

SELF-CHECK: Assess your understanding of the Practical Perspective by trying Chapter Problems 12.57 and 12.58.

Summary

- The **Laplace transform** is a tool for converting time-domain equations into frequency-domain equations, according to the following general definition:

$$\mathcal{L}\{f(t)\} = \int_0^\infty f(t)e^{-st} dt = F(s),$$

where $f(t)$ is the time-domain expression and $F(s)$ is the frequency-domain expression. (See page 446.)

- The **step function** $Ku(t)$ describes a function that experiences a discontinuity from one constant level to another at some point in time. K is the magnitude of the jump; if $K = 1$, $Ku(t)$ is the **unit step function**. (See page 447.)
- The **impulse function** $K\delta(t)$ is defined as

$$\int_{-\infty}^{\infty} K\delta(t)dt = K,$$

$$\delta(t) = 0, \quad t \neq 0.$$

K is the strength of the impulse; if $K = 1$, $K\delta(t)$ is the **unit impulse function**. (See page 449.)

- A **functional transform** is the Laplace transform of a specific function. Important functional transform pairs are summarized in Table 12.1. (See page 453.)
- Operational transforms** define the general mathematical properties of the Laplace transform. Important operational transform pairs are summarized in Table 12.2. (See page 458.)
- In linear lumped-parameter circuits, $F(s)$ is a rational function of s . (See page 460.)
- If $F(s)$ is a proper rational function, the inverse transform is found by a partial fraction expansion. (See page 461.)
- If $F(s)$ is an improper rational function, it can be inverse-transformed by first expanding it into a sum of a polynomial and a proper rational function. (See page 469.)
- $F(s)$ can be expressed as the ratio of two factored polynomials. The roots of the denominator are called **poles** and are plotted as X's on the complex s plane. The roots of the numerator are called **zeros** and are plotted as O's on the complex s plane. (See page 470.)

- The initial-value theorem states that

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{s \rightarrow \infty} sF(s).$$

The theorem assumes that $f(t)$ contains no impulse functions. (See page 472.)

- The final-value theorem states that

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0^+} sF(s).$$

The theorem is valid only if the poles of $F(s)$, except for a first-order pole at the origin, lie in the left half of the s plane. (See page 472.)

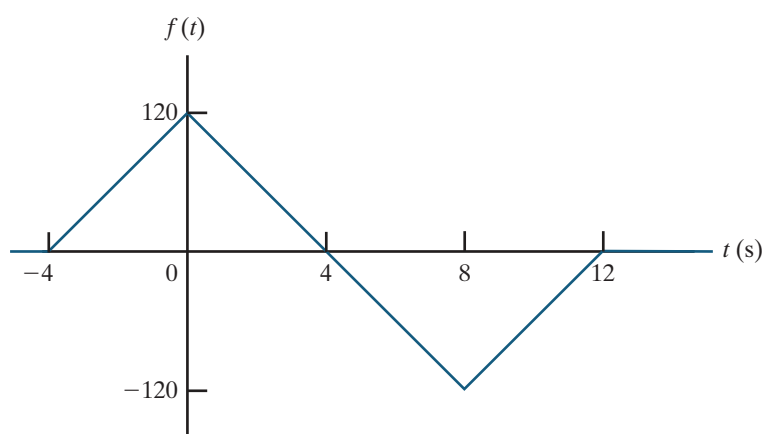
- The initial- and final-value theorems allow us to predict the initial and final values of $f(t)$ from an s -domain expression. (See page 474.)

Problems

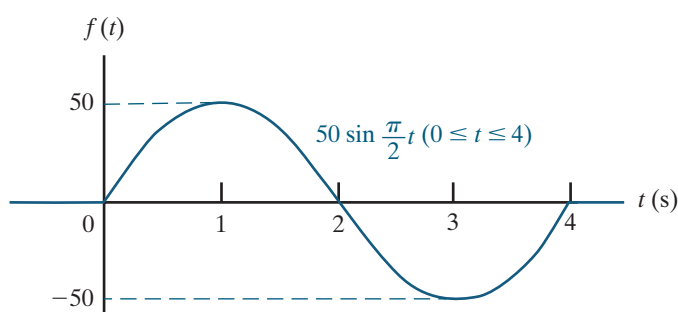
Section 12.2

- 12.1** Use step functions to write the expression for each of the functions shown in Fig. P12.1.

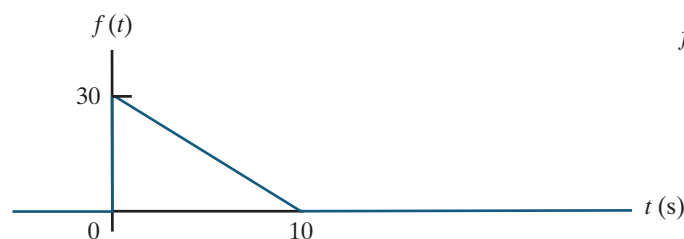
Figure P12.1



(a)



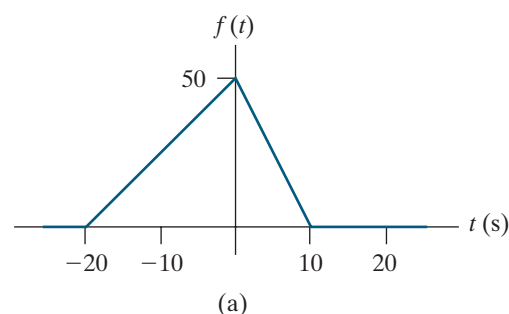
(b)



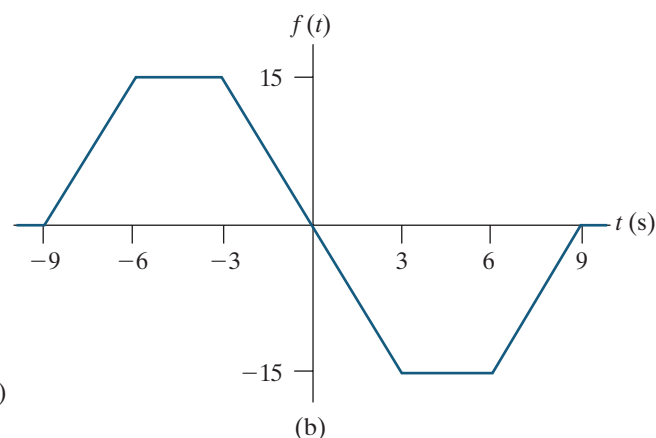
(c)

- 12.2** Use step functions to write the expression for each function shown in Fig. P12.2.

Figure P12.2



(a)



(b)

- 12.3** Make a sketch of $f(t)$ for $-25 \text{ s} \leq t \leq 25 \text{ s}$ when $f(t)$ is given by the following expression:

$$\begin{aligned} f(t) = & -(20t + 400)u(t + 20) + (40t + 400)u(t + 10) \\ & + (400 - 40t)u(t - 10) \\ & + (20t - 400)u(t - 20). \end{aligned}$$

12.4 Step functions can be used to define a *window* function. Thus $u(t - 1) - u(t - 4)$ defines a window 1 unit high and 3 units wide located on the time axis between 1 and 4.

A function $f(t)$ is defined as follows:

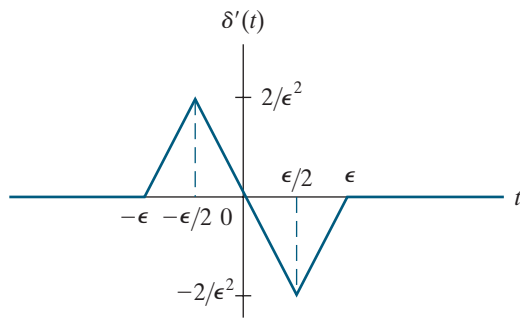
$$\begin{aligned} f(t) &= 0, & t &\leq 0; \\ &= 30t, & 0 \leq t \leq 2 \text{ s}; \\ &= 60, & 2 \text{ s} \leq t \leq 4 \text{ s}; \\ &= 60 \cos\left(\frac{\pi}{4}t - \pi\right), & 4 \text{ s} \leq t \leq 8 \text{ s}; \\ &= 30t - 300, & 8 \text{ s} \leq t \leq 10 \text{ s}; \\ &= 0, & 10 \text{ s} \leq t < \infty. \end{aligned}$$

- Sketch $f(t)$ over the interval $-2 \text{ s} \leq t \leq 12 \text{ s}$.
- Use the concept of the window function to write an expression for $f(t)$.

Section 12.3

12.5 The triangular pulses shown in Fig. P12.5 are equivalent to the rectangular pulses in Fig. 12.12(b), because they both enclose the same area ($1/\epsilon$) and they both approach infinity proportional to $1/\epsilon^2$ as $\epsilon \rightarrow 0$. Use this triangular-pulse representation for $\delta'(t)$ to find the Laplace transform of $\delta''(t)$.

Figure P12.5



12.6 Explain why the following function generates an impulse function as $\epsilon \rightarrow 0$:

$$f(t) = \frac{\epsilon/\pi}{\epsilon^2 + t^2}, \quad -\infty \leq t \leq \infty.$$

12.7 a) Find the area under the function shown in Fig. 12.12(a).

- What is the duration of the function when $\epsilon = 0$?
- What is the magnitude of $f(0)$ when $\epsilon = 0$?

12.8 In Section 12.3, we used the sifting property of the impulse function to show that $\mathcal{L}\{\delta(t)\} = 1$. Show that we can obtain the same result by finding the

Laplace transform of the rectangular pulse that exists between $\pm\epsilon$ in Fig. 12.9 and then finding the limit of this transform as $\epsilon \rightarrow 0$.

12.9 Evaluate the following integrals:

$$\text{a) } I = \int_{-2}^4 (t^3 + 4)[\delta(t) + 4\delta(t - 2)] dt;$$

$$\text{b) } I = \int_{-3}^4 t^2[\delta(t) + \delta(t + 2.5) + \delta(t - 5)] dt.$$

12.10 Find $f(t)$ if

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{j\omega t} d\omega,$$

and

$$F(\omega) = \frac{3 + j\omega}{4 + j\omega} \pi \delta(\omega).$$

12.11 Show that

$$\mathcal{L}\{\delta^{(n)}(t)\} = s^n.$$

12.12 a) Show that

$$\int_{-\infty}^{\infty} f(t) \delta'(t - a) dt = -f'(a).$$

(Hint: Integrate by parts.)

b) Use the formula in (a) to show that

$$\mathcal{L}\{\delta'(t)\} = s.$$

Sections 12.4–12.5

12.13 Find the Laplace transform of each of the following functions:

- $f(t) = t$;
- $f(t) = te^{-at}$;
- $f(t) = \sin \omega t$;
- $f(t) = \cosh \beta t$;
- $f(t) = \sinh \beta t$.

12.14 a) Use the first derivative (time) operational transform given in Table 12.2 to find the Laplace transform of $\frac{d}{dt}(te^{-at})$.

b) Check your result in part (a) by first differentiating and then transforming the resulting expression.

12.15 Show that

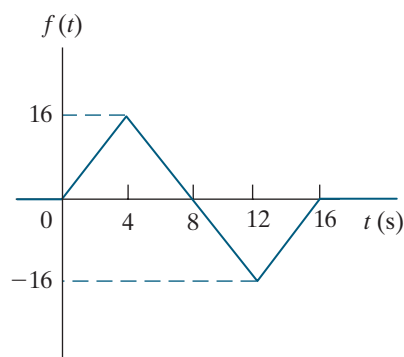
$$\mathcal{L}\{e^{-at}f(t)\} = F(s + a).$$

12.16 Show that

$$\mathcal{L}\{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right).$$

- 12.17** a) Find $\mathcal{L}\left\{\int_{0^-}^t e^{-ax} dx\right\}$.
 b) Check the results of (a) by first integrating and then transforming.
- 12.18** a) Find $\mathcal{L}\left\{\frac{d}{dt} \sin \omega t\right\}$.
 b) Find $\mathcal{L}\left\{\frac{d}{dt} \cos \omega t\right\}$.
 c) Find $\mathcal{L}\left\{\frac{d^3}{dt^3} t^2 u(t)\right\}$.
 d) Check the results of parts (a), (b), and (c) by first differentiating and then transforming.
- 12.19** a) Find the Laplace transform of the function illustrated in Fig. P12.19.
 b) Find the Laplace transform of the first derivative of the function illustrated in Fig. P12.19.
 c) Find the Laplace transform of the second derivative of the function illustrated in Fig. P12.19.

Figure P12.19



- 12.20** Find the Laplace transform of each of the following functions:
 a) $f(t) = -20e^{-5(t-2)}u(t-2)$.
 b) $f(t) = (8t-8)[u(t-1)-u(t-2)] + (24-8t)[u(t-2)-u(t-4)] + (8t-40)[u(t-4)-u(t-5)]$.
- 12.21** a) Find the Laplace transform of $\int_{0^-}^t x dx$ by first integrating and then transforming.
 b) Check the result obtained in (a) by using the time integral operational transform given in Table 12.2.
- 12.22** Find the Laplace transform (when $\epsilon \rightarrow 0$) of the derivative of the exponential function illustrated in Fig. 12.9, using each of the following two methods:
 a) First differentiate the function and then find the transform of the resulting function.
 b) Use the first derivative (time) operational transform given in Table 12.2.

12.23 Find the Laplace transform for (a) and (b).

a) $f(t) = \frac{d}{dt} (e^{-at} \sinh \beta t)$.

b) $f(t) = \int_{0^-}^t e^{-ax} \cos \omega x dx$.

c) Verify the results obtained in (a) and (b) by first carrying out the indicated mathematical operation and then finding the Laplace transform.

12.24 a) Given that $F(s) = \mathcal{L}\{f(t)\}$, show that

$$-\frac{dF(s)}{ds} = \mathcal{L}\{t f(t)\}.$$

b) Show that

$$(-1)^n \frac{d^n F(s)}{ds^n} = \mathcal{L}\{t^n f(t)\}.$$

c) Use the result of (b) to find $\mathcal{L}\{t^5\}$, $\mathcal{L}\{t \sin \beta t\}$, and $\mathcal{L}\{te^{-t} \cosh t\}$.

12.25 a) Show that if $F(s) = \mathcal{L}\{f(t)\}$, and $\{f(t)/t\}$ is Laplace-transformable, then

$$\int_s^\infty F(u) du = \mathcal{L}\left\{\frac{f(t)}{t}\right\}.$$

(Hint: Use the defining integral to write

$$\int_s^\infty F(u) du = \int_s^\infty \left(\int_{0^-}^\infty f(t)e^{-ut} dt \right) du$$

and then reverse the order of integration.)

b) Starting with the Laplace transform of $t \sin \beta t$, from Problem 12.24(c), use the operational transform given in (a) of this problem to find $\mathcal{L}\{\sin \beta t\}$.

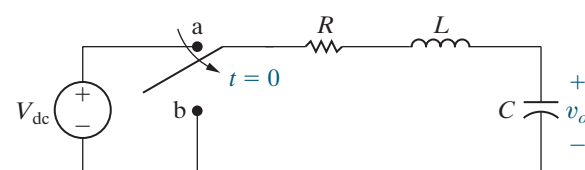
Section 12.6

12.26 The switch in the circuit in Fig. P12.26 has been in position a for a long time. At $t = 0$, the switch moves instantaneously to position b.

- a) Derive the integrodifferential equation that governs the behavior of the voltage v_o for $t \geq 0^+$.
 b) Show that

$$V_o(s) = \frac{V_{dc}[s + (R/L)]}{[s^2 + (R/L)s + (1/LC)]}.$$

Figure P12.26

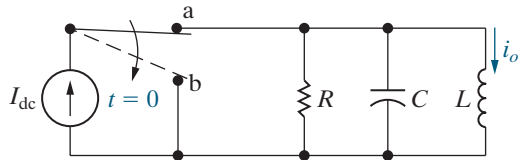


12.27 The switch in the circuit in Fig. P12.27 has been in position a for a long time. At $t = 0$, the switch moves instantaneously to position b.

- Derive the integrodifferential equation that governs the behavior of the current i_o for $t \geq 0^+$.
- Show that

$$I_o(s) = \frac{I_{dc}[s + (1/RC)]}{[s^2 + (1/RC)s + (1/LC)]}.$$

Figure P12.27



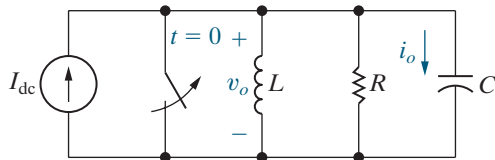
12.28 There is no energy stored in the circuit shown in Fig. P12.28 at the time the switch is opened. In Section 12.6, we derived the integrodifferential equation that governs the behavior of the voltage v_o . We also showed that the Laplace transform of v_o is

$$V_o(s) = \frac{I_{dc}/C}{s^2 + (1/RC)s + (1/LC)}.$$

Use $V_o(s)$ to show that the Laplace transform of i_o is

$$I_o(s) = \frac{sI_{dc}}{s^2 + (1/RC)s + (1/LC)}.$$

Figure P12.28



12.29 The switch in the circuit in Fig. P12.29 has been open for a long time. At $t = 0$, the switch closes.

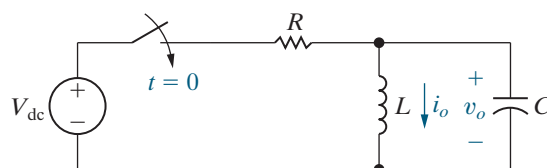
- Derive the integrodifferential equation that governs the behavior of the voltage v_o for $t \geq 0$.
- Show that

$$V_o(s) = \frac{V_{dc}/RC}{s^2 + (1/RC)s + (1/LC)}.$$

- Show that

$$I_o(s) = \frac{V_{dc}/RLC}{s[s^2 + (1/RC)s + (1/LC)]}.$$

Figure P12.29

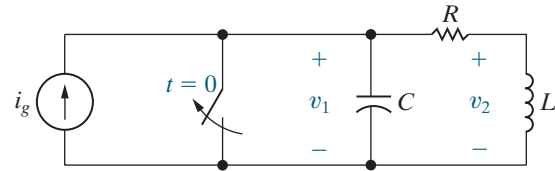


12.30 There is no energy stored in the circuit shown in Fig. P12.30 at the time the switch is opened.

- Derive the integrodifferential equations that govern the behavior of the node voltages v_1 and v_2 .
- Show that

$$V_2(s) = \frac{sI_g(s)}{C[s^2 + (R/L)s + (1/LC)]}.$$

Figure P12.30



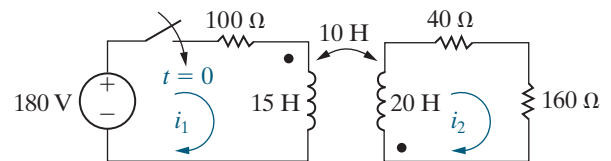
12.31 a) Write the two simultaneous differential equations that describe the circuit shown in Fig. P12.31 in terms of the mesh currents i_1 and i_2 .

PSPICE
MULTISIM

- Laplace-transform the equations derived in (a). Assume that the initial energy stored in the circuit is zero.

- Solve the equations in (b) for $I_1(s)$ and $I_2(s)$.

Figure P12.31



12.32 In the circuit shown in Fig. 12.16, the dc current source is replaced with a sinusoidal source that delivers a current of $5\cos 10t$ A. The circuit components are $R = 1 \Omega$, $C = 25$ mF, and $L = 625$ mH. Find the numerical expression for $V(s)$.

Section 12.7

12.33 The circuit parameters in the circuit in Fig. P12.26 are $R = 5$ k Ω , $L = 1$ H, and $C = 250$ nF. If $V_{dc} = 15$ V, find $v_o(t)$ for $t \geq 0$.

12.34 The circuit parameters in the circuit in Fig. P12.27 are $R = 50 \Omega$, $L = 31.25$ mH, and $C = 2 \mu$ F. If $I_{dc} = 100$ mA, find $i_o(t)$ for $t \geq 0$.

PSPICE
MULTISIM

12.35 The parameter values for the circuit in Fig. P12.28 are as follows: $R = 4$ k Ω , $L = 2.5$ H, $C = 25$ nF, and $I_{dc} = 3$ mA.

PSPICE
MULTISIM

- Find $v_o(t)$ for $t \geq 0$.
- Find $i_o(t)$ for $t \geq 0$.
- Does your solution for $i_o(t)$ make sense when $t = 0$? Explain.

12.36 The circuit parameters in the circuit in Fig. P12.29 are $R = 10\text{ k}\Omega$; $L = 800\text{ mH}$; and $C = 100\text{ nF}$. If V_{dc} is 70 V , find

- a) $v_o(t)$ for $t \geq 0$;
b) $i_o(t)$ for $t \geq 0$.

12.37 The circuit parameters in the circuit in Fig. P12.30 are $R = 2500\text{ }\Omega$; $L = 500\text{ mH}$; and $C = 500\text{ nF}$. If $i_g(t) = 15\text{ mA}$, find $v_2(t)$.

12.38 Use the results from Problem 12.31 and the circuit shown in Fig P12.31 to

- a) Find $i_1(t)$ and $i_2(t)$.
b) Find $i_1(\infty)$ and $i_2(\infty)$.
c) Do the solutions for i_1 and i_2 make sense? Explain.

12.39 Find $v(t)$ in Problem 12.32.

12.40 Derive the transform pair given by Eq. 12.10.

12.41 a) Derive the transform pair given by Eq. 12.11.

b) Derive the transform pair given by Eq. 12.12.

12.42 Find $f(t)$ for each of the following functions:

- a) $F(s) = \frac{6s^2 + 26s + 26}{(s+1)(s+2)(s+3)}$;
b) $F(s) = \frac{13s^3 + 134s^2 + 392s + 288}{s(s+2)(s^2 + 10s + 24)}$;
c) $F(s) = \frac{10(s^2 + 119)}{(s+5)(s^2 + 10s + 169)}$;
d) $F(s) = \frac{56s^2 + 112s + 5000}{s(s^2 + 14s + 625)}$.

12.43 Find $f(t)$ for each of the following functions.

- a) $F(s) = \frac{480}{(s^2 + 12s + 100)}$;
b) $F(s) = \frac{s^2 + 15s + 30}{s(s^2 + 10s + 50)}$;
c) $F(s) = \frac{10s^2 + 30s + 400}{(s+5)(s^2 + 20s + 325)}$;
d) $F(s) = \frac{10(s+3)^2}{(s^2 + 6s + 34)(s^2 + 12s + 52)}$.

12.44 Find $f(t)$ for each of the following functions.

- a) $F(s) = \frac{8(s^2 - 5s + 50)}{s^2(s+10)}$;
b) $F(s) = \frac{4s^2 + 7s + 1}{s(s+1)^2}$;
c) $F(s) = \frac{s^3 - 6s^2 + 15s + 50}{s^2(s^2 + 4s + 5)}$;
d) $F(s) = \frac{16s^3 + 72s^2 + 216s - 128}{(s^2 + 2s + 5)^2}$.

12.45 Find $f(t)$ for each of the following functions.

- a) $F(s) = \frac{100(s+1)}{s^2(s^2 + 2s + 5)}$;

b) $F(s) = \frac{40(s+2)}{s(s+1)^3}$;

c) $F(s) = \frac{5s^2 + 29s + 32}{(s+2)(s+4)}$;

d) $F(s) = \frac{2s^3 + 8s^2 + 2s - 4}{(s^2 + 5s + 4)}$.

Sections 12.8–12.9

12.46 Find the poles and zeros for the s -domain functions in Problems 12.43(a) and 12.43(d).

12.47 Find the poles and zeros for the s -domain functions in Problems 12.45(a) and 12.45(b).

12.48 Use the initial- and final-value theorems to check the initial and final values of the current and voltage in Problem 12.27.

12.49 Use the initial- and final-value theorems to check the initial and final values of the current in Problem 12.28.

12.50 Use the initial- and final-value theorems to check the initial and final values of the current and voltage in Problem 12.29.

12.51 a) Use the initial-value theorem to find the initial value of v in Problem 12.32.

b) Can the final-value theorem be used to find the steady-state value of v ? Why or why not?

12.52 Apply the initial- and final-value theorems to each transform pair in Problem 12.42.

12.53 Apply the initial- and final-value theorems to each transform pair in Problem 12.43.

12.54 Apply the initial- and final-value theorems to each transform pair in Problem 12.44.

12.55 Apply the initial- and final-value theorems to each transform pair in Problem 12.45.

Sections 12.1–12.9

12.56 a) Use phasor circuit analysis techniques from Chapter 9 to determine the steady-state expression for the inductor current in Fig. 12.19.

b) How does your result in part (a) compare to the complete response for inductor current calculated in the Practical Perspective?

12.57 Find the maximum magnitude of the sinusoidal source in Fig. 12.19 such that the complete response of the inductor current does not exceed the 40 mA current rating at $t = 1\text{ ms}$.

12.58 Suppose the input to the circuit in Fig 12.19 is a damped ramp of the form $Kte^{-500t}\text{ V}$. Find the largest value of K such that the inductor current does not exceed the 40 mA current rating.

CHAPTER 13

The Laplace Transform in Circuit Analysis

CHAPTER CONTENTS

- 13.1 **Circuit Elements in the s Domain** *p. 484*
- 13.2 **Circuit Analysis in the s Domain** *p. 486*
- 13.3 **Applications** *p. 488*
- 13.4 **The Transfer Function** *p. 500*
- 13.5 **The Transfer Function in Partial Fraction Expansions** *p. 502*
- 13.6 **The Transfer Function and the Convolution Integral** *p. 505*
- 13.7 **The Transfer Function and the Steady-State Sinusoidal Response** *p. 511*
- 13.8 **The Impulse Function in Circuit Analysis** *p. 514*

CHAPTER OBJECTIVES

- 1 Be able to transform a circuit into the s domain using Laplace transforms; be sure you understand how to represent the initial conditions on energy-storage elements in the s domain.
- 2 Know how to analyze a circuit in the s domain and be able to transform an s -domain solution back to the time domain.
- 3 Understand the definition and significance of the transfer function and be able to calculate the transfer function for a circuit using s -domain techniques.
- 4 Know how to use a circuit's transfer function to calculate the circuit's unit impulse response, its unit step response, and its steady-state response to a sinusoidal input.

The **Laplace transform** has two characteristics that make it an attractive tool for circuit analysis.

- It transforms a set of linear constant-coefficient differential equations into a set of linear polynomial equations, which are easier to manipulate and solve.
- It automatically introduces into the polynomial equations the initial values of the current and voltage variables. Thus, initial conditions are an inherent part of the transform process. (This contrasts with the classical approach to the solution of differential equations, in which initial conditions are considered when the unknown coefficients are evaluated.)

We begin this chapter by showing how to avoid writing time-domain integrodifferential equations and transforming them into the s domain. In Section 13.1, we'll develop the s -domain circuit models for resistors, inductors, and capacitors. Then we can transform entire circuits into the s domain and write the s -domain equations directly. Section 13.2 reviews Ohm's and Kirchhoff's laws for s -domain circuits. After establishing these fundamentals, we apply the Laplace transform method to a variety of circuit problems in Section 13.3, using the circuit analysis and simplification tools first introduced for resistive circuits.

After solving for the circuit response in the s domain, we inverse-transform back to the time domain, using partial fraction expansion (as demonstrated in Chapter 12). As before, checking the final time-domain equations in terms of the initial conditions and final values is an important step in the solution process.

We introduce a new concept, the transfer function, in Section 13.4. The transfer function for a circuit is the ratio of the Laplace transform of its output to the Laplace transform of its input. In Chapters 14 and 15, we'll use the transfer function in circuit design, but here we focus on using it for circuit analysis. We continue this chapter with a look at the role of partial fraction expansion (Section 13.5) and the convolution integral (Section 13.6) when using the transfer function in circuit analysis. We conclude with a discussion of the impulse function in circuit analysis.

■ Practical Perspective

Surge Suppressors

When using personal computers and other sensitive electronic equipment, we need to provide protection from voltage surges. These surges can occur during a lightning storm or just by switching an electrical device on or off. A commercially available surge suppressor is shown in the accompanying figure.

How can flipping a switch to turn on a light or turn off a hair dryer cause a voltage surge? At the end of this chapter, we will answer that question using Laplace transform techniques to analyze a circuit. We will illustrate how a voltage surge can be created by switching off a resistive load in a circuit operating in the sinusoidal steady state.



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Jhaz Photography/Shutterstock

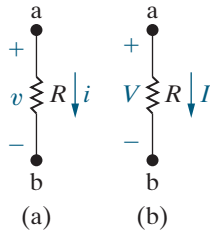


Figure 13.1 ▲ The resistor. (a) Time domain. (b) Frequency domain.

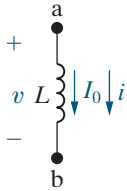


Figure 13.2 ▲ An inductor of L henrys carrying an initial current of I_0 amperes.

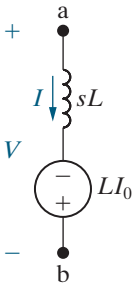


Figure 13.3 ▲ The series equivalent circuit for an inductor of L henrys carrying an initial current of I_0 amperes.

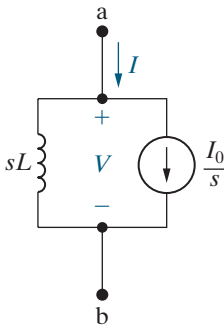


Figure 13.4 ▲ The parallel equivalent circuit for an inductor of L henrys carrying an initial current of I_0 amperes.

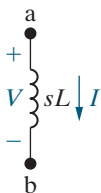


Figure 13.5 ▲ The s -domain circuit for an inductor when the initial current is zero.

13.1 Circuit Elements in the s Domain

A three-step procedure transforms each circuit element into an s -domain equivalent circuit.

1. Write the time-domain equation relating the terminal voltage to the terminal current.
2. Laplace transform the time-domain equation to generate an algebraic relationship between the s -domain current and voltage.
3. Construct a circuit model that satisfies the relationship between the s -domain current and voltage.

Note the s -domain dimensions: the s -domain voltage dimension is volt-seconds [V-s], the s -domain current dimension is ampere-seconds [A-s], and thus the s -domain voltage-to-current ratio dimension is volts per ampere, or ohms. An impedance in the s domain is measured in ohms, and an admittance is measured in siemens. We use the passive sign convention in all the derivations.

A Resistor in the s Domain

We begin with the resistor. From Ohm's law,

$$v = Ri.$$

Because R is a constant, the Laplace transform of Ohm's law is

$$V = RI, \quad (13.1)$$

where

$$V = \mathcal{L}\{v\} \quad \text{and} \quad I = \mathcal{L}\{i\}.$$

From Eq. 13.1 we see that the s -domain equivalent circuit of a resistor is a resistance of R ohms that carries a current of I ampere-seconds and has a terminal voltage of V volt-seconds.

Figure 13.1 shows the time- and frequency-domain circuits of the resistor. Note that going from the time domain to the frequency domain does not change the resistance element.

An Inductor in the s Domain

Figure 13.2 shows an inductor carrying an initial current of I_0 amperes. The time-domain equation relating the terminal voltage to the terminal current is

$$v = L \frac{di}{dt}.$$

The Laplace transform of the inductor equation gives

$$V = L[sI - i(0^-)] = sLI - LI_0. \quad (13.2)$$

One circuit configuration that satisfies Eq. 13.2 is an impedance of sL ohms in series with an independent voltage source of LI_0 volt-seconds, as shown in Fig. 13.3. Note that the polarity of the voltage source LI_0 agrees with the minus sign in Eq. 13.2. Note also that LI_0 carries

its own algebraic sign; that is, if the initial value of i is opposite to the reference direction for i , then I_0 has a negative value.

We can also solve Eq. 13.2 for the current I and then construct the circuit to satisfy the resulting equation. The current I is given by

$$I = \frac{V + LI_0}{sL} = \frac{V}{sL} + \frac{I_0}{s}.$$

The s -domain equivalent circuit that satisfies this equation is an impedance of sL ohms in parallel with an independent current source of I_0/s ampere-seconds, as seen in Fig. 13.4.

There are two other ways to construct the s domain circuit in Fig. 13.4: (1) find the Norton equivalent of the circuit shown in Fig. 13.3 or (2) start with the inductor current as a function of the inductor voltage in the time domain and then Laplace transform the resulting integral equation. We leave these two approaches to Problems 13.1 and 13.2.

If the initial energy stored in the inductor is zero, that is, if $I_0 = 0$, the s -domain equivalent circuit is an inductor with an impedance of sL ohms. Figure 13.5 shows this circuit.

A Capacitor in the s Domain

A capacitor with initial stored energy also has two s -domain equivalent circuits. Figure 13.6 shows a capacitor initially charged to V_0 volts. The capacitor current is

$$i = C \frac{dv}{dt}.$$

Transforming the capacitor equation yields

$$I = C[sV - v(0^-)] = sCV - CV_0, \quad (13.3)$$

so the s -domain current I is the sum of two branch currents. One branch contains an admittance of sC siemens, and the second branch contains an independent current source of CV_0 ampere-seconds. Figure 13.7 shows this parallel equivalent circuit.

To derive the other equivalent circuit for the charged capacitor, solve Eq. 13.3 for V :

$$V = \left(\frac{1}{sC}\right)I + \frac{V_0}{s}.$$

Figure 13.8 shows the circuit that satisfies the equation for capacitor voltage, which is a series combination of an impedance and an independent voltage source.

In the equivalent circuits shown in Figs. 13.7 and 13.8, V_0 carries its own algebraic sign. In other words, if the polarity of V_0 is opposite to the reference polarity for v , V_0 is a negative quantity. If the initial voltage on the capacitor is zero, both equivalent circuits reduce to an impedance of $1/sC$ ohms, as shown in Fig. 13.9.

In Chapter 9 we used the phasor transform to turn a time-domain circuit into a frequency-domain circuit. In this chapter, we use the s -domain equivalent circuits, summarized in Table 13.1, to transform a time-domain circuit into the s domain. When the time-domain circuit contains inductors and capacitors with initial stored energy, you need to decide whether to use the parallel or series s -domain equivalent circuit. With a little forethought and some experience, the best choice is often evident.

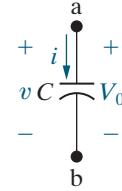


Figure 13.6 ▲ A capacitor of C farads initially charged to V_0 volts.

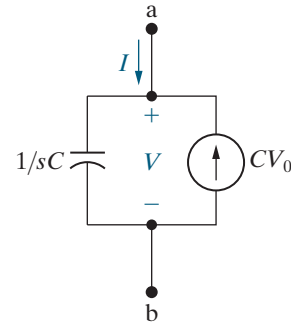


Figure 13.7 ▲ The parallel equivalent circuit for a capacitor initially charged to V_0 volts.

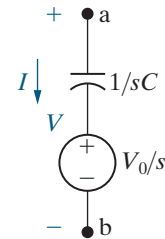


Figure 13.8 ▲ The series equivalent circuit for a capacitor initially charged to V_0 volts.

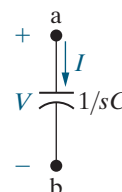
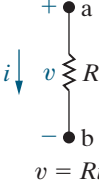
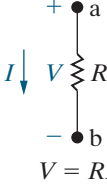
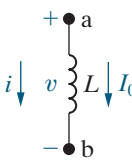
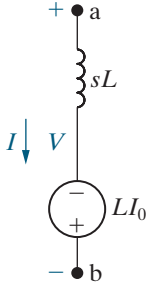
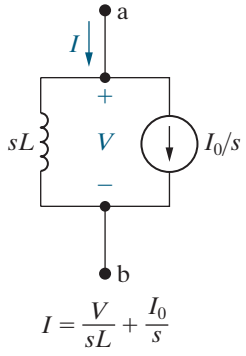
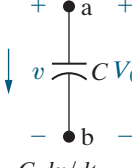
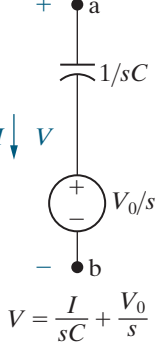
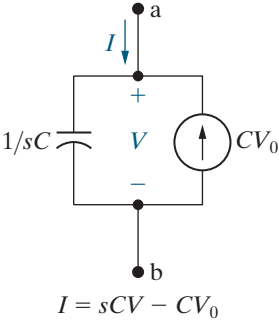


Figure 13.9 ▲ The s -domain circuit for a capacitor when the initial voltage is zero.

TABLE 13.1 Summary of the s -Domain Equivalent Circuits

Time Domain	Frequency Domain	
 $v = Ri$	 $V = RI$	
 $v = L \frac{di}{dt},$ $i = \frac{1}{L} \int_{0^-}^t v dx + I_0$	 $V = sLI - LI_0$	 $I = \frac{V}{sL} + \frac{I_0}{s}$
 $i = C \frac{dv}{dt},$ $v = \frac{1}{C} \int_{0^-}^t i dx + V_0$	 $V = \frac{I}{sC} + \frac{V_0}{s}$	 $I = sCV - CV_0$

13.2 Circuit Analysis in the s Domain

Before presenting a method for using the s -domain equivalent circuits in analysis, we make some important observations.

- If no energy is stored in the inductor or capacitor, the relationship between the s -domain voltage and current for each passive element is:

OHM’S LAW IN THE s DOMAIN

$$V = ZI, \tag{13.4}$$

where Z is the s -domain impedance of the element. Thus, a resistor has an impedance of R ohms, an inductor has an impedance of sL ohms, and a capacitor has an impedance of $1/sC$ ohms. The relationship contained in Eq. 13.4 also appears in Figs. 13.1(b), 13.5, and 13.9. Equation 13.4 is known as Ohm’s law in the s domain. The reciprocal of the impedance is admittance. Therefore, the s -domain admittance of a resistor is $1/R$ siemens, an inductor has an admittance of $1/sL$ siemens, and a capacitor has an admittance of sC siemens.

- The rules for combining impedances and admittances in the s domain are the same as those for frequency-domain circuits. Thus, series-parallel simplifications and Δ -to- Y conversions also are applicable to s -domain analysis.
- Kirchhoff's laws apply to s -domain currents and voltages because an operational Laplace transform states that the Laplace transform of a sum of time-domain functions equals the sum of the Laplace transforms of the individual functions (see Table 12.2). The algebraic sum of the currents at a node is zero in the time domain, so the algebraic sum of the Laplace-transformed currents is also zero. A similar statement holds for the algebraic sum of the Laplace-transformed voltages around a closed path. The s -domain version of Kirchhoff's laws is

$$\text{at every node in a circuit, } \sum I = 0, \quad (13.5)$$

$$\text{around every closed path in a circuit, } \sum V = 0. \quad (13.6)$$

Therefore, because Ohm's law, KVL, and KCL hold in the s domain, all of the circuit-analysis techniques developed in Chapters 2–4 for resistive circuits can be used to analyze circuits in the s domain. These techniques include combining impedances in series and parallel to find equivalent impedances, voltage division and current division, the node-voltage method and the mesh-current method, source transformation, and Thévenin and Norton equivalent circuits. This leads us to the following step-by-step procedure for using Laplace transform techniques to analyze circuits.

Step 1 determines the initial current in each inductor and the initial voltage across each capacitor by analyzing the time-domain circuit for $t < 0$.

Step 2 transforms each independent voltage or current source defined by time-domain functions into the s domain using the functional and operational transforms in Tables 12.1 and 12.2.

Step 3 transforms voltages and currents represented by time-domain symbols such as $v(t)$ and $i(t)$ into corresponding s -domain symbols such as V and I .

Step 4 transforms any remaining components in the time-domain circuit into the s domain using the circuits in Table 13.1. When inductors and capacitors have nonzero initial values, calculated in Step 1, these initial conditions are represented by independent sources in series or parallel with the component impedances.

Step 5 analyzes the resulting s -domain circuit using the techniques developed for resistive circuits in Chapters 2–4. The analysis produces s -domain voltages and currents that should each be represented as a ratio of two polynomials in s .

Step 6 applies the initial- and final-value theorems to the s -domain functions from Step 5 to check the values of the corresponding time-domain functions at $t = 0$ and $t = \infty$. Note that it might not be possible to apply one or both of these theorems, depending on the form of the s -domain function.

Step 7 represents each s -domain voltage and current of interest as a partial fraction expansion and then uses Table 12.3 to inverse-Laplace-transform the s -domain voltages and currents back to the time domain.

This analysis method yields the complete response to any circuit whose voltage and current sources have Laplace transforms. This method represents the most comprehensive circuit analysis technique presented in this text and is summarized in Analysis Method 13.1. Example 13.1 applies the first five steps in Analysis Method 13.1 to a time-domain circuit.

LAPLACE TRANSFORM METHOD

1. **Determine the initial conditions** for inductors and capacitors.
2. **Laplace-transform independent voltage and current functions** using Tables 12.1 and 12.2.
3. **Transform symbolic time-domain voltages and currents** into s -domain symbols.
4. **Transform remaining circuit components** into the s domain using Table 13.1.
5. **Analyze the s -domain circuit** using resistive circuit analysis techniques; represent the resulting s -domain voltages and currents as ratios of polynomials in s .
6. **Use the initial- and final-value theorems** to check the s -domain voltages and currents.
7. **Inverse-Laplace-transform the s -domain voltages and currents** using partial fraction expansion and Table 12.3.

Analysis Method 13.1 Laplace-transform circuit-analysis method.

EXAMPLE 13.1 Transforming a Circuit into the s Domain

A $500\ \Omega$ resistor, a 16 mH inductor, and a 25 nF capacitor are connected in parallel.

- Express the admittance of this parallel combination of elements as a rational function of s .
- Compute the numerical values of the zeros and poles.

Solution

- We use Analysis Method 13.1 to transform the three parallel-connected components from the time domain to the s domain.

Step 1: There are no initial conditions for the inductor and capacitor.

Step 2: There are no voltage or current sources.

Step 3: There are no symbolic voltages or currents.

Step 4: Because there are no initial conditions, each component is represented by its s -domain impedance:

$$Z_R = 500\ \Omega;$$

$$Z_L = 0.016s\ \Omega;$$

$$Z_C = \frac{1}{25 \times 10^{-9}s} = \frac{40 \times 10^6}{s}\ \Omega.$$

The s -domain circuit is shown in Fig. 13.10.

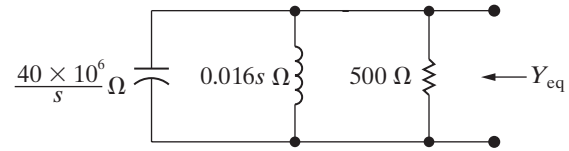


Figure 13.10 ▲ The s domain circuit for Example 13.1.

Step 5: Find the equivalent admittance by adding the inverse of the three impedances:

$$\begin{aligned} Y_{eq} &= \frac{1}{500} + \frac{1}{0.016s} + \frac{s}{40 \times 10^6} \\ &= \frac{s^2 + 80,000s + 25 \times 10^8}{40 \times 10^6 s}\text{ S.} \end{aligned}$$

- The numerator factors are $(s + 40,000 + j30,000)$ and $(s + 40,000 - j30,000)$. Therefore the zeros are $-40,000 + j30,000$ and $-40,000 - j30,000$. There is a pole at 0.

ASSESSMENT PROBLEM

Objective 1 — Be able to transform a circuit into the s domain using Laplace transforms

13.1 A $2\text{ k}\Omega$ resistor, a 6.25 H inductor, and a 250 nF capacitor are in parallel.

- Express the impedance of this parallel combination as a rational function of s .
- Compute the numerical values of the zeros and poles.

Answer: a) $4 \times 10^6 s / (s^2 + 2000s + 64 \times 10^4)$;

- $-z_1 = 0$,
 $-p_1 = -400\text{ rad/s}$,
 $-p_2 = -1600\text{ rad/s}$.

SELF-CHECK: Also try Chapter Problems 13.5 and 13.6.

13.3 Applications

We now use Analysis Method 13.1 to find the complete response of several example circuits. We start with two familiar circuits from Chapters 7 and 8 to show that the Laplace transform approach yields the same results found using the first- and second-order circuit analysis techniques. Example 13.2 solves an RC circuit, and Example 13.3 solves an RLC circuit.

EXAMPLE 13.2 The Natural Response of an RC Circuit

The circuit in Fig. 13.11 was analyzed in Example 7.3 using first-order circuit analysis techniques. Use the Laplace transform method to find $v_o(t)$ for $t \geq 0^+$.

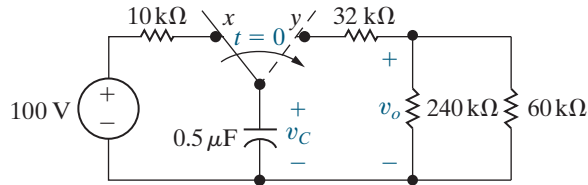


Figure 13.11 ▲ The RC circuit for Example 13.2.

Solution

Apply the Laplace transform method using Analysis Method 13.1.

Step 1: Determine the initial voltage across the capacitor by analyzing the circuit in Fig. 13.11 for $t < 0$. Because the switch has been in position x for a long time, the capacitor behaves like an open circuit. The voltage across the open circuit $V_0 = 100$ V.

Step 2: For $t \geq 0$, there are no voltage or current sources in the circuit, so we can skip this step.

Step 3: The voltage across the $240 \text{ k}\Omega$ resistor is represented in the s domain as V , as shown in Fig. 13.12.

Step 4: The impedance of the three resistors is their resistance. The impedance of the capacitor is

$$Z_C = \frac{1}{sC} = \frac{1}{0.5 \times 10^{-6}s} = \frac{2 \times 10^6}{s} \Omega.$$

Because the capacitor has an initial condition, we must decide whether to represent it using a series-connected voltage source or a parallel-connected current source, as shown in Table 13.1. Here we use the series-connected voltage source, whose value is $V_0/s = 100/s$ V-s. The s -domain circuit that results from Steps 1–4 is shown in Fig. 13.12.

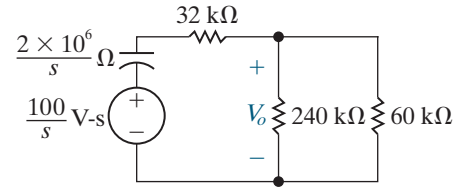


Figure 13.12 ▲ The circuit in Fig. 13.11 for $t \geq 0$, transformed into the s domain.

Step 5: Begin by combining the parallel-connected $240 \text{ k}\Omega$ and $60 \text{ k}\Omega$ resistors into a single equivalent $48 \text{ k}\Omega$ resistor whose voltage is V_o . Now use voltage division to find V_o :

$$\begin{aligned} V_o &= \frac{48,000}{(2 \times 10^6/s) + 32,000 + 48,000} \left(\frac{100}{s} \right) \\ &= \frac{60}{s + 25}. \end{aligned}$$

Step 6: Use the initial- and final-value theorems to show that the initial value of v_o is 60 V and final value of v_o is zero, as we expect from the circuit in Fig. 13.11.

$$\begin{aligned} \lim_{s \rightarrow \infty} sV_o &= \lim_{s \rightarrow \infty} \frac{60s}{s + 25} = \lim_{(1/s) \rightarrow 0} \frac{60}{1 + (25/s)} \\ &= \frac{60}{1 + 0} = 60 = \lim_{t \rightarrow 0} v_o(t); \end{aligned}$$

$$\lim_{s \rightarrow 0} sV_o = \lim_{s \rightarrow 0} \frac{60s}{s + 25} = \frac{60(0)}{0 + 25} = 0 = \lim_{t \rightarrow \infty} v_o(t).$$

Step 7: Since V_o is already a partial fraction, we can use the transforms in Table 12.3 to find v_o :

$$v_o(t) = \mathcal{L}^{-1} \left\{ \frac{60}{s + 25} \right\} = 60e^{-25t}u(t) \text{ V}.$$

This matches the voltage found in Example 7.3 using first-order circuit-analysis methods.

EXAMPLE 13.3 The Step Response of an RLC Circuit

Consider the circuit in Fig. 13.13, where the initial current in the inductor is 29 mA and the initial voltage across the capacitor is 50 V. This circuit was analyzed in Example 8.10 using second-order circuit analysis techniques. Use the Laplace transform method to find $v(t)$ for $t \geq 0$.

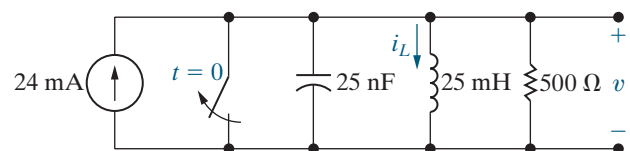


Figure 13.13 ▲ The parallel RLC circuit for Example 13.3.

Solution

Apply the Laplace transform method using Analysis Method 13.1.

Step 1: Both initial conditions are given in the problem statement, so we can skip this step.

Step 2: We can describe the parallel combination of the current source and the switch in Fig. 13.13 as $24u(t)$ mA. The Laplace transform of this function is $24/s$ mA-s, which is the current source value in the s -domain circuit in Fig. 13.14.

Step 3: The voltage across the $240\text{ k}\Omega$ resistor is represented in the s domain as V , as shown in Fig. 13.12.

Step 4: The impedance of the resistor is its resistance. The impedance of the inductor is

$$Z_L = sL = 0.025s = \frac{s}{40} \Omega$$

and the impedance of the capacitor is

$$Z_C = \frac{1}{sC} = \frac{1}{25 \times 10^{-9}s} = \frac{40 \times 10^6}{s} \Omega.$$

Both the inductor and capacitor have non-zero initial conditions. We must decide whether to represent them as series-connected voltage sources or as parallel-connected current sources, as shown in Table 13.1. Here we use parallel-connected current sources because the resulting circuit has only parallel-connected components. The value of the current source in parallel with the inductor is $I_0/s = 0.029/s$ A-s, with the current arrow directed down. The value of the current source in parallel with the capacitor is $CV_0 = (50)(25 \times 10^{-9}) = 1.25 \mu\text{A-s}$, with the current arrow directed up. The s -domain circuit that results from Steps 1–4 is shown in Fig. 13.14.

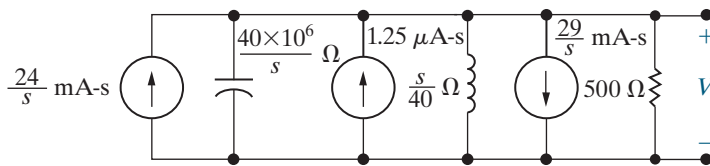


Figure 13.14 ▲ The circuit in Fig. 13.13 for $t \geq 0$, transformed into the s domain.

Step 5: Begin by combining the three parallel-connected impedances into a single equivalent impedance:

$$\begin{aligned} Z_{\text{eq}} &= \left(\frac{s}{40 \times 10^6} + \frac{40}{s} + \frac{1}{500} \right)^{-1} \\ &= \frac{40 \times 10^6 s}{s^2 + 80,000s + 16 \times 10^8} \Omega. \end{aligned}$$

To find the voltage across this equivalent impedance, multiply by the sum of the three parallel-connected currents:

$$\begin{aligned} V &= Z_{\text{eq}} \left(\frac{0.024}{s} + 1.25 \times 10^{-6} - \frac{0.029}{s} \right) \\ &= \frac{50s - 20 \times 10^4}{s^2 + 80,000s + 16 \times 10^8} \text{ V-s}. \end{aligned}$$

Step 6: Use the initial- and final-value theorems to predict the initial and final values of v :

$$\begin{aligned} \lim_{s \rightarrow \infty} sV &= \lim_{s \rightarrow \infty} \frac{50s^2 - 20 \times 10^4 s}{s^2 + 80,000s + 16 \times 10^8} \\ &= \lim_{(1/s) \rightarrow 0} \frac{50 - 20 \times 10^4 (1/s)}{1 + 80,000(1/s) + 16 \times 10^8 (1/s)^2} \\ &= \frac{50}{1} = 50 = \lim_{t \rightarrow 0} v(t); \end{aligned}$$

$$\begin{aligned} \lim_{s \rightarrow 0} sV &= \lim_{s \rightarrow 0} \frac{50s^2 - 20 \times 10^4 s}{s^2 + 80,000s + 16 \times 10^8} \\ &= \frac{(0)}{16 \times 10^8} = 0 = \lim_{t \rightarrow \infty} v(t). \end{aligned}$$

The initial-value theorem predicts the correct initial voltage from the problem statement, $V_0 = 50\text{ V}$. To confirm the final value of the voltage, envision the circuit in Fig. 13.13 as $t \rightarrow \infty$. The inductor is behaving like a short circuit, and the capacitor is behaving like an open circuit in parallel with the short circuit. Therefore, the final value of capacitor voltage in the circuit is zero, as predicted by the final-value theorem.

Step 7: The partial fraction expansion of V is

$$\begin{aligned} V &= \frac{50s - 20 \times 10^4}{(s + 40,000)^2} \\ &= \frac{-2.2 \times 10^6}{(s + 40,000)^2} + \frac{50}{(s + 40,000)}. \end{aligned}$$

Now use the transforms in Table 12.3 to find v :

$$\begin{aligned} v(t) &= \mathcal{L}^{-1} \left\{ \frac{-2.2 \times 10^6}{(s + 40,000)^2} + \frac{50}{(s + 40,000)} \right\} \\ &= (-2.2 \times 10^6 te^{-40,000t} + 50e^{-40,000t})u(t) \text{ V}. \end{aligned}$$

This matches the voltage found in Example 8.10 using second-order circuit analysis methods.

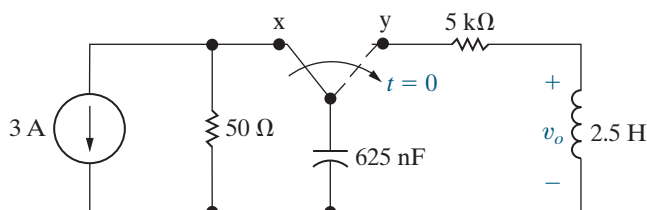
ASSESSMENT PROBLEMS

Objective 2—Know how to analyze a circuit in the s domain and be able to transform an s domain solution to the time domain

13.2 The switch in the circuit shown has been in position x for a long time. At $t = 0$, the switch is thrown to position y.

- Find V_o as a rational function of s .
- Find the time-domain expression for v_o .

Answer: a) $V_o = -150/(s + 400)(s + 1600)$;
b) $v_o = (50e^{-400t} - 200e^{-1600t})u(t)$ V.



13.3 There is no energy stored in the circuit shown at the time the current source is turned on.

- Find the s -domain expression for V .
- Find the s -domain expression for I .

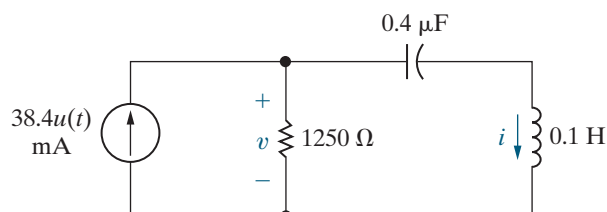
- Find the time-domain expression for v when $t > 0$.
- Find the time-domain expression for i when $t > 0$.

Answer: a) $V = \frac{48(s^2 + 25 \times 10^6)}{s(s + 2500)(s + 10,000)}$;

b) $I = \frac{480}{(s + 2500)(s + 10,000)}$;

c) $v(t) = (48 - 80e^{-2500t} + 80e^{-10,000t})u(t)$ V;

d) $i(t) = (64e^{-2500t} - 64e^{-10,000t})u(t)$ mA.



SELF-CHECK: Also try Chapter Problems 13.11, 13.12, and 13.22.

Next, we analyze a circuit with a sinusoidal source using Laplace methods. Phasor methods, presented in Chapter 9, were used to analyze circuits with sinusoidal sources but have an important limitation—they only produce the steady-state response of the circuit, not the complete response. Recall that the complete response consists of both the steady-state and the transient response. Laplace methods do not have this limitation, so they produce a circuit's complete response to a sinusoidal source, as seen in Example 13.4.

EXAMPLE 13.4 Analyzing a Circuit with a Sinusoidal Source

The circuit in Fig. 13.15 has no initial stored energy. At $t = 0$, the switch closes and the circuit is driven by a sinusoidal source $v(t) = 15 \cos 40,000t$ V. Use Laplace methods to find $i(t)$ for $t \geq 0$.

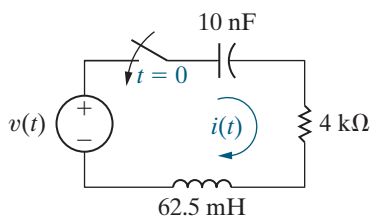


Figure 13.15 ▲ The circuit for Example 13.4.

Solution

Apply the Laplace transform method using Analysis Method 13.1.

Step 1: There is no initial stored energy, so both initial conditions are zero.

Step 2: Using the functional and operational transform tables (Tables 12.1 and 12.2), we see that the Laplace transform of $v(t)$ is

$$V = \mathcal{L}\{15 \cos 40,000t\} = \frac{15s}{s^2 + 40,000^2} \text{ V-s.}$$

This is the value of the voltage source for the s -domain circuit in Fig. 13.16.

Step 3: The current is represented in the s domain as I , as shown in Fig. 13.16.

Step 4: The impedance of the resistor is its resistance. The impedance of the inductor is

$$Z_L = sL = 0.0625s = \frac{s}{16} \Omega$$

and the impedance of the capacitor is

$$Z_C = \frac{1}{sC} = \frac{1}{10 \times 10^{-9} s} = \frac{10^8}{s} \Omega.$$

The s -domain circuit that results from Steps 1–4 is shown in Fig. 13.16.

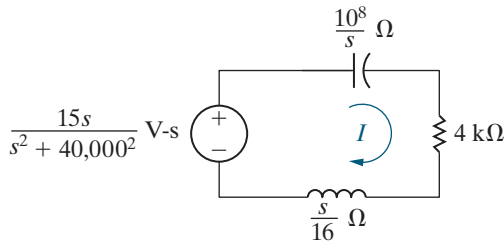


Figure 13.16 ▲ The circuit in Fig 13.16 for $t \geq 0$, transformed into the s domain.

Step 5: To find the current I , divide the source voltage by the sum of the impedances:

$$\begin{aligned} I &= \frac{\frac{15s}{s^2 + 40,000^2}}{4000 + \frac{10^8}{s} + \frac{s}{16}} \\ &= \frac{240s^2}{(s^2 + 40,000^2)(s^2 + 64,000s + 16 \times 10^8)} \text{ A-s.} \end{aligned}$$

Step 6: Use the initial-value theorem to predict the initial value of i :

$$\begin{aligned} \lim_{s \rightarrow \infty} sI &= \lim_{s \rightarrow \infty} \frac{240s^3}{(s^2 + 40,000^2)(s^2 + 64,000s + 16 \times 10^8)} \\ &= \lim_{(1/s) \rightarrow 0} \frac{240\left(\frac{1}{s}\right)}{\left[1 + \left(\frac{40,000}{s}\right)^2\right]\left[1 + \left(\frac{60,000}{s}\right) + \left(\frac{16 \times 10^8}{s^2}\right)\right]} \\ &= \frac{0}{(1)(1)} = 0 = \lim_{t \rightarrow 0} i(t). \end{aligned}$$

The initial-value theorem predicts correctly that the initial current is zero. We cannot use the final-value theorem to predict the final current because I has two poles on the imaginary axis at $\pm j40,000$ rad/s.

Step 7: The partial fraction expansion of I is

$$\begin{aligned} I &= \frac{240s^2}{(s^2 + 40,000^2)(s^2 + 64,000s + 16 \times 10^8)} \\ &= \frac{K_1}{s - j40,000} + \frac{K_1^*}{s + j40,000} \\ &\quad + \frac{K_2}{s + 32,000 - j24,000} + \frac{K_2^*}{s + 32,000 + j24,000}. \end{aligned}$$

Finding K_1 and K_2 ,

$$\begin{aligned} K_1 &= \left. \frac{240s^2}{(s + j40,000)(s^2 + 64,000s + 16 \times 10^8)} \right|_{s=j40,000} \\ &= 1.875 \times 10^{-3}; \\ K_2 &= \left. \frac{240s^2}{(s^2 + 40,000^2)(s + 32,000 + j24,000)} \right|_{s=-32,000+j24,000} \\ &= 3.125 \times 10^{-3} \angle -126.87^\circ. \end{aligned}$$

Now use the transforms in Table 12.3 to find i :

$$\begin{aligned} i(t) &= \mathcal{L}^{-1} \left\{ \frac{1.875 \times 10^{-3}}{s - j40,000} + \frac{1.875 \times 10^{-3}}{s + j40,000} \right. \\ &\quad \left. + \frac{3.125 \times 10^{-3} \angle -126.87^\circ}{s + 32,000 - j24,000} + \frac{3.125 \times 10^{-3} \angle 126.87^\circ}{s + 32,000 + j24,000} \right\} \\ &= [3.75 \cos 40,000t \\ &\quad + 6.25e^{-32,000t} \cos(24,000t - 126.87^\circ)] u(t) \text{ mA.} \end{aligned}$$

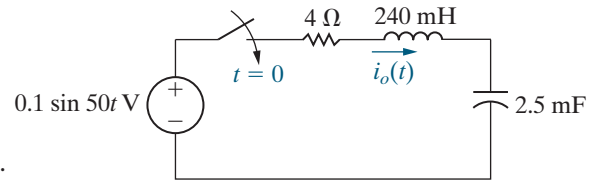
The first term in the expression for $i(t)$ is the steady-state response. Its frequency matches the frequency of the source, and this term persists for all time. You should use the phasor methods from Chapter 9 to verify this result. The second term in the expression for $i(t)$ is the transient response, or the natural response. Note that it decays to zero as $t \rightarrow \infty$. This part of the response is independent of the voltage source and is based only on the passive component values and their interconnections.

ASSESSMENT PROBLEM

Objective 2—Know how to analyze a circuit in the s domain and be able to transform an s -domain solution to the time domain

13.4 The energy stored in the circuit shown is zero at the time when the switch is closed. Find the current $i_o(t)$ for $t \geq 0$.

Answer: $[17.68 \cos(50t - 135^\circ) + 18.06e^{-8.33t} \cos(39.965t + 46.194^\circ)]u(t)$ mA.



SELF-CHECK: Also try Chapter Problem 13.24.

Until now, we avoided analyzing circuits with inductors and capacitors that have two or more meshes, or three or more essential nodes. Such circuits are described by two or more simultaneous differential equations, and the techniques for solving these systems of equations are beyond the scope of this text. However, using Laplace techniques, we can transform a circuit into the s domain and write a set of simultaneous algebraic equations, whose solution is much more manageable. Example 13.5 illustrates this by solving a circuit with two meshes.

EXAMPLE 13.5 Analyzing a Circuit with Multiple Meshes

The circuit in Fig. 13.17 has no initial stored energy. At $t = 0$, the switch closes. Use Laplace methods to find $i_1(t)$ and $i_2(t)$ for $t \geq 0$.

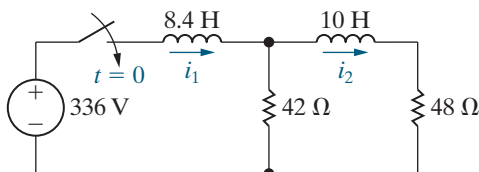


Figure 13.17 ▲ A multiple-mesh RL circuit.

Solution

Apply the Laplace transform method using Analysis Method 13.1.

Step 1: There is no initial stored energy, so both initial conditions are zero.

Step 2: The series connection of the 336 V dc voltage source and the switch can be described in the time domain as $336u(t)$ V. Using the functional and operational transform tables (Tables 12.1 and 12.2), the Laplace transform of this voltage is

$$\mathcal{L}\{336u(t)\} = \frac{336}{s} \text{ V}\cdot\text{s}.$$

This is the value of the voltage source for the s -domain circuit in Fig. 13.18.

Step 3: The currents are represented in the s domain as I_1 and I_2 , as shown in Fig. 13.18.

Step 4: The impedance of the resistors is their resistance. The impedance of the inductors is

$$Z_{L1} = sL_1 = 8.4s \Omega; \quad Z_{L2} = sL_2 = 10s \Omega.$$

The s -domain circuit that results from Steps 1–4 is shown in Fig. 13.18.

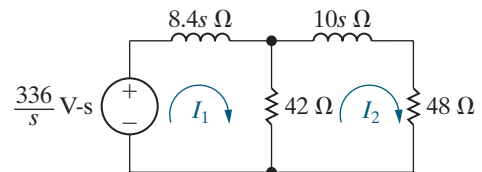


Figure 13.18 ▲ The s -domain equivalent circuit for the circuit shown in Fig. 13.17.

Step 5: The two KVL equations are

$$\frac{336}{s} = (42 + 8.4s)I_1 - 42I_2,$$

$$0 = -42I_1 + (90 + 10s)I_2.$$

Using Cramer's method, we get

$$\begin{aligned}\Delta &= \begin{vmatrix} 42 + 8.4s & -42 \\ -42 & 90 + 10s \end{vmatrix} \\ &= 84(s^2 + 14s + 24) \\ &= 84(s + 2)(s + 12), \\ N_1 &= \begin{vmatrix} 336/s & -42 \\ 0 & 90 + 10s \end{vmatrix} \\ &= \frac{3360(s + 9)}{s}, \\ N_2 &= \begin{vmatrix} 42 + 8.4s & 336/s \\ -42 & 0 \end{vmatrix} \\ &= \frac{14,112}{s}.\end{aligned}$$

Therefore,

$$\begin{aligned}I_1 &= \frac{N_1}{\Delta} = \frac{40(s + 9)}{s(s + 2)(s + 12)}, \\ I_2 &= \frac{N_2}{\Delta} = \frac{168}{s(s + 2)(s + 12)}.\end{aligned}$$

Step 6: Use the initial-value theorem to predict the initial values of i_1 and i_2 :

$$\begin{aligned}\lim_{s \rightarrow \infty} sI_1 &= \lim_{s \rightarrow \infty} \frac{40(s + 9)}{(s + 2)(s + 12)} \\ &= \lim_{(1/s) \rightarrow 0} \frac{40[(1/s) + 9(1/s)^2]}{(1 + (2/s))(1 + (12/s))} \\ &= \frac{0}{(1)(1)} = 0 = \lim_{t \rightarrow 0} i_1(t); \\ \lim_{s \rightarrow \infty} sI_2 &= \lim_{s \rightarrow \infty} \frac{168}{(s + 2)(s + 12)} \\ &= \lim_{(1/s) \rightarrow 0} \frac{168(1/s)^2}{(1 + (2/s))(1 + (12/s))} \\ &= \frac{0}{(1)(1)} = 0 = \lim_{t \rightarrow 0} i_2(t).\end{aligned}$$

The initial-value theorem predicts correctly that the initial currents are zero. Now use the final-value theorem to predict the values of i_1 and i_2 as $t \rightarrow \infty$:

$$\begin{aligned}\lim_{s \rightarrow 0} sI_1 &= \lim_{s \rightarrow 0} \frac{40(s + 9)}{(s + 2)(s + 12)} = \frac{40(9)}{(2)(12)} \\ &= 15 \text{ A} = \lim_{t \rightarrow \infty} i_1(t);\end{aligned}$$

$$\begin{aligned}\lim_{s \rightarrow 0} sI_2 &= \lim_{s \rightarrow 0} \frac{168}{(s + 2)(s + 12)} \\ &= \frac{168}{(2)(12)} = 7 \text{ A} = \lim_{t \rightarrow \infty} i_2(t).\end{aligned}$$

To verify these values, consider the circuit in Fig. 13.17 as $t \rightarrow \infty$. The inductors now behave like short circuits, as shown in Fig. 13.19.

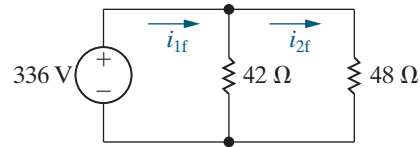


Figure 13.19 ▲ The circuit in Fig 13.17 as $t \rightarrow \infty$.

The two KVL equations that describe this circuit are

$$\begin{aligned}42i_{1f} - 42i_{2f} &= 336; \\ -42i_{1f} + 90i_{2f} &= 0.\end{aligned}$$

Solving, we find the final value of i_1 is 15 A and the final value of i_2 is 7 A, as predicted by the final-value theorem.

Step 7: Expanding I_1 and I_2 into a sum of partial fractions gives

$$\begin{aligned}I_1 &= \frac{15}{s} - \frac{14}{s + 2} - \frac{1}{s + 12}, \\ I_2 &= \frac{7}{s} - \frac{8.4}{s + 2} + \frac{1.4}{s + 12}.\end{aligned}$$

We obtain the expressions for i_1 and i_2 by inverse-transforming I_1 and I_2 , using Table 12.3:

$$\begin{aligned}i_1 &= (15 - 14e^{-2t} - e^{-12t})u(t) \text{ A}, \\ i_2 &= (7 - 8.4e^{-2t} + 1.4e^{-12t})u(t) \text{ A}.\end{aligned}$$

One final test involves calculating the voltage drop across the 42 Ω resistor using three different methods. From the circuit, the voltage across the 42 Ω resistor (positive at the top) is

$$v = 42(i_1 - i_2) = 336 - 8.4 \frac{di_1}{dt} = 48i_2 + 10 \frac{di_2}{dt}.$$

You should verify that regardless of which expression is used, the voltage is

$$v = (336 - 235.2e^{-2t} - 100.80e^{-12t})u(t) \text{ V}.$$

We are thus confident that the solutions for i_1 and i_2 are correct.

ASSESSMENT PROBLEM

Objective 2—Know how to analyze a circuit in the s domain and be able to transform an s -domain solution to the time domain

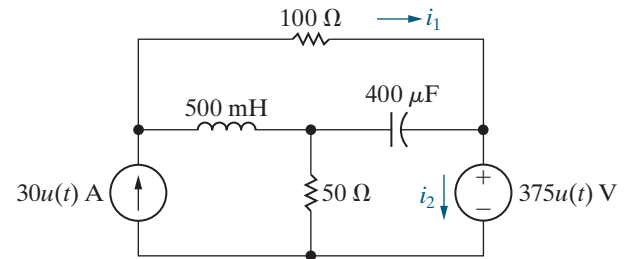
13.5 There is no energy stored in the circuit shown at the time the sources are energized.

- Find $I_1(s)$ and $I_2(s)$.
- Find $i_1(t)$ and $i_2(t)$ for $t \geq 0$.

Answer: a) $I_1 = \frac{30(s^2 + 50s + 3750)}{s(s + 100)(s + 150)}$,
 $I_2 = \frac{22.5s^2 + 6000s + 112,500}{s(s + 100)(s + 150)}$;

b) $i_1(t) = (7.5 - 52.5e^{-100t} + 75e^{-150t})u(t)$ A,
 $i_2(t) = (7.5 + 52.5e^{-100t} - 37.5e^{-150t})u(t)$ A.

SELF-CHECK: Also try Chapter Problems 13.16 and 13.20.



We can use Thévenin and Norton equivalents to simplify circuits in the s domain. When one part of the circuit changes frequently while another part remains constant, we can find a simpler equivalent for the constant part of the circuit that makes analysis of the entire circuit easier. Example 13.6 creates a Thévenin equivalent to simplify part of an s -domain circuit.

EXAMPLE 13.6

Creating a Thévenin Equivalent in the s Domain

We want to find the capacitor current, i_C , for the circuit in Fig. 13.20. The circuit has no initial stored energy, and at $t = 0$, the switch closes. Find the Thévenin equivalent for the circuit to the left of the terminals a and b in the s domain, using Laplace methods. Then analyze the simplified circuit to find $i_C(t)$ for $t \geq 0$.

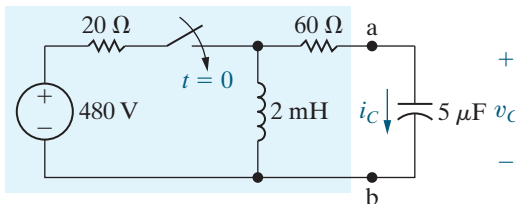


Figure 13.20 ▲ The circuit for Example 13.6.

Solution

Apply the Laplace transform method using Analysis Method 13.1.

Step 1: There is no initial stored energy, so both initial conditions are zero.

Step 2: The series connection of the 480 V dc voltage source and the switch can be described in the time domain as $480u(t)$ V. Using the functional and operational transform tables (Tables 12.1 and 12.2), the Laplace transform of this voltage is

$$\mathcal{L}\{480u(t)\} = \frac{480}{s} \text{ V}\cdot\text{s}.$$

This is the value of the voltage source for the s -domain circuit in Fig. 13.21.

Step 3: The capacitor current and voltage are represented in the s domain as I_C and V_C , as shown in Fig. 13.21.

Step 4: The impedance of the resistors is their resistance. The impedances of the inductor and capacitor are

$$Z_L = sL = 0.002s = \frac{s}{500} \Omega;$$

$$Z_C = \frac{1}{sC} = \frac{1}{5 \times 10^{-6}s} = \frac{2 \times 10^5}{s} \Omega.$$

The s -domain circuit that results from Steps 1–4 is shown in Fig. 13.21.

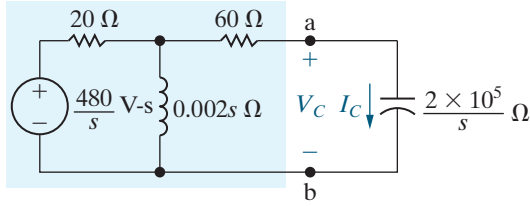


Figure 13.21 ▲ The Laplace transform of the circuit shown in Fig. 13.20.

Step 5: The Thévenin voltage is the open-circuit voltage across terminals a and b. Under open-circuit conditions, there is no voltage across the $60\ \Omega$ resistor. Using voltage division,

$$V_{Th} = \frac{(480/s)(0.002s)}{20 + 0.002s} = \frac{480}{s + 10^4}.$$

The Thévenin impedance seen from terminals a and b equals the $60\ \Omega$ resistor in series with the parallel combination of the $20\ \Omega$ resistor and the inductive impedance. Thus

$$Z_{Th} = 60 + \frac{0.002s(20)}{20 + 0.002s} = \frac{80(s + 7500)}{s + 10^4}.$$

Using the Thévenin equivalent, we reduce the circuit shown in Fig. 13.21 to the one shown in Fig. 13.22.

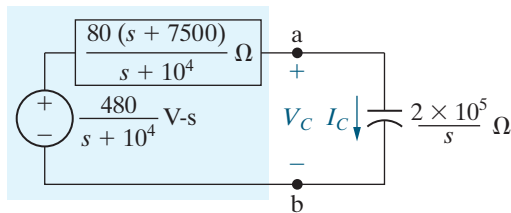


Figure 13.22 ▲ A simplified version of the circuit shown in Fig. 13.21, using a Thévenin equivalent.

In this circuit, the capacitor current I_C equals the Thévenin voltage divided by the total series impedance. Thus,

$$I_C = \frac{\frac{480}{s + 10^4}}{\frac{80(s + 7500)}{s + 10^4} + \frac{2 \times 10^5}{s}} = \frac{6s}{(s + 5000)^2}.$$

Step 6: Use the initial- and final-value theorems to predict the initial and final values of i_C :

$$\begin{aligned} \lim_{s \rightarrow \infty} sI_C &= \lim_{s \rightarrow \infty} \frac{6s^2}{(s + 5000)^2} = \lim_{(1/s) \rightarrow 0} \frac{6}{(1 + (5000/s))^2} \\ &= \frac{6}{(1)^2} = 6\text{ A} = \lim_{t \rightarrow 0} i_C(t); \end{aligned}$$

$$\lim_{s \rightarrow 0} sI_C = \lim_{s \rightarrow 0} \frac{6s^2}{(s + 5000)^2} = \frac{0}{(5000)^2} = 0 = \lim_{t \rightarrow \infty} i_C(t).$$

Let's calculate the initial capacitor current from the circuit in Fig. 13.20. The initial inductor current is zero and the initial capacitor voltage is zero, so the initial capacitor current is $480/(20 + 60)$ or 6 A, which agrees with the prediction of the initial-value theorem. The final value of the capacitor current is zero because as $t \rightarrow \infty$ in the circuit shown in Fig. 13.20, the capacitor behaves like an open circuit. This final capacitor current also agrees with the prediction of the final-value theorem.

Step 7: The partial fraction expansion of I_C is

$$I_C = \frac{-30,000}{(s + 5000)^2} + \frac{6}{s + 5000},$$

and its inverse transform is

$$i_C = (-30,000te^{-5000t} + 6e^{-5000t})u(t)\text{ A}.$$

Suppose we also want to find the voltage drop across the capacitor, v_C . Once we know i_C , we could find v_C by integration in the time domain:

$$v_C = 2 \times 10^5 \int_{0^-}^t (6 - 30,000x)e^{-5000x} dx.$$

Although the integration is not difficult, we can avoid it altogether by first finding the s -domain expression for V_C and then using the inverse transform to find v_C . Thus

$$\begin{aligned} V_C &= \frac{1}{sC} I_C = \frac{2 \times 10^5}{s} \frac{6s}{(s + 5000)^2} \\ &= \frac{12 \times 10^5}{(s + 5000)^2}, \end{aligned}$$

and

$$v_C = 12 \times 10^5 te^{-5000t} u(t)\text{ V}.$$

You can explore this circuit's behavior further in Problem 13.36.

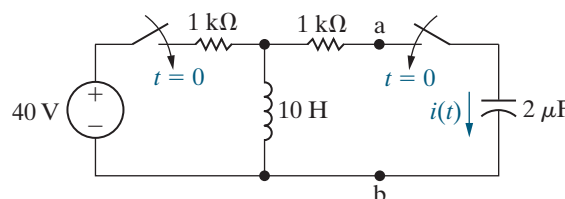
ASSESSMENT PROBLEM

Objective 2—Know how to analyze a circuit in the s domain and be able to transform an s -domain solution to the time domain

13.6 The two switches in the circuit shown operate simultaneously. There is no energy stored in the circuit at the instant the switches close. Find $i(t)$ for $t \geq 0^+$ by first finding the Thévenin equivalent of the circuit to the left of the terminals a,b.

Answer: $63.25e^{-150t} \cos(50t + 71.57^\circ)u(t)$ mA.

SELF-CHECK: Also try Chapter Problem 13.34.



We can also use Laplace transform methods to analyze circuits with mutually coupled coils. Example 13.7 illustrates this process.

EXAMPLE 13.7 Analyzing a Circuit with Mutual Inductance

Consider the circuit in Fig. 13.23. The make-before-break switch has been in position a for a long time. At $t = 0$, the switch moves instantaneously to position b. Use Laplace methods to find $i_2(t)$ for $t \geq 0$.

Solution

Apply the Laplace transform method using Analysis Method 13.1.

Step 1: Because the switch has been in position a for a long time, both inductors behave like short circuits. The current in the 2 H inductor is $60/(9 + 3) = 5$ A, and the current in the 8 H inductor is zero. Therefore, $i_1(0^-) = 5$ A and $i_2(0^-) = 0$.

Step 2: For $t \geq 0$, there is no independent source in the circuit, so we can skip this step.

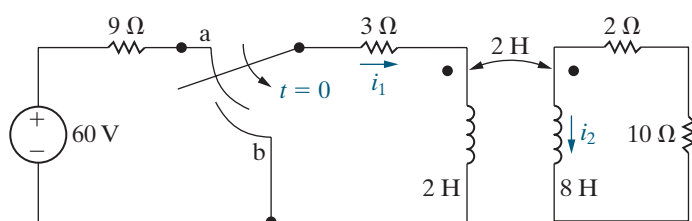


Figure 13.23 ▲ The circuit for Example 13.7, containing magnetically coupled coils.

Step 3: The currents in the two coils are represented in the s domain as I_1 and I_2 .

Step 4: Before calculating the impedance of the resistors and inductors, we replace the magnetically coupled coils with a T-equivalent circuit.¹ Figure 13.24 shows the new circuit.

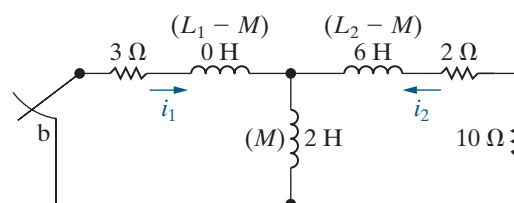


Figure 13.24 ▲ The circuit shown in Fig. 13.23, with the magnetically coupled coils replaced by a T-equivalent circuit.

The impedance of the resistors is their resistance, and the impedance of the inductors is sL . Because we plan to use the mesh-current method in the s domain, we use the series-equivalent circuit for inductors carrying initial current. We place a voltage source in the vertical leg of the tee, to represent the initial value of the current in that

¹ See Appendix C.

vertical leg, which is $i_1(0^-) + i_2(0^-)$, or 5 A. The s -domain circuit resulting from Steps 1–4 is shown in Fig. 13.25.

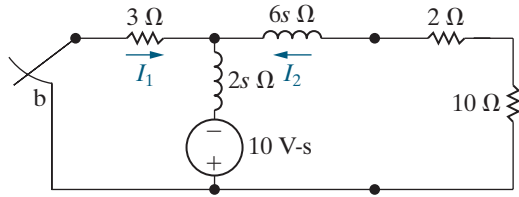


Figure 13.25 ▲ The s -domain equivalent circuit for the circuit shown in Fig. 13.24.

Step 5: Write the two s -domain mesh equations for the circuit in Fig. 13.25:

$$(3 + 2s)I_1 + 2sI_2 = 10;$$

$$2sI_1 + (12 + 8s)I_2 = 10.$$

Solving for I_2 yields

$$I_2 = \frac{2.5}{(s + 1)(s + 3)}.$$

Step 6: Use the initial- and final-value theorems to predict the initial and final values of i_2 :

$$\begin{aligned} \lim_{s \rightarrow \infty} sI_2 &= \lim_{s \rightarrow \infty} \frac{2.5s}{(s + 1)(s + 3)} \\ &= \lim_{(1/s) \rightarrow 0} \frac{2.5(1/s)}{(1 + (1/s))(1 + (3/s))} \\ &= \frac{0}{(1)(1)} = 0 = \lim_{t \rightarrow 0} i_2(t); \end{aligned}$$

$$\begin{aligned} \lim_{s \rightarrow 0} sI_2 &= \lim_{s \rightarrow 0} \frac{2.5s}{(s + 1)(s + 3)} = \frac{0}{(1)(3)} \\ &= 0 = \lim_{t \rightarrow \infty} i_2(t). \end{aligned}$$

The initial- and final-value theorems correctly predict the initial and final values of the current i_2 for the circuit in Fig. 13.23.

Step 7: The partial fraction expansion of I_2 is

$$I_2 = \frac{1.25}{s + 1} - \frac{1.25}{s + 3}.$$

Using Table 12.3, the inverse Laplace transform of I_2 is

$$i_2 = (1.25e^{-t} - 1.25e^{-3t})u(t) \text{ A}.$$

Figure 13.26 shows a plot of i_2 versus t . This response makes sense in terms of the known physical behavior of the magnetically coupled coils. A current can exist in the L_2 inductor only if there is a time-varying current in the L_1 inductor. As i_1 decreases from its initial value of 5 A, i_2 increases from zero and then approaches zero as i_1 approaches zero.

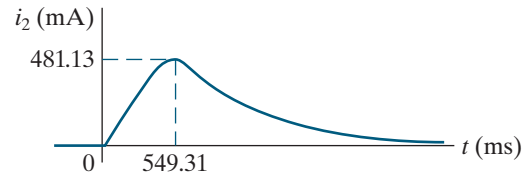


Figure 13.26 ▲ The plot of i_2 versus t for the circuit shown in Fig. 13.23.

ASSESSMENT PROBLEM

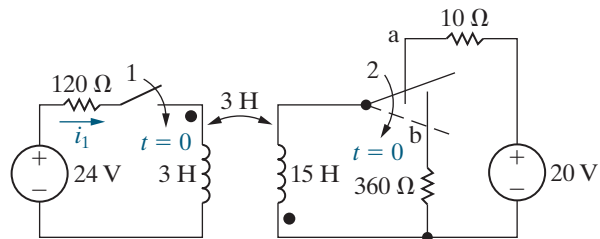
Objective 2—Know how to analyze a circuit in the s domain and be able to transform an s -domain solution to the time domain

- 13.7** In the circuit shown, switch 1 closes at $t = 0$ and the make-before-break switch moves instantaneously from position a to position b.
- Find $I_1(s)$.
 - Find $i_1(t)$.

Answer: a) $\frac{-50(s - 4.8)}{s(s + 20)(s + 60)}$;

b) $(0.2 - 1.55e^{-20t} + 1.35e^{-60t})u(t) \text{ A}.$

SELF-CHECK: Also try Chapter Problems 13.40 and 13.41.



Because we are analyzing linear lumped-parameter circuits, we can use superposition to divide the response into components that can be identified with particular sources and initial conditions. We need to identify these components in order to use the transfer function, which we introduce in the next section. Example 13.8 illustrates superposition in the s domain by revisiting the circuit from Example 13.3 and separating its output voltage, v , into three components, two associated with the circuit's initial conditions and one associated with the circuit's independent source.

EXAMPLE 13.8 Applying Superposition in the s Domain

Repeat the analysis of the s -domain circuit for Example 13.3, shown in Fig. 13.14. Use superposition to separate the time-domain voltage v into three components, each one being the response to one of the three sources in Fig. 13.14.

Solution

Since each source in Fig. 13.14 is a current source, we will use Ohm's law to find each component of V , using the equivalent impedance of the three parallel-connected impedances. This equivalent impedance, calculated in Example 13.3, is

$$Z_{\text{eq}} = \frac{40 \times 10^6 s}{s^2 + 80,000s + 16 \times 10^8} \Omega.$$

Define V' as the component of V in Fig. 13.14 due to the initial capacitor energy. Find V' using the circuit in Fig. 13.27, where the only source is the one that represents the initial condition for the capacitor.

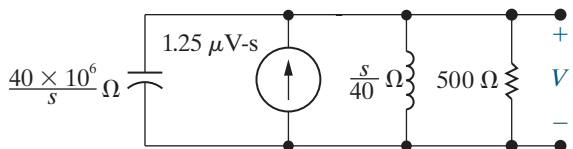


Figure 13.27 ▲ The response of the circuit in Fig. 13.14 due to the initial capacitor voltage.

$$V' = Z_{\text{eq}} (1.25 \times 10^{-6}) = \frac{50s}{s^2 + 80,000s + 16 \times 10^8}.$$

The partial fraction expansion of V' is

$$V' = \frac{-2 \times 10^6}{(s + 40,000)^2} + \frac{50}{(s + 40,000)}.$$

Use Table 12.3 to find the inverse Laplace transform of V' :

$$v' = \mathcal{L}^{-1}\{V'\} = (50e^{-40,000t} - 2 \times 10^6 te^{-40,000t})u(t) \text{ V}.$$

Therefore, v' is the component of v that is due only to the initial energy of the capacitor in the circuit of Fig. 13.13.

Next, define V'' as the component of V in Fig. 13.14 due to the initial inductor energy. Find V'' using the circuit in Fig. 13.28, where the only source is the one that represents the initial condition for the inductor.

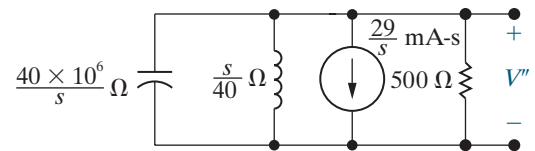


Figure 13.28 ▲ The response of the circuit in Fig. 13.14 due to the initial inductor current.

$$V'' = Z_{\text{eq}} \left(\frac{-0.029}{s} \right) = \frac{-1.16 \times 10^6}{s^2 + 80,000s + 16 \times 10^8}.$$

The partial fraction expansion of V'' is

$$V'' = \frac{-1.16 \times 10^6}{(s + 40,000)^2}.$$

Use Table 12.3 to find the inverse Laplace transform of V'' :

$$v'' = \mathcal{L}^{-1}\{V''\} = (-1.16 \times 10^6 te^{-40,000t})u(t) \text{ V}.$$

Therefore, v'' is the component of v that is due only to the initial energy of the inductor in the circuit of Fig. 13.13.

Finally, define V''' as the component of V in Fig. 13.14 due to the 24 mA dc current source. Find V''' using the circuit in Fig. 13.29, where the only source is the one that represents the 24 mA dc current source.

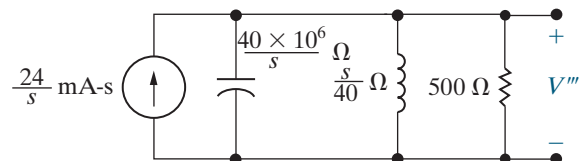


Figure 13.29 ▲ The response of the circuit in Fig. 13.14 due to the independent dc current source.

$$V''' = Z_{\text{eq}} \left(\frac{0.024}{s} \right) = \frac{0.96 \times 10^6}{s^2 + 80,000s + 16 \times 10^8}.$$

The partial fraction expansion of V''' is

$$V''' = \frac{0.96 \times 10^6}{(s + 40,000)^2}.$$

Use Table 12.3 to find the inverse Laplace transform of V''' :

$$v''' = \mathcal{L}^{-1}\{V'''\} = (0.96 \times 10^6 te^{-40,000t})u(t) \text{ V}.$$

Therefore, v''' is the component of v that is due only to the 24 mA current source in the circuit of Fig. 13.13.

The voltage v in the circuit of Fig. 13.13 is the sum of the three component voltages we just found:

$$\begin{aligned} v &= v' + v'' + v''' \\ &= (50e^{-40,000t} - 2.2 \times 10^6 te^{-40,000t})u(t) \text{ V}. \end{aligned}$$

This is the result found in Example 13.3.

ASSESSMENT PROBLEM

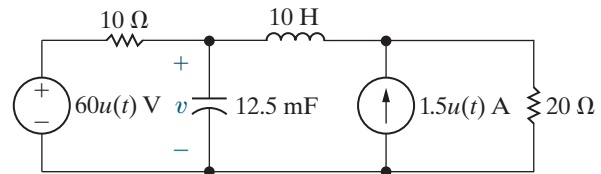
Objective 2—Know how to analyze a circuit in the s domain and be able to transform an s -domain solution to the time domain

13.8 The energy stored in the circuit shown is zero at the instant the two sources are turned on.

- Find the component of v for $t > 0$ owing to the voltage source.
- Find the component of v for $t > 0$ owing to the current source.
- Find the expression for v when $t > 0$.

Answer: a) $(40 + 120e^{-4t} - 160e^{-6t})u(t) \text{ V}$;
 b) $(10 - 30e^{-4t} + 20e^{-6t})u(t) \text{ V}$;
 c) $(50 + 90e^{-4t} - 140e^{-6t})u(t) \text{ V}.$

SELF-CHECK: Also try Chapter Problem 13.44.



13.4 The Transfer Function

The **transfer function** is defined as the s -domain ratio of the Laplace transform of the output (response) to the Laplace transform of the input (source). As we will see, the transfer function characterizes a circuit's behavior in a single s -domain expression, without revealing what components make up the circuit or how those components are interconnected. In computing the transfer function, we only consider circuits where all initial conditions are zero. If a circuit has multiple independent sources, we can find the transfer function for each source and use superposition to find the response to all sources.

The transfer function is

DEFINITION OF A TRANSFER FUNCTION

$$H(s) = \frac{Y(s)}{X(s)}, \quad (13.7)$$

where $Y(s)$ is the Laplace transform of the output signal, and $X(s)$ is the Laplace transform of the input signal. Note that the transfer function depends on what is defined as the output signal. Consider, for example, the series circuit shown in Fig. 13.30. If the current is defined as the output signal of the circuit,

$$H(s) = \frac{I}{V_g} = \frac{1}{R + sL + 1/sC} = \frac{sC}{s^2LC + RCs + 1}.$$

In deriving $H(s)$, we recognized that I corresponds to the output $Y(s)$ and V_g corresponds to the input $X(s)$.

If, instead, the capacitor voltage is defined as the output signal of the circuit shown in Fig. 13.30, then the transfer function is

$$H(s) = \frac{V}{V_g} = \frac{1/sC}{R + sL + 1/sC} = \frac{1}{s^2LC + RCs + 1}.$$

Thus, because circuits may have multiple sources and because the definition of the output signal of interest can vary, a single circuit can generate many transfer functions. Remember that when multiple sources are involved, no single transfer function can represent the total output; transfer functions associated with each source must be combined using superposition to yield the total response. Example 13.9 illustrates the computation of a transfer function for known numerical values of R , L , and C .

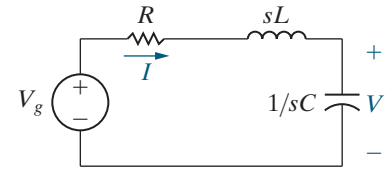


Figure 13.30 ▲ A series RLC circuit.

EXAMPLE 13.9

Deriving the Transfer Function of a Circuit

The voltage source v_g drives the circuit shown in Fig. 13.31. The output signal is the voltage across the capacitor, v_o .

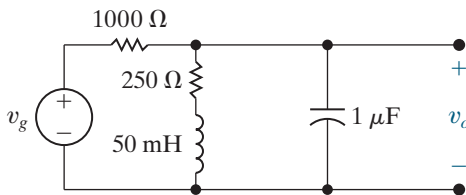


Figure 13.31 ▲ The circuit for Example 13.9.

- Find the transfer function for this circuit.
- Calculate the numerical values for the poles and zeros of the transfer function.

Solution

- Use Analysis Method 13.1 to construct and analyze the s -domain circuit.

Step 1: The circuit in Fig. 13.31 has no initial stored energy, an assumption we always make when calculating a circuit's transfer function. Therefore, we can skip this step.

Step 2: There are no independent sources described by a time-domain function in this circuit, so we can skip this step.

Step 3: Represent the source and output voltages using V_g and V_o , respectively, as shown in Fig. 13.32.

Step 4: The impedance of the resistors is their resistance. The impedances of the inductor and capacitor are

$$Z_L = sL = 0.05s \, \Omega;$$

$$Z_C = \frac{1}{sC} = \frac{1}{s(10^{-6})} = \frac{10^6}{s} \, \Omega.$$

The s -domain circuit resulting from Steps 1–4 is shown in Fig. 13.32.

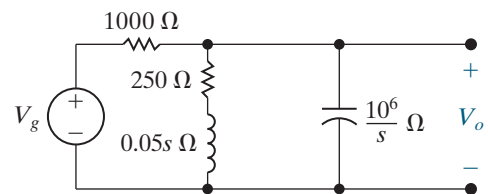


Figure 13.32 ▲ The s -domain equivalent circuit for the circuit shown in Fig. 13.31.

Step 5: From the problem statement, the transfer function is the ratio of V_o/V_g . Writing a node-voltage equation at the upper node and summing the currents leaving the node gives

$$\frac{V_o - V_g}{1000} + \frac{V_o}{250 + 0.05s} + \frac{V_o s}{10^6} = 0.$$

Solving for V_o yields

$$V_o = \frac{1000(s + 5000)V_g}{s^2 + 6000s + 25 \times 10^6}.$$

Hence, the transfer function is

$$H(s) = \frac{V_o}{V_g} = \frac{1000(s + 5000)}{s^2 + 6000s + 25 \times 10^6}.$$

b) The poles of $H(s)$ are the roots of the denominator polynomial. Therefore

$$-p_1 = -3000 - j4000,$$

$$-p_2 = -3000 + j4000.$$

The zeros of $H(s)$ are the roots of the numerator polynomial; thus, $H(s)$ has a zero at

$$-z_1 = -5000.$$

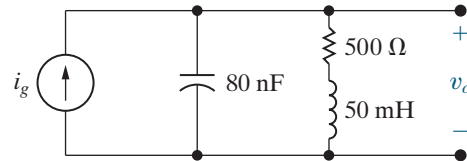
ASSESSMENT PROBLEM

Objective 3—Understand the definition and significance of the transfer function; be able to derive a transfer function

- 13.9** a) Derive the numerical expression for the transfer function V_o/I_g for the circuit shown.
b) Give the numerical value of each pole and zero of $H(s)$.

Answer: a) $H(s) = \frac{12.5 \times 10^6(s + 10,000)}{s^2 + 10,000s + 250 \times 10^6};$

b) $-p_1 = -5000 + j15,000,$
 $-p_2 = -5000 - j15,000, -z = -10,000.$



SELF-CHECK: Also try Chapter Problem 13.53.

The Location of Poles and Zeros of $H(s)$

For linear lumped-parameter circuits, $H(s)$ is always a rational function of s . Complex poles and zeros always occur in conjugate pairs. The poles of $H(s)$ must lie in the left half of the s plane if the response to a bounded source (one whose values lie within some finite bounds) is to be bounded. The zeros of $H(s)$ may lie in either the right half or the left half of the s plane.

With these general characteristics in mind, we next discuss the role that $H(s)$ plays in determining the circuit's output.

13.5 The Transfer Function in Partial Fraction Expansions

Using the definition of a circuit's transfer function (Eq. 13.7), we can find the circuit's output by multiplying the transfer function and the circuit's input:

$$Y(s) = H(s)X(s). \quad (13.8)$$

We have already noted that $H(s)$ is a rational function of s ; $X(s)$ also is a rational function of s for the time-domain functions of most interest in circuit analysis (see Table 12.1).

Expanding the right-hand side of Eq. 13.8 into a sum of partial fractions produces a term for each pole of $H(s)$ and $X(s)$. The terms generated by the poles of $H(s)$ correspond to the transient components of the total response. The terms generated by the poles of $X(s)$ correspond to the steady-state components of the response, which exist after the transient components have become negligible. Example 13.10 illustrates these general observations.

EXAMPLE 13.10 Analyzing the Transfer Function of a Circuit

The circuit in Example 13.9 (Fig. 13.31) is driven by a voltage source whose voltage increases linearly with time, namely, $v_g = 50tu(t)$ V.

- Use the transfer function to find v_o .
- Identify the transient component of the response.
- Identify the steady-state component of the response.
- Sketch v_o versus t for $0 \leq t \leq 1.5$ ms.

Solution

- From Example 13.9,

$$H(s) = \frac{1000(s + 5000)}{s^2 + 6000s + 25 \times 10^6}.$$

The Laplace transform of the source voltage is $50/s^2$; therefore, the s -domain expression for the output voltage is

$$V_o = \frac{1000(s + 5000)}{(s^2 + 6000s + 25 \times 10^6)} \frac{50}{s^2}.$$

The partial fraction expansion of V_o is

$$V_o = \frac{K_1}{s + 3000 - j4000} + \frac{K_1^*}{s + 3000 + j4000} + \frac{K_2}{s^2} + \frac{K_3}{s}.$$

We evaluate the coefficients K_1 , K_2 , and K_3 by using the techniques described in Section 12.7:

$$K_1 = 5\sqrt{5} \times 10^{-4} \angle 79.70^\circ;$$

$$K_1^* = 5\sqrt{5} \times 10^{-4} \angle -79.70^\circ;$$

$$K_2 = 10;$$

$$K_3 = -4 \times 10^{-4}.$$

Using Table 12.3, the time-domain expression for v_o is

$$v_o = [10\sqrt{5} \times 10^{-4} e^{-3000t} \cos(4000t + 79.70^\circ) + 10t - 4 \times 10^{-4}] u(t) \text{ V.}$$

- The transient component of v_o is

$$10\sqrt{5} \times 10^{-4} e^{-3000t} \cos(4000t + 79.70^\circ) \text{ V.}$$

Note that this term is generated by the poles $(-3000 + j4000)$ and $(-3000 - j4000)$ of the transfer function.

- The steady-state component of the response is

$$(10t - 4 \times 10^{-4}) u(t) \text{ V.}$$

These two terms are generated by the second-order pole (K/s^2) of the input voltage.

- Figure 13.33 shows a sketch of v_o versus t . Note that the deviation from the steady-state solution $10,000t - 0.4$ mV is imperceptible after approximately 1 ms.

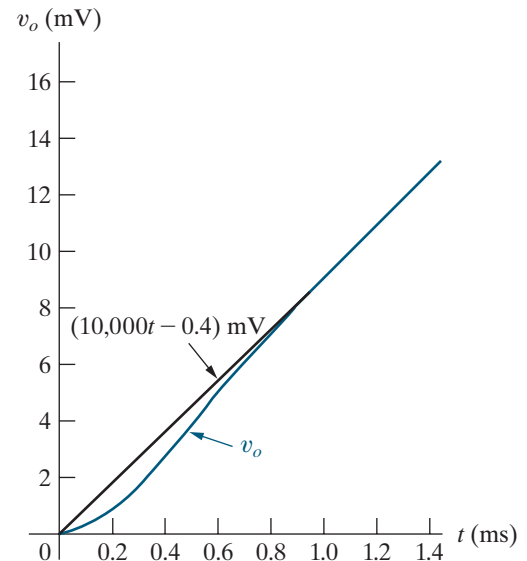


Figure 13.33 ▲ The graph of v_o versus t for Example 13.10.

Observations on the Use of $H(s)$ in Circuit Analysis

Example 13.10 related the components that make up a circuit's response to the poles of the transfer function, $H(s)$, and the poles of the circuit's input in the s domain, using a partial fraction expansion. However, the example raises questions about driving a circuit with an increasing ramp voltage that generates an increasing ramp response. Eventually, excessive voltage will cause the circuit components to fail, and then our linear model is invalid. But some practical applications have input ramp functions that increase to some maximum value over a finite time interval, so the response to a ramp input is important. If the time it takes for the ramp to reach its maximum value is long compared with the time constants of the circuit, the solution assuming an unbounded ramp is valid for this finite time interval.

Here are some additional observations about a circuit's transfer function, defined in Eq. 13.7.

- If the circuit's input is delayed by a seconds,

$$\mathcal{L}\{x(t-a)u(t-a)\} = e^{-as}X(s),$$

then from Eq. 13.8, the circuit's output is

$$Y(s) = H(s)X(s)e^{-as}.$$

If $y(t) = \mathcal{L}^{-1}\{H(s)X(s)\}$, then

$$y(t-a)u(t-a) = \mathcal{L}^{-1}\{H(s)X(s)e^{-as}\}.$$

Therefore, delaying the input by a seconds delays the output by a seconds. A circuit that exhibits this characteristic is **time invariant**.

- If the circuit's input is a unit impulse, the circuit's output equals the inverse transform of the transfer function. Thus, if

$$x(t) = \delta(t), \quad \text{then } X(s) = 1$$

and

$$Y(s) = H(s).$$

Hence,

$$y(t) = h(t),$$

so the inverse transform of the transfer function equals the unit impulse response of the circuit.

- A circuit's unit impulse response is also its natural response because applying an impulsive source is equivalent to instantaneously storing energy in the circuit (see Section 13.8). The subsequent release of this stored energy is the circuit's natural response (see Problem 13.87).
- A circuit's unit impulse response, $h(t)$, contains enough information to compute the response to any source that drives the circuit. We can extract a circuit's response to an arbitrary source from the circuit's unit impulse response, using the convolution integral. This technique is demonstrated in the next section.

ASSESSMENT PROBLEMS

Objective 4—Know how to use a circuit's transfer function to calculate the circuit's impulse response, unit step response, and steady-state response to sinusoidal input

13.10 Find (a) the unit step and (b) the unit impulse response of the circuit shown in Assessment Problem 13.9.

Answer:

- a) $[500 + 2500e^{-5000t} \cos(15,000t - 126.87^\circ)]u(t)$ V;
 b) $[13.2e^{-5000t} \cos(15,000t - 18.43^\circ)]u(t)$ MV.

13.11 The unit impulse response of a circuit is

$$v_o(t) = (\sqrt{1000}/3)e^{-t} \cos(3t + \theta) \text{ V},$$

where $\tan \theta = -\frac{1}{3}$.

- a) Find the transfer function of the circuit.
 b) Find the unit step response of the circuit.

Answer: a) $10(s + 2)/(s^2 + 2s + 10)$;
 b) $[2 + (10/3)e^{-t} \cos(3t - 126.87^\circ)]u(t)$ V.

SELF-CHECK: Also try Chapter Problems 13.56 and 13.57

13.6 The Transfer Function and the Convolution Integral

The **convolution integral** relates the output $y(t)$ of a linear time-invariant circuit to the input $x(t)$ of the circuit and the circuit's impulse response $h(t)$. The integral relationship can be expressed in two ways:

CONVOLUTION INTEGRAL

$$y(t) = \int_{-\infty}^{\infty} h(\lambda)x(t - \lambda) d\lambda = \int_{-\infty}^{\infty} h(t - \lambda)x(\lambda) d\lambda. \quad (13.9)$$

We are interested in the convolution integral for several reasons.

- We can find a circuit's output for any input by working entirely in the time domain. This is beneficial when $x(t)$ and $h(t)$ are known only through experimental data. In such cases, using Laplace transform methods may be awkward or even impossible, as we would need to compute the Laplace transform of experimental data.
- The convolution integral introduces the concepts of memory and the weighting function into analysis. The concept of memory enables us to predict, to some degree, how closely the output waveform replicates the input waveform, using the impulse response (or the weighting function) $h(t)$.
- The convolution integral provides a formal procedure for finding the inverse transform of products of Laplace transforms.

To derive Eq. 13.9, we assume that the circuit is linear and time invariant. Because the circuit is linear, the principle of superposition is valid, and because it is time invariant, the response delay and the input delay are the same. Consider Fig. 13.34, in which the block containing $h(t)$ represents any linear time-invariant circuit whose impulse response

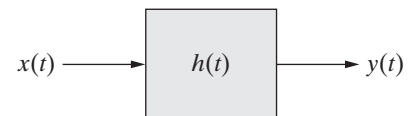


Figure 13.34 ▲ A block diagram of a general circuit.

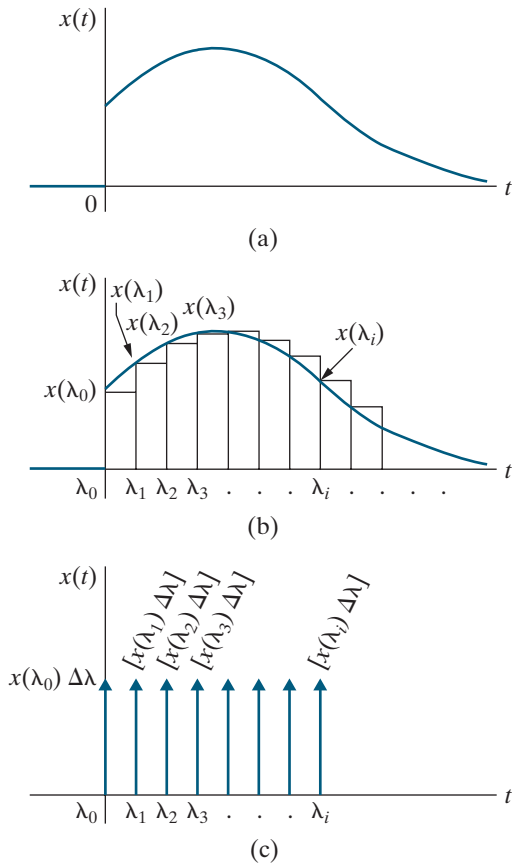


Figure 13.35 ▲ The excitation signal of $x(t)$. (a) A general excitation signal. (b) Approximating $x(t)$ with a series of pulses. (c) Approximating $x(t)$ with a series of impulses.

is known, $x(t)$ represents the input signal and $y(t)$ represents the output signal. We also assume the following:

- $x(t)$ is the general signal shown in Fig. 13.35(a).
- $x(t) = 0$ for $t < 0^-$. Once we derive the convolution integral assuming $x(t) = 0$ for $t < 0^-$, extending the integral to include excitation functions that exist for all time is straightforward.
- A discontinuity in $x(t)$ at the origin (between 0^- and 0^+) is permitted.

Begin by approximating $x(t)$ with a series of rectangular pulses of uniform width $\Delta\lambda$ as shown in Fig. 13.35(b). Thus

$$x(t) = x_0(t) + x_1(t) + \dots + x_i(t) + \dots,$$

where $x_i(t)$ is a rectangular pulse that equals $x(\lambda_i)$ between λ_i and λ_{i+1} and is zero elsewhere. Note that the i th pulse can be expressed using step functions; that is,

$$x_i(t) = x(\lambda_i)\{u(t - \lambda_i) - u[t - (\lambda_i + \Delta\lambda)]\}.$$

Continue to approximate $x(t)$ by making the pulse width $\Delta\lambda$ so small that we can approximate the i th component using an impulse function of strength $x(\lambda_i)\Delta\lambda$. Figure 13.35(c) shows this impulse representation, where the brackets beside each arrow represent the impulse strength. The impulse representation of $x(t)$ is

$$\begin{aligned} x(t) &= x(\lambda_0)\Delta\lambda\delta(t - \lambda_0) + x(\lambda_1)\Delta\lambda\delta(t - \lambda_1) + \dots \\ &\quad + x(\lambda_i)\Delta\lambda\delta(t - \lambda_i) + \dots \end{aligned}$$

When we represent $x(t)$ with a series of impulse functions (which occur at equally spaced intervals of time, that is, at $\lambda_0, \lambda_1, \lambda_2, \dots$), the output function $y(t)$ is the sum of uniformly delayed impulse responses. The strength of each response depends on the strength of the impulse driving the circuit. For example, let's assume that the unit impulse response of the circuit represented by the box in Fig. 13.34 is the exponential decay function shown in Fig. 13.36(a). Then the approximation of $y(t)$ is the sum of the impulse responses shown in Fig. 13.36(b).

Analytically, the expression for $y(t)$ is

$$\begin{aligned} y(t) &= x(\lambda_0)\Delta\lambda h(t - \lambda_0) + x(\lambda_1)\Delta\lambda h(t - \lambda_1) \\ &\quad + x(\lambda_2)\Delta\lambda h(t - \lambda_2) + \dots \\ &\quad + x(\lambda_i)\Delta\lambda h(t - \lambda_i) + \dots \end{aligned}$$

As $\Delta\lambda \rightarrow 0$, the sum approaches a continuous integral, or

$$\sum_{i=-\infty}^{\infty} x(\lambda_i)h(t - \lambda_i)\Delta\lambda \rightarrow \int_0^{\infty} x(\lambda)h(t - \lambda) d\lambda.$$

Therefore,

$$y(t) = \int_0^{\infty} x(\lambda)h(t - \lambda) d\lambda.$$

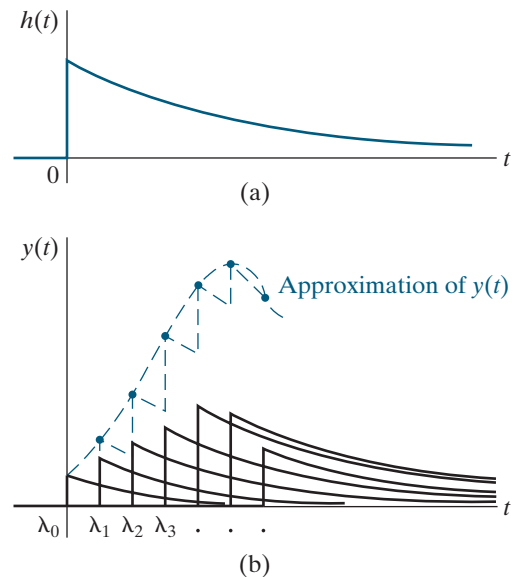


Figure 13.36 ▲ The approximation of $y(t)$. (a) The impulse response of the box shown in Fig. 13.34. (b) Summing the impulse responses.

If $x(t)$ exists over all time, then the lower limit on the integral is $-\infty$. Thus, in general,

$$y(t) = \int_{-\infty}^{\infty} x(\lambda)h(t - \lambda) d\lambda,$$

which is the second form of the convolution integral given in Eq. 13.9.

To derive the first form of the integral in Eq. 13.9, we change the variable of integration in the second form of the integral. Let $u = t - \lambda$, and then $du = -d\lambda$, $u = -\infty$ when $\lambda = \infty$, and $u = +\infty$ when $\lambda = -\infty$. Now we can write

$$y(t) = \int_{\infty}^{-\infty} x(t - u)h(u)(-du)$$

or

$$y(t) = \int_{-\infty}^{\infty} x(t - u)h(u)(du).$$

Since u is a symbol of integration, this integral is equivalent to the first form of the convolution integral, Eq. 13.9.

The convolution integral relating $y(t)$, $h(t)$, and $x(t)$ (Eq. 13.9), is often written using a shorthand notation:

$$y(t) = h(t) * x(t) = x(t) * h(t), \quad (13.10)$$

where the asterisk represents the integral relationship between $h(t)$ and $x(t)$. Thus, $h(t) * x(t)$ is read as “ $h(t)$ is convolved with $x(t)$ ” and implies that

$$h(t) * x(t) = \int_{-\infty}^{\infty} h(\lambda)x(t - \lambda) d\lambda,$$

whereas $x(t) * h(t)$ is read as “ $x(t)$ is convolved with $h(t)$ ” and implies that

$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\lambda)h(t - \lambda) d\lambda.$$

The integrals in Eq. 13.9 give the most general relationship for the convolution of two functions. However, when we apply the convolution integral, we can change the lower limit to zero and the upper limit to t . Then we can write Eq. 13.9 as

$$y(t) = \int_0^t h(\lambda)x(t - \lambda) d\lambda = \int_0^t x(\lambda)h(t - \lambda) d\lambda.$$

We change the limits for two reasons. First, for physically realizable circuits, $h(t)$ is zero for $t < 0$ because there is no impulse response before you apply an impulse. Second, we start measuring time at the instant we turn on the input $x(t)$; therefore $x(t) = 0$ for $t < 0^-$.

A graphic interpretation of the convolution integrals (Eq. 13.9) helps us use convolution as a computational tool. We begin with the first integral and assume that the circuit’s impulse response is the exponential decay function shown in Fig. 13.37(a) and its input function has the waveform shown in Fig. 13.37(b). In each of the plots, we replace t with λ , the symbol

of integration. Replacing λ with $-\lambda$ folds the excitation function over the vertical axis, as shown in Fig. 13.37(c). The folding operation explains why Eq. 13.9 is called the *convolution* integral. Replacing $-\lambda$ with $t - \lambda$ slides the folded function to the right, as shown in Fig. 13.37(d).

At any specified value of t , the output function $y(t)$ is the area under the product function $h(\lambda)x(t - \lambda)$, as shown in Fig. 13.37(e). This plot explains why we can set the lower limit on the convolution integral to zero and the upper limit to t . For $\lambda < 0$, the product $h(\lambda)x(t - \lambda)$ is zero because $h(\lambda)$ is zero. For $\lambda > t$, the product $h(\lambda)x(t - \lambda)$ is zero because $x(t - \lambda)$ is zero.

Figure 13.38 shows the second form of the convolution integral. Once again, the product function in Fig. 13.38(e) explains why we use zero for the lower limit and t for the upper limit.

Example 13.11 illustrates how to use the convolution integral and the unit impulse response to find a circuit's output in response to a given input.

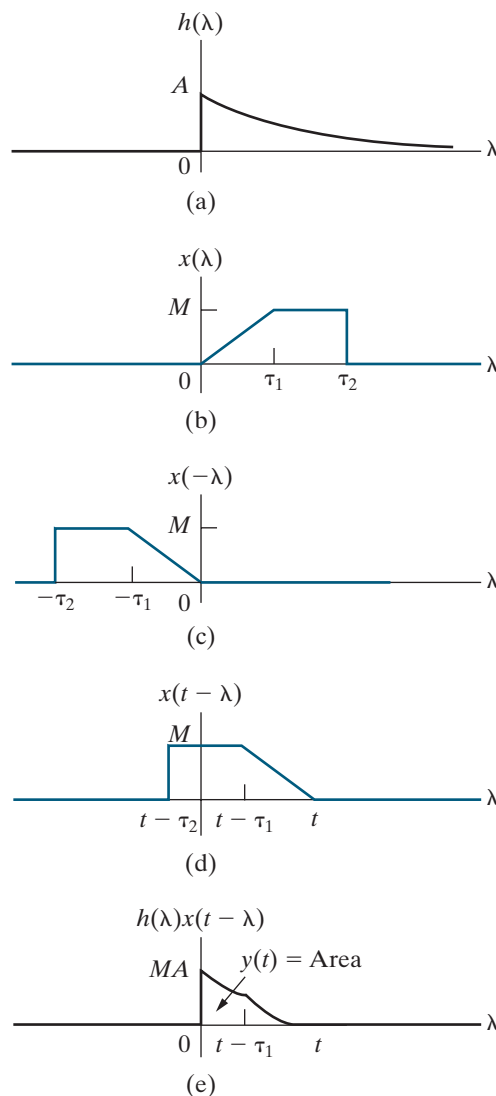


Figure 13.37 ▲ A graphic interpretation of the convolution integral $\int_0^t h(\lambda)x(t - \lambda) d\lambda$. (a) The impulse response. (b) The excitation function. (c) The folded excitation function. (d) The folded excitation function displaced t units. (e) The product $h(\lambda)x(t - \lambda)$.

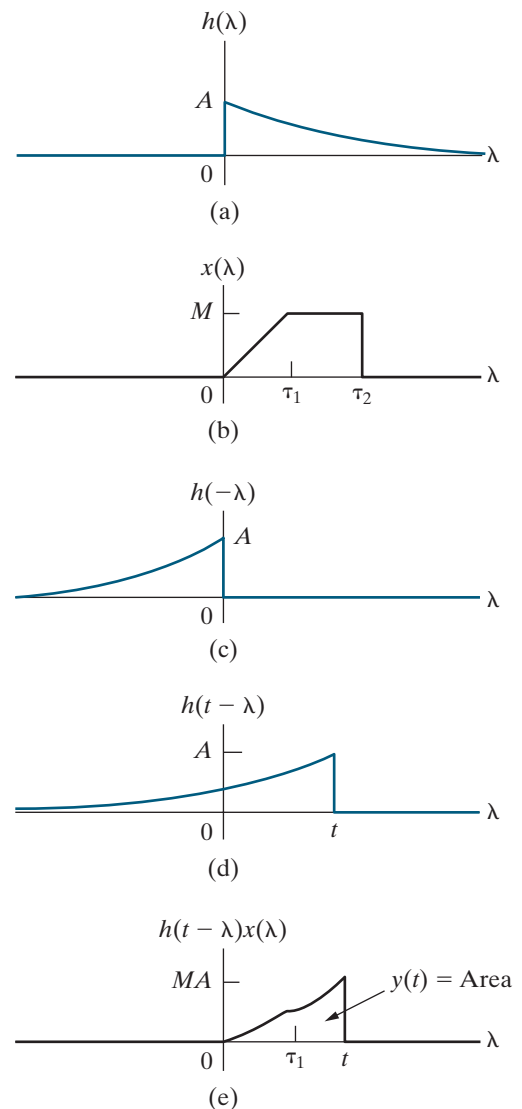


Figure 13.38 ▲ A graphic interpretation of the convolution integral $\int_0^t h(t - \lambda)x(\lambda) d\lambda$. (a) The impulse response. (b) The excitation function. (c) The folded impulse response. (d) The folded impulse response displaced t units. (e) The product $h(t - \lambda)x(\lambda)$.

EXAMPLE 13.11 Using the Convolution Integral to Find an Output Signal

The input voltage v_i for the circuit shown in Fig. 13.39(a) is shown in Fig. 13.39(b).

- Use the convolution integral to find v_o .
- Plot v_o over the range of $0 \leq t \leq 15$ s.

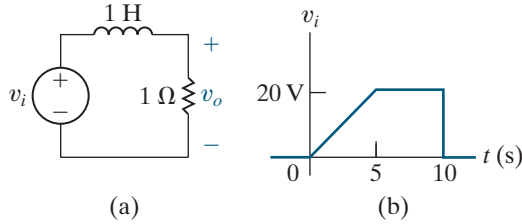


Figure 13.39 ▲ The circuit and excitation voltage for Example 13.11. (a) The circuit. (b) The input voltage.

Solution

- Begin by finding the unit impulse response of the circuit. The s -domain equivalent of the circuit in Fig. 13.39(a) replaces the inductor with an impedance of $s \Omega$ and the two voltages with V_o and V_i . Using voltage division,

$$V_o = \frac{1}{s+1} V_i.$$

When v_i is a unit impulse function $\delta(t)$, $V_i = 1$ and $H(s) = V_o/V_i = 1/(s+1)$. Then,

$$v_o = h(t) = e^{-t}u(t),$$

from which

$$h(\lambda) = e^{-\lambda}u(\lambda).$$

Using the first form of the convolution integral in Eq. 13.9, we construct the impulse response and folded input function shown in Fig. 13.40, which helps us select the convolution integral limits. As we slide the folded input function to the right, we break the integration into three intervals: $0 \leq t \leq 5$; $5 \leq t \leq 10$; and $10 \leq t \leq \infty$. The integration intervals correspond to points where the input function's definition changes. Figure 13.41 depicts the position of the folded input function for each interval.

The expression for v_i in the time interval $0 \leq t \leq 5$ is

$$v_i = 4t, \quad 0 \leq t \leq 5 \text{ s.}$$

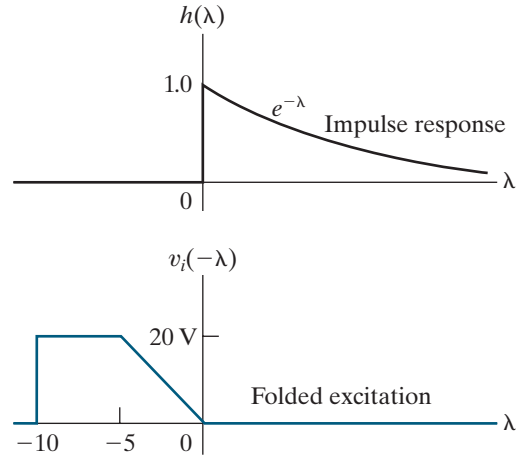


Figure 13.40 ▲ The impulse response and the folded input function for Example 13.11.

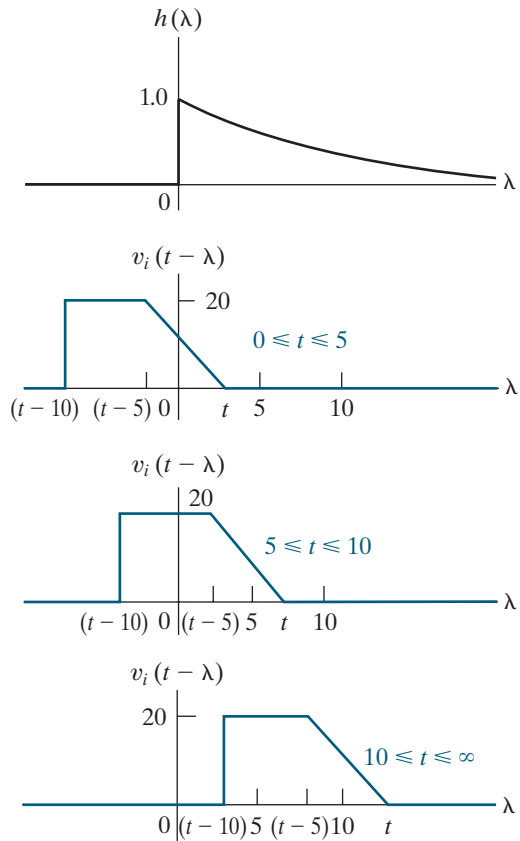


Figure 13.41 ▲ The displacement $v_i(t-\lambda)$ for three different time intervals.

Hence, the expression for the folded input function in the interval $t - 5 \leq \lambda \leq t$ is

$$v_i(t - \lambda) = 4(t - \lambda), \quad t - 5 \leq \lambda \leq t.$$

In the other two time intervals, v_i is a constant.

We can now set up the three convolution integrals to find v_o . For $0 \leq t \leq 5$ s :

$$\begin{aligned} v_o &= \int_0^t 4(t - \lambda)e^{-\lambda} d\lambda \\ &= 4(e^{-t} + t - 1) \text{ V.} \end{aligned}$$

For $5 \leq t \leq 10$ s,

$$\begin{aligned} v_o &= \int_0^{t-5} 20e^{-\lambda} d\lambda + \int_{t-5}^t 4(t - \lambda)e^{-\lambda} d\lambda \\ &= 4(5 + e^{-t} - e^{-(t-5)}) \text{ V.} \end{aligned}$$

For $10 \leq t \leq \infty$ s,

$$\begin{aligned} v_o &= \int_{t-10}^{t-5} 20e^{-\lambda} d\lambda + \int_{t-5}^t 4(t - \lambda)e^{-\lambda} d\lambda \\ &= 4(e^{-t} - e^{-(t-5)} + 5e^{-(t-10)}) \text{ V.} \end{aligned}$$

b) Values for v_o are shown graphically in Fig. 13.42.

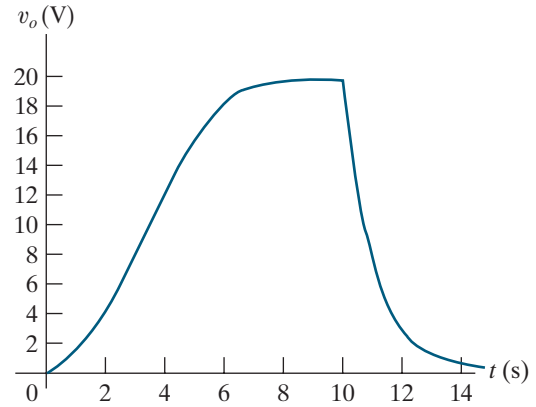


Figure 13.42 ▲ The voltage response versus time for Example 13.11.

SELF-CHECK: Assess your understanding of convolution by trying Chapter Problems 13.64 and 13.65.

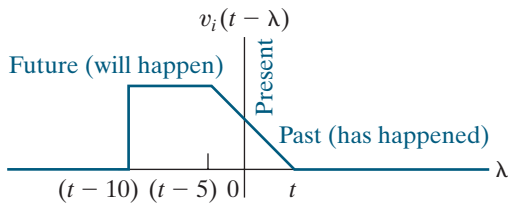


Figure 13.43 ▲ The past, present, and future values of the input function.

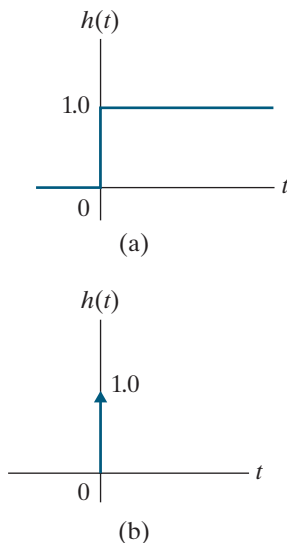


Figure 13.44 ▲ Weighting functions. (a) Perfect memory. (b) No memory.

The Concepts of Memory and the Weighting Function

We mentioned at the beginning of this section that the convolution integral introduces the concepts of memory and the weighting function into circuit analysis. The graphical interpretation of the convolution integral is the easiest way to understand these concepts. We can view the folding and sliding of a circuit's input function on a time scale divided into past, present, and future time intervals. The vertical axis represents the present value of $x(t)$, with past values to the right of the vertical axis and future values to the left. Figure 13.43 shows this description of $x(t)$, using the input function from Example 13.11.

When we combine the past, present, and future views of $x(t - \tau)$ with the impulse response of the circuit, we see that the impulse response weights $x(t)$ according to present and past values. For example, Fig. 13.41 shows that the impulse response in Example 13.11 gives less weight to past values of $x(t)$ than to the present value of $x(t)$. In other words, the circuit “remembers” less and less about past input values. Therefore, in Fig. 13.42, v_o quickly approaches zero when the present value of the input is zero (that is, when $t > 10$ s). Since the present value of the input receives more weight than the past values, the output quickly approaches the present value of the input.

Because the convolution integral uses the product of $x(t - \lambda)$ and $h(\lambda)$, the impulse response is considered the circuit's **weighting function**. The weighting function, in turn, determines how much memory the circuit has. **Memory** represents how accurately the circuit's response matches its input. For example, if the impulse response, or weighting function, is flat, as shown in Fig. 13.44(a), it gives equal weight to all values of $x(t)$, past and present. Such a circuit has a perfect memory. However, if the impulse response is an impulse function, as shown in Fig. 13.44(b), it gives no weight to past values of $x(t)$ and the circuit has no memory.

The more memory a circuit has, the more distortion exists between the circuit's input waveform and its output. We can demonstrate this by assuming that the circuit has no memory, that is, $h(t) = A\delta(t)$, and then noting from the convolution integral that

$$\begin{aligned} y(t) &= \int_0^t h(\lambda)x(t-\lambda) d\lambda \\ &= \int_0^t A\delta(\lambda)x(t-\lambda) d\lambda \\ &= Ax(t). \end{aligned}$$

The expression for $y(t)$ shows that, if the circuit has no memory, the output is a scaled replica of the input.

The circuit shown in Example 13.11 illustrates the distortion between input and output for a circuit that has some memory. This distortion is clear when we plot the input and output waveforms on the same graph, as in Fig. 13.45.

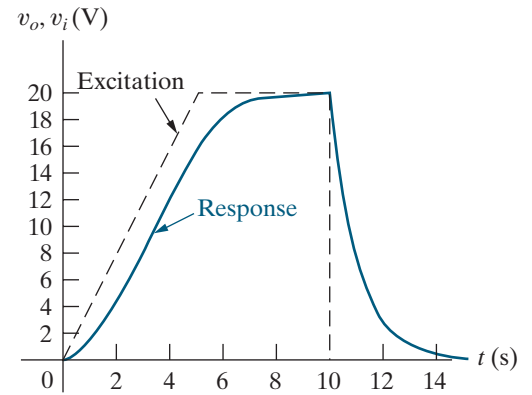


Figure 13.45 ▲ The input and output waveforms for Example 13.11.

13.7 The Transfer Function and the Steady-State Sinusoidal Response

Once we have computed a circuit's transfer function, we can use it to find the steady-state response to a sinusoidal input. To show this, we assume that

$$x(t) = A \cos(\omega t + \phi),$$

and then we use Eq. 13.8 to find the steady-state solution of $y(t)$. To find the Laplace transform of $x(t)$, we first write $x(t)$ as

$$x(t) = A \cos \omega t \cos \phi - A \sin \omega t \sin \phi,$$

from which

$$\begin{aligned} X(s) &= \frac{(A \cos \phi)s}{s^2 + \omega^2} - \frac{(A \sin \phi)\omega}{s^2 + \omega^2} \\ &= \frac{A(s \cos \phi - \omega \sin \phi)}{s^2 + \omega^2}. \end{aligned}$$

Substituting the expression for $X(s)$ into Eq. 13.8 gives the s -domain expression for the response:

$$Y(s) = H(s) \frac{A(s \cos \phi - \omega \sin \phi)}{s^2 + \omega^2}.$$

Think about the partial fraction expansion of $Y(s)$. The number of terms in the expansion depends on the number of poles of $H(s)$. Because $H(s)$ is not specified, we write the expansion of $Y(s)$ as

$$Y(s) = \frac{K_1}{s - j\omega} + \frac{K_1^*}{s + j\omega} + \sum \text{terms generated by the poles of } H(s).$$

The first two terms in the partial fraction expansion correspond to the complex conjugate poles of the sinusoidal input because $s^2 + \omega^2 = (s - j\omega)(s + j\omega)$. But the terms corresponding to the poles of $H(s)$ do not contribute to the steady-state response of $y(t)$, because all these poles lie in the left half of the s plane. Consequently, the corresponding time-domain terms approach zero as t increases. Thus, only the first two partial fraction terms determine the steady-state response, so we only have to calculate a single partial fraction coefficient, K_1 :

$$\begin{aligned} K_1 &= \left. \frac{H(s)A(s \cos \phi - \omega \sin \phi)}{s + j\omega} \right|_{s=j\omega} \\ &= \frac{H(j\omega)A(j\omega \cos \phi - \omega \sin \phi)}{2j\omega} \\ &= \frac{H(j\omega)A(\cos \phi + j \sin \phi)}{2} = \frac{1}{2} H(j\omega)Ae^{j\phi}. \end{aligned}$$

In general, $H(j\omega)$ is a complex number, which we can write in polar form as

$$H(j\omega) = |H(j\omega)|e^{j\theta(\omega)}.$$

We see that the transfer function's magnitude, $|H(j\omega)|$, and its phase angle, $\theta(\omega)$, vary with the frequency ω . Substituting the polar form for $H(j\omega)$ into the equation for K_1 and simplifying, we see that the expression for K_1 becomes

$$K_1 = \frac{A}{2}|H(j\omega)|e^{j[\theta(\omega)+\phi]}.$$

We find the steady-state component for $y(t)$ by inverse-transforming the first two terms in the partial fraction expansion of $Y(s)$, ignoring the terms generated by the poles of $H(s)$. Thus

SINUSOIDAL STEADY-STATE RESPONSE COMPUTED USING A TRANSFER FUNCTION

$$y_{ss}(t) = A|H(j\omega)|\cos[\omega t + \phi + \theta(\omega)], \quad (13.11)$$

which tells us how to find a circuit's steady-state response to a sinusoidal input using the circuit's transfer function:

- Determine the input sinusoid's magnitude, A , frequency, ω , and phase angle, ϕ .
- Evaluate the circuit's transfer function, $H(s)$, for $s = j\omega$.
- Transform $H(j\omega)$ into polar form, with a magnitude $|H(j\omega)|$ and a phase angle θ .
- Write the steady-state output as a cosine with the amplitude $A|H(j\omega)|$, a phase angle of $\phi + \theta$, and a frequency of ω .

Example 13.12 uses a circuit's transfer function to find its sinusoidal steady-state response.

EXAMPLE 13.12**Using the Transfer Function to Find the Steady-State Sinusoidal Response**

The circuit from Example 13.9 is shown in Fig. 13.46. The sinusoidal source voltage is

$$v_g = 120 \cos(5000t + 30^\circ) \text{ V.}$$

Find the steady-state expression for v_o .

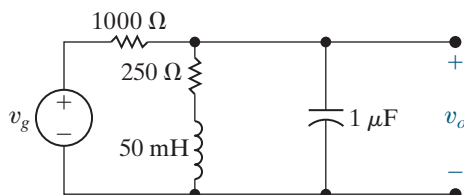


Figure 13.46 ▲ The circuit for Example 13.12.

Solution

From Example 13.9,

$$H(s) = \frac{1000(s + 5000)}{s^2 + 6000s + 25 \times 10^6}.$$

The frequency of the voltage source is 5000 rad/s, so we evaluate $H(j5000)$:

$$\begin{aligned} H(j5000) &= \frac{1000(5000 + j5000)}{-25 \times 10^6 + j5000(6000) + 25 \times 10^6} \\ &= \frac{1 + j1}{j6} = \frac{1 - j1}{6} = \frac{\sqrt{2}}{6} \angle -45^\circ. \end{aligned}$$

Then, from Eq. 13.11,

$$\begin{aligned} v_{oss} &= \frac{(120)\sqrt{2}}{6} \cos(5000t + 30^\circ - 45^\circ) \\ &= 20\sqrt{2} \cos(5000t - 15^\circ) \text{ V.} \end{aligned}$$

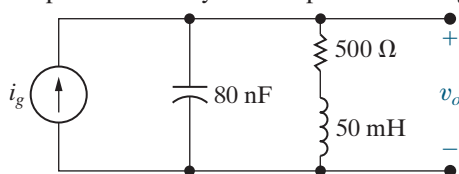
The relationship between $H(s)$ and $H(j\omega)$ provides a link between the time domain and the frequency domain. Using Eq. 13.11, we can find the sinusoidal steady-state response of a circuit by evaluating $H(j\omega)$. Theoretically, we can reverse the process; instead of using $H(s)$ to find $H(j\omega)$, we can use $H(j\omega)$ to find $H(s)$. To do so, we determine $H(j\omega)$ experimentally and then construct $H(s)$ from the data. Once we know $H(s)$, we can find the circuit's response to other inputs. While this experimental approach is not always possible, in some cases it does provide a way to find $H(s)$ for a circuit whose components and their values are unknown.

The transfer function is also used to find a circuit's frequency response, a concept we introduce in the next chapter.

ASSESSMENT PROBLEMS

Objective 4—Know how to use a circuit's transfer function to calculate the circuit's impulse response, unit step response, and steady-state response to sinusoidal input

13.12 The current source in the circuit from Assessment Problem 13.9, shown here, is delivering $10 \cos 20,000t$ mA. Use the transfer function to compute the steady-state expression for v_o .

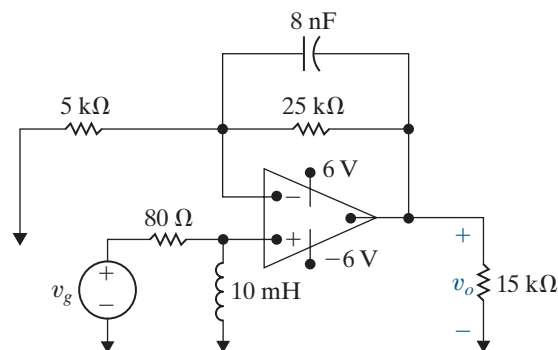


Answer: $11.18 \cos(20,000t - 63.43^\circ) \text{ V.}$

13.13 The op amp in the circuit shown is ideal.

- Find the transfer function V_o/V_g .
- Find v_o if $v_g = 0.6u(t)$ V.
- Find the steady-state expression for v_o if $v_g = 2 \cos 10,000t$ V.

SELF-CHECK: Also try Chapter Problems 13.80 and 13.81.



- Answer:**
- $\frac{s(s + 30,000)}{(s + 5000)(s + 8000)}$;
 - $(5e^{-5000t} - 4.4e^{-8000t})u(t)$ V;
 - $4.42 \cos(10,000t - 6.34^\circ) \text{ V.}$

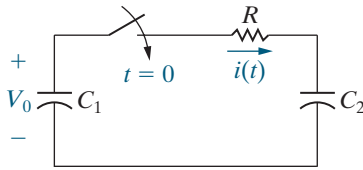


Figure 13.47 ▲ A circuit showing the creation of an impulsive current.

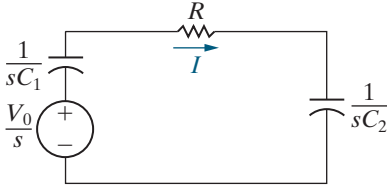


Figure 13.48 ▲ The s-domain equivalent circuit for the circuit shown in Fig. 13.47.

13.8 The Impulse Function in Circuit Analysis

Circuit analysis uses impulse functions to represent a switching operation or because a circuit's input is an impulsive source. We can use Laplace transform methods to determine the impulsive currents and voltages created during switching, and to find a circuit's response to an impulsive source. To begin, we create an impulse function with a switching operation.

Switching Operations

We use two different circuits to create an impulse function with a switching operation: a capacitor circuit, and a series inductor circuit.

Capacitor Circuit

In the circuit shown in Fig. 13.47, the capacitor C_1 is charged to an initial voltage of V_0 at the time the switch is closed. The initial charge on C_2 is zero. We want to find the expression for $i(t)$ as $R \rightarrow 0$. Figure 13.48 shows the s-domain equivalent circuit, and we use Ohm's law to find the s-domain current I :

$$I = \frac{V_0/s}{R + (1/sC_1) + (1/sC_2)}$$

$$= \frac{V_0/R}{s + (1/RC_e)},$$

where the equivalent capacitance $C_1C_2/(C_1 + C_2)$ is replaced by C_e .

We inverse-transform the expression for I using Table 12.3 to get

$$i = \left(\frac{V_0}{R} e^{-t/RC_e} \right) u(t),$$

which indicates that as R decreases, the initial current (V_0/R) increases and the time constant (RC_e) decreases. Thus, as R gets smaller, the current starts from a larger initial value and then drops off more rapidly. Figure 13.49 shows these characteristics of i .

As R approaches zero, the initial value of i approaches infinity and the duration of i approaches zero. If the area under the current function is independent of R , i approaches an impulse function. Physically, the total area under the i versus t curve represents the total charge transferred to C_2 after the switch is closed. Thus

$$\text{Area} = q = \int_0^\infty \frac{V_0}{R} e^{-t/RC_e} dt = V_0C_e,$$

so the total charge transferred to C_2 equals V_0C_e coulombs and is independent of R . Thus, as R approaches zero, the current approaches an impulse with strength V_0C_e ; that is,

$$i \rightarrow V_0C_e\delta(t).$$

The physical interpretation of this expression says that when $R = 0$, a finite amount of charge is transferred to C_2 instantaneously. Set $R = 0$ in the circuit of Fig. 13.47 to see why we get an instantaneous transfer of charge. With $R = 0$, closing the switch creates a contradiction: we apply a

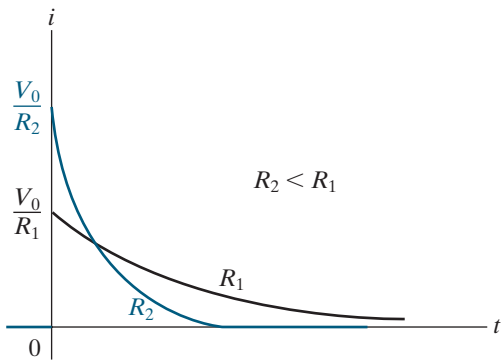


Figure 13.49 ▲ The plot of $i(t)$ versus t for two different values of R .

voltage across a capacitor whose initial voltage is zero. The capacitor voltage can change instantaneously only if there is an instantaneous transfer of charge. When the switch is closed, the voltage across C_2 does not jump to V_0 but to its final value of

$$v_2 = \frac{C_1 V_0}{C_1 + C_2}.$$

We leave the derivation of v_2 to you (see Problem 13.85).

If we set $R = 0$ at the outset, Laplace transform analysis predicts the impulsive current response. Thus,

$$I = \frac{V_0/s}{(1/sC_1) + (1/sC_2)} = \frac{C_1 C_2 V_0}{C_1 + C_2} = C_e V_0.$$

In writing this equation for I , we use the capacitor voltages at $t = 0^-$. The inverse transform of I , which is constant, is the constant times the impulse function; therefore,

$$i = C_e V_0 \delta(t).$$

Example 13.13 uses Laplace methods to analyze an inductor circuit whose output contains an impulse.

EXAMPLE 13.13 A Series Inductor Circuit with an Impulsive Response

The switch in the circuit shown in Fig. 13.50 has been closed for a long time. At $t = 0$ it opens. Use Laplace methods to find the output voltage, v_o , and the current in the 3 H inductor, i_1 .

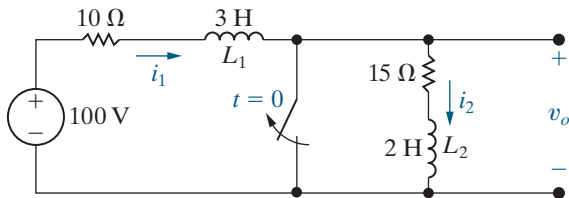


Figure 13.50 ▲ The circuit for Example 13.13.

Solution

Use Analysis Method 13.1 to construct and analyze the s -domain circuit.

Step 1: For $t < 0$, the switch is closed, and the inductors behave like short circuits. The current in the 2 H inductor is zero, since there is no source in that part of the circuit. The current in the 3 H inductor is $100/10 = 10$ A.

Step 2: The combination of the dc voltage source and the switch is defined by the time-domain function $100u(t)$. The Laplace transform of this function is $100/s$ V-s, which labels the voltage source in the s -domain circuit in Fig. 13.51.

Step 3: Represent the output voltage and current using V_o and I_1 , respectively, as shown in Fig. 13.51. Note that in the s -domain circuit, the current in both inductors is the same.

Step 4: The impedance of the resistors is their resistance. The impedance of the inductors is sL . We also need to represent the nonzero initial current in the 3 H inductor using either a series-connected voltage source or a parallel-connected current source. Here, we choose the voltage source, so the resulting s -domain circuit has a single mesh. The voltage source has the value $LI_0 = 3(10) = 30$ V-s. The s -domain circuit resulting from Steps 1–4 is shown in Fig. 13.51.

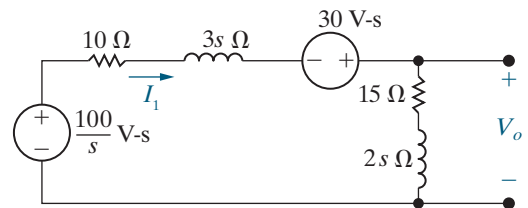


Figure 13.51 ▲ The s -domain equivalent circuit for the circuit shown in Fig. 13.50.

Step 5: From the circuit in Fig. 13.51, we find V_o using voltage division:

$$V_o = \frac{2s + 15}{5s + 25} \left(\frac{100}{s} + 30 \right) = \frac{12s^2 + 130s + 300}{s^2 + 5s}.$$

Now use Ohm's law to find I_1 :

$$I_1 = \frac{(100/s) + 30}{5s + 25} = \frac{6s + 20}{s(s + 5)}.$$

Step 6: We will discuss initial and final values in this circuit after we find the output voltage and current, so for now we skip this step.

Step 7: The expression for V_o is an improper rational function, so before we find its partial fraction expansion we must divide numerator by denominator to get

$$V_o = \frac{12s^2 + 130s + 300}{s^2 + 5s} = 12 + \frac{70s + 300}{s(s + 5)}.$$

The second term is a proper rational function of s , so we can find the partial fraction expansion:

$$V_o = 12 + \frac{60}{s} + \frac{10}{(s + 5)}.$$

Using Tables 12.1–12.3, we can inverse-Laplace-transform V_o to get

$$v_o = 12\delta(t) + (60 + 10e^{-5t})u(t) \text{ V}.$$

The partial fraction expansion for I_1 is

$$I_1 = \frac{6s + 20}{s(s + 5)} = \frac{4}{s} + \frac{2}{(s + 5)}.$$

Using Table 12.3, we can inverse-Laplace-transform I_1 to get

$$i_1 = (4 + 2e^{-5t})u(t) \text{ A}.$$

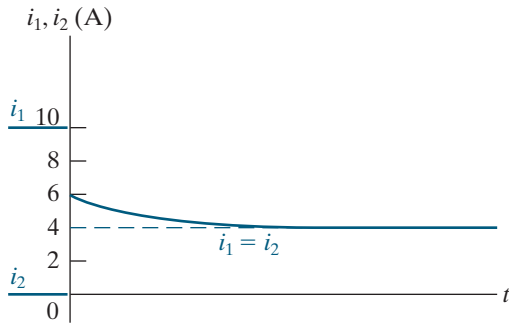


Figure 13.52 ▲ The inductor currents versus t for the circuit shown in Fig. 13.50.

Do the solutions for the output voltage and current in the circuit of Fig. 13.50 make sense? Before the switch is opened, the current in the 3 H inductor is 10 A, and the current in the 2 H inductor is 0 A. From the equation for i_1 derived in Example 13.13, we know that at $t = 0^+$, the current in both inductors is 6 A. This means the current in the 3 H inductor changes instantaneously from 10 to 6 A, while the current in the 2 H inductor changes instantaneously from 0 to 6 A. Then, the current decreases exponentially from 6 A to a final value of 4 A. This final value is easily verified from the circuit; that is, it should equal $100/25$, or 4 A. Figure 13.52 shows these characteristics of i_1 and i_2 .

Do these instantaneous jumps in the inductor current make sense in terms of the physical behavior of the circuit? First, note that when the switch opens in Fig. 13.50, the two inductors are in series. Any impulsive voltage appearing across the 3 H inductor must be exactly balanced by an impulsive voltage across the 2 H inductor because the sum of the impulsive voltages around a closed path must equal zero. Faraday's law states that the induced voltage is proportional to the change in flux linkage ($v = d\lambda/dt$). Therefore, the change in flux linkage must sum to zero. In other words, the total flux linkage immediately after switching is the same as that before switching. For the circuit here, the flux linkage before switching is

$$\lambda = L_1 i_1 + L_2 i_2 = 3(10) + 2(0) = 30 \text{ Wb-turns}.$$

Immediately after switching, it is

$$\lambda = (L_1 + L_2)i(0^+) = 5i(0^+).$$

Therefore,

$$i_1(0^+) = 30/5 = 6 \text{ A}.$$

Thus, the solution for i_1 in Example 13.13 agrees with the principle of the conservation of flux linkage.

We now test the validity of v_o from Example 13.13. First, we check the impulsive term $12\delta(t)$. The instantaneous jump of i_2 from 0 to 6 A at $t = 0$ means the derivative of i_2 contains the impulse $6\delta(t)$. This current impulse results in a $12\delta(t)$ impulse in the voltage across the 2 H inductor. For $t > 0$, $di_2/dt = di_1/dt = -10e^{-5t}$ A/s. The voltage v_o is the sum of the voltage across the 15 Ω resistor and the voltage across the 2 H inductor:

$$\begin{aligned} v_o &= 15(4 + 2e^{-5t}) + 2(-10e^{-5t}) \\ &= (60 + 10e^{-5t})u(t) \text{ V.} \end{aligned}$$

This expression agrees with the last two terms of v_o from Example 13.13. Thus, the expression for v_o does make sense in terms of known circuit behavior.

We can also check the instantaneous drop from 10 to 6 A in the current i_1 . This drop means the derivative of i_1 contains the impulse $-4\delta(t)$. Therefore, the voltage across L_1 includes the impulse $-12\delta(t)$. This impulse exactly balances the impulse included in the voltage across L_2 ; that is, the sum of the impulsive voltages around a closed path equals zero.

Impulsive Sources

Impulse functions can appear in circuit sources as well as circuit outputs. These sources are called **impulsive sources**. An impulsive source creates a finite amount of energy in the circuit instantaneously. A mechanical analogy is striking a bell with an impulsive clapper blow. After the energy has been transferred to the bell, the natural response of the bell determines the tone emitted (that is, the frequency of the resulting sound waves) and the tone's duration.

In the circuit shown in Fig. 13.53, an impulsive voltage source with a strength of V_0 volt-seconds is applied to a series connection of a resistor and an inductor. When the voltage source is applied, the initial energy in the inductor is zero, so the initial current is zero. There is no voltage drop across R , so the impulsive voltage source appears directly across L . The impulsive voltage across the inductor creates an instantaneous current. The current is

$$i = \frac{1}{L} \int_{0^-}^t V_0 \delta(x) dx.$$

The integral of $\delta(t)$ over any interval that includes zero is 1, so

$$i(0^+) = \frac{V_0}{L} \text{ A.}$$

Thus, in an infinitesimal moment, the impulsive voltage source has stored energy in the inductor, given by

$$w = \frac{1}{2} L \left(\frac{V_0}{L} \right)^2 = \frac{V_0^2}{2L} \text{ J.}$$

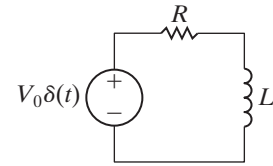


Figure 13.53 ▲ An RL circuit excited by an impulsive voltage source.

The initial current V_0/L now decays to zero due to the natural response of the circuit; that is,

$$i = \frac{V_0}{L} e^{-t/\tau} u(t),$$

where $\tau = L/R$. Remember from Chapter 7 that the natural response occurs as passive elements release or store energy. When a circuit is driven by an impulsive source, the total response is completely defined by the natural response; the duration of the impulsive source is so infinitesimal that it does not create a forced response.

We can also derive the inductor current using the Laplace transform method. Figure 13.54 shows the s -domain equivalent of the circuit in Fig. 13.53. From this circuit,

$$I = \frac{V_0}{R + sL} = \frac{V_0/L}{s + (R/L)}.$$

The inverse Laplace transform is

$$i = \frac{V_0}{L} e^{-(R/L)t} = \frac{V_0}{L} e^{-t/\tau} u(t).$$

Thus, the Laplace transform method gives the correct solution for $i \geq 0^+$.

Finally, we consider the case in which internally generated impulses and externally applied impulses occur simultaneously. As we see in Example 13.14, the Laplace transform method automatically ensures the correct solution for $t > 0^+$ if inductor currents and capacitor voltages at $t = 0^-$ are used in constructing the s -domain equivalent circuit and if externally applied impulses are represented by their transforms.

EXAMPLE 13.14

A Circuit with Both Internally Generated and Externally Applied Impulses

The switch in the circuit shown in Fig. 13.55 has been closed for a long time. At $t = 0$ it opens. Use Laplace methods to find the output voltage, v_o , and the current in the 3 H inductor, i_1 .

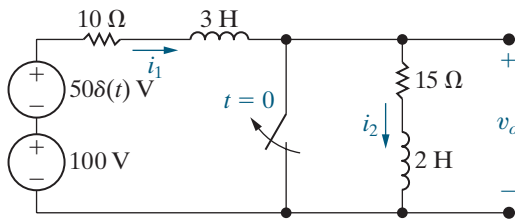


Figure 13.55 ▲ The circuit shown in Fig. 13.50 with an impulsive voltage source added in series with the 100 V source.

Solution

Note that the circuit in Fig. 13.55 was created from the circuit for Example 13.13 (Fig. 13.50) by adding an impulsive voltage source of $50\delta(t)$ in series with the 100 V source. In Example 13.13, we found the s -domain circuit that corresponds to the time-domain circuit in Fig. 13.50. The s -domain circuit

is shown in Fig. 13.51. To find the s -domain circuit corresponding to the circuit in Fig. 13.55, we add another voltage source in series, whose value is $\mathcal{L}\{50\delta(t)\} = 50 \text{ V}\cdot\text{s}$. The s -domain equivalent circuit is shown in Fig. 13.56.

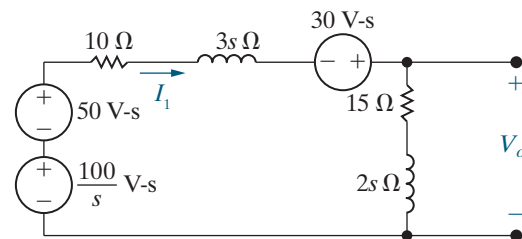


Figure 13.56 ▲ The s -domain equivalent circuit for the circuit shown in Fig. 13.55.

From this circuit, the expression for I_1 is

$$I_1 = \frac{(100/s) + 50 + 30}{5s + 25} = \frac{16s + 20}{s(s + 5)}.$$

The partial fraction expansion of I_1 is

$$I_1 = \frac{4}{s} + \frac{12}{s + 5},$$

and the inverse Laplace transform of I_1 is

$$i_1 = (4 + 12e^{-5t})u(t) \text{ A.}$$

The expression for V_o is

$$V_o = (2s + 15)I_1 = \frac{32s^2 + 280s + 300}{s(s + 5)},$$

which is an improper rational function. Before we can find the partial fraction expansion, we

divide the numerator of V_o by its denominator to get

$$V_o = 32 + \frac{120s + 300}{s(s + 5)}.$$

The partial fraction expansion of V_o is then

$$V_o = 32 + \frac{60}{s} + \frac{60}{s + 5}$$

and the inverse Laplace transform of V_o is

$$v_o = 32\delta(t) + (60 + 60e^{-5t})u(t) \text{ V.}$$

As before, we test the results of Example 13.14 to see whether they make sense. From the expression for i_1 , we see that the current in L_1 and L_2 is 16 A at $t = 0^+$. In Example 13.13, at the instant the switch opens, i_1 decreases from 10 to 6 A and i_2 increases from 0 to 6 A. Superimposed on these changes is a 10 A current in L_1 and L_2 , due to the impulsive voltage source; that is,

$$i_1 = \frac{1}{3 + 2} \int_{0^-}^t 50\delta(x) dx = 10 \text{ A.}$$

Therefore, i_1 increases suddenly from 10 to 16 A, while i_2 increases suddenly from 0 to 16 A. The final value of both currents is 4 A. Figure 13.57 shows i_1 and i_2 graphically.

We can also find the abrupt changes in i_1 and i_2 without using superposition. The sum of the impulsive voltages across the 3 H and 2 H inductors equals $50\delta(t)$. Thus, the change in flux linkage must sum to 50; that is,

$$\Delta\lambda_1 + \Delta\lambda_2 = 50.$$

Because $\lambda = Li$, we get

$$3\Delta i_1 + 2\Delta i_2 = 50.$$

But because i_1 and i_2 must be equal after the switch opens,

$$i_1(0^-) + \Delta i_1 = i_2(0^-) + \Delta i_2.$$

Thus,

$$10 + \Delta i_1 = 0 + \Delta i_2.$$

Solving for Δi_1 and Δi_2 yields

$$\begin{aligned}\Delta i_1 &= 6 \text{ A,} \\ \Delta i_2 &= 16 \text{ A.}\end{aligned}$$

These expressions agree with the previous check.

Figure 13.57 also indicates that the derivatives of i_1 and i_2 will contain an impulse at $t = 0$. Specifically, the derivative of i_1 will have an impulse of $6\delta(t)$, and the derivative of i_2 will have an impulse of $16\delta(t)$. Figure 13.58 illustrates the derivatives of i_1 and i_2 .

Now let's turn to the expression for the output voltage, v_o , found in Example 13.14. The impulsive component $32\delta(t)$ agrees with the impulse

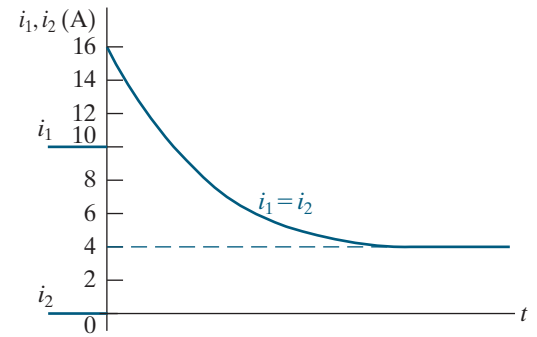


Figure 13.57 ▲ The inductor currents versus t for the circuit shown in Fig. 13.55.

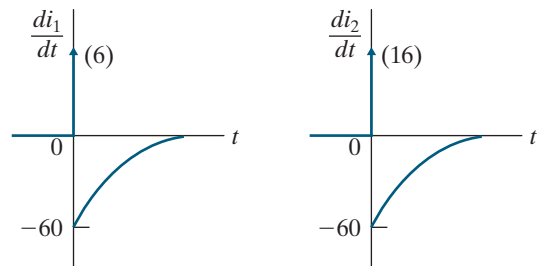


Figure 13.58 ▲ The derivatives of i_1 and i_2 .

$16\delta(t)$ that characterizes di_2/dt at the origin. The term $(60e^{-5t} + 60)$ agrees with the fact that for $t > 0^+$,

$$v_o = 15i_2 + 2\frac{di_2}{dt} = 15i_1 + 2\frac{di_1}{dt}.$$

We test the impulsive component of di_1/dt by noting that it produces an impulsive voltage of $(3)6\delta(t)$, or $18\delta(t)$, across L_1 . This voltage, added to $32\delta(t)$ across L_2 , gives us $50\delta(t)$. Thus, the algebraic sum of the impulsive voltages around the mesh is zero.

To summarize, the Laplace transform will correctly predict the impulsive currents and voltages created by switching. However, the s -domain equivalent circuits must be based on initial conditions at $t = 0^-$, that is, on the initial conditions that exist prior to the switching. The Laplace transform will correctly predict the response to impulsive input sources by representing these sources in the s domain by their Laplace transforms.

SELF-CHECK: Assess your understanding of the impulse function in circuit analysis by trying Chapter Problems 13.89 and 13.93.

■ Practical Perspective

Surge Suppressors

As mentioned at the beginning of this chapter, voltage surges can occur in a circuit that is operating in the sinusoidal steady state. We use Laplace transform methods to see how a voltage surge is created between the line and neutral conductors of a household circuit when a load is switched off during sinusoidal steady-state operation.

Consider the phasor-transformed circuit shown in Fig. 13.59, which models a household circuit with three loads, one of which is switched off at $t = 0$. To simplify the analysis, we assume that the line-to-neutral voltage, $\mathbf{V}_g$, is $120\angle 0^\circ$ V(rms), a standard household voltage, and that when the load is switched off at $t = 0$, the value of $\mathbf{V}_g$ does not change. After the switch is opened, we can construct the s -domain circuit, as shown in Fig. 13.60. Note that because the phase angle of the voltage across the inductive load is 0° , the initial current through the inductive load is 0. Therefore, only the line inductance has a nonzero initial condition, which is modeled in the s -domain circuit as a voltage source with the value $L_\ell I_0$, as seen in Fig. 13.60.

Just before the switch is opened at $t = 0$, each of the loads has a steady-state sinusoidal voltage with a peak magnitude of $120\sqrt{2} = 169.7$ V. All of the current in the line from the voltage source divides among the three loads. When the switch is opened at $t = 0$, all of the current in the line appears in the remaining resistive load because the inductive load current is 0 at $t = 0$ and the inductor current cannot change instantaneously. Therefore, the voltage drop across the remaining loads can experience a surge as the line current is directed through the resistive load.

For example, if the initial current in the line is 25 A(rms) and the impedance of the resistive load is $12\ \Omega$, the voltage drop across the resistor surges from 169.7 V to $(25)(\sqrt{2})(12) = 424.3$ V when the switch is opened. If the resistive load cannot handle this amount of voltage, it needs to be protected with a surge suppressor such as the one shown at the beginning of the chapter.

SELF-CHECK: Assess your understanding of this Practical Perspective by trying Chapter Problems 13.94 and 13.95.

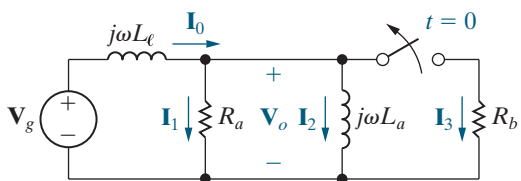


Figure 13.59 ▲ Phasor-transformed circuit used to introduce a switching surge voltage.

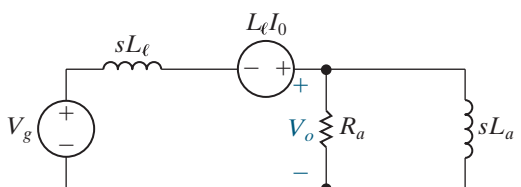


Figure 13.60 ▲ The circuit in Fig. 13.59, transformed to the s domain.

Summary

- We can represent each of the circuit elements as an s -domain equivalent circuit by Laplace-transforming the voltage-current equation for each element:

- Resistor: $V = RI$
- Inductor: $V = sLI - LI_0$
- Capacitor: $V = (1/sC)I + V_0/s$

In these equations, $V = \mathcal{L}\{v\}$, $I = \mathcal{L}\{i\}$, I_0 is the initial current through the inductor, and V_0 is the initial voltage across the capacitor. (See pages 484–485.)

- Using Steps 1–4 in Analysis Method 13.1, we can transform the time-domain circuit into the s domain. Table 13.1 summarizes the equivalent circuits for resistors, inductors, and capacitors in the s domain. (See page 486.)
- Solve the s -domain equivalent circuit in Analysis Method 13.1, Step 5, by writing algebraic equations using the circuit analysis techniques from resistive circuits. Step 6 checks the resulting s -domain voltages and currents using the initial- and final-value theorems, where possible. Step 7 finds the partial fraction expansion for the s -domain voltages and currents, and uses Table 12.3 to find the inverse Laplace transforms. (See page 487.)
- Circuit analysis in the s domain is particularly advantageous for solving transient response problems in linear lumped parameter circuits when initial conditions are known. It is also useful for problems involving multiple simultaneous mesh-current or node-voltage equations because it reduces problems to algebraic rather than differential equations. (See page 489.)
- The **transfer function** is the s -domain ratio of a circuit's output to its input. It is defined as

$$H(s) = \frac{Y(s)}{X(s)},$$

where $Y(s)$ is the Laplace transform of the output signal, and $X(s)$ is the Laplace transform of the input signal. (See page 500.)

- The partial fraction expansion of the product $H(s)X(s)$ yields a term for each pole of $H(s)$ and $X(s)$. The $H(s)$

terms correspond to the transient component of the total response, while the $X(s)$ terms correspond to the steady-state component. (See page 503.)

- If a circuit is driven by a unit impulse, $x(t) = \delta(t)$, then the response of the circuit equals the inverse Laplace transform of the transfer function, $y(t) = \mathcal{L}^{-1}\{H(s)\} = h(t)$. (See pages 504–505.)
- When a circuit is **time-invariant**, delaying the input by a seconds delays the output by a seconds. (See page 504.)
- The output of a circuit, $y(t)$, can be computed by convolving the input, $x(t)$, with the impulse response of the circuit, $h(t)$:

$$\begin{aligned} y(t) &= h(t) * x(t) = \int_0^t h(\lambda)x(t - \lambda) d\lambda \\ &= x(t) * h(t) = \int_0^t x(\lambda)h(t - \lambda) d\lambda. \end{aligned}$$

A graphical interpretation of the convolution integral usually helps you to compute $y(t)$. (See pages 505 and 508.)

- We can use the transfer function of a circuit to compute its steady-state response to a sinusoidal source. To do so, make the substitution $s = j\omega$ in $H(s)$ and represent the resulting complex number as a magnitude and phase angle. If

$$\begin{aligned} x(t) &= A \cos(\omega t + \phi), \\ H(j\omega) &= |H(j\omega)|e^{j\theta(\omega)}, \end{aligned}$$

then

$$y_{ss}(t) = A|H(j\omega)|\cos[\omega t + \phi + \theta(\omega)].$$

(See page 512.)

- Laplace transform analysis correctly predicts impulsive currents and voltages arising from switching and impulsive sources. You must ensure that the s -domain equivalent circuits are based on initial conditions prior to the switching, that is, at $t = 0^-$. (See page 520.)

Problems

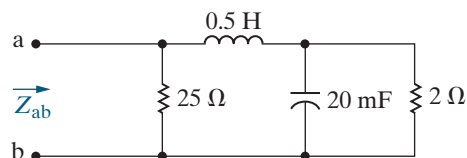
Section 13.1

- 13.1** Derive the s -domain equivalent circuit shown in Fig. 13.4 by expressing the inductor current i as a function of the terminal voltage v and then finding the Laplace transform of this time-domain integral equation.
- 13.2** Find the Norton equivalent of the circuit shown in Fig. 13.3.
- 13.3** Find the Thévenin equivalent of the circuit shown in Fig. 13.7.

Section 13.2

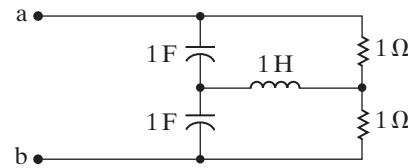
- 13.4** A $1\text{ k}\Omega$ resistor is in series with an 500 mH inductor. This series combination is in parallel with a 400 nF capacitor.
- Express the equivalent s -domain impedance of these parallel branches as a rational function.
 - Determine the numerical values of the poles and zeros.
- 13.5** A $10\text{ k}\Omega$ resistor, a 5 H inductor, and a 20 nF capacitor are in series.
- Express the s -domain impedance of this series combination as a rational function.
 - Give the numerical value of the poles and zeros of the impedance.
- 13.6** The parallel circuit in Example 13.1 is placed in series with a $2\text{ k}\Omega$ resistor.
- Express the s -domain impedance of this series combination as a rational function.
 - Give the numerical values of the poles and zeros of the impedance.
- 13.7** Find the poles and zeros of the impedance seen looking into the terminals a,b of the circuit shown in Fig. P13.7.

Figure P13.7



- 13.8** Find the poles and zeros of the impedance seen looking into the terminals a,b of the circuit shown in Fig. P13.8.

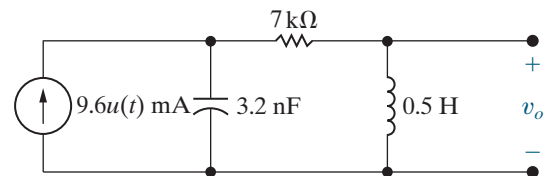
Figure P13.8



Section 13.3

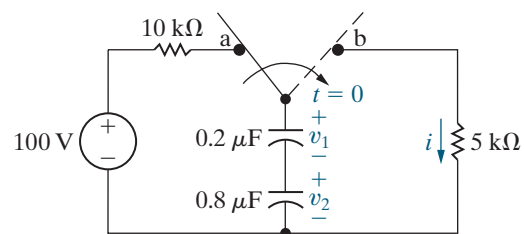
- 13.9** a) Find the s -domain expression for V_o in the circuit in Fig. P13.9.
 b) Use the s -domain expression derived in (a) to predict the initial and final values of v_o .
 c) Find the time-domain expression for v_o .

Figure P13.9



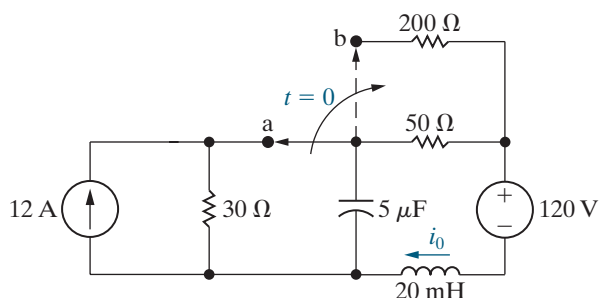
- 13.10** Find the time-domain expression for the current in the capacitor in Fig. P13.9. Assume the reference direction for i_C is down.
- 13.11** The switch in the circuit in Fig. P13.11 has been in position a for a long time. At $t = 0$, the switch is thrown to position b.
- Construct an s -domain circuit for $t > 0$.
 - Find I , V_1 , and V_2 as rational functions of s .
 - Find the time-domain expressions for i , v_1 , and v_2 .

Figure P13.11



- 13.12** The switch in the circuit in Fig. P13.12 has been in position a for a long time. At $t = 0$, it moves instantaneously from a to b.
- Construct the s -domain circuit for $t > 0$.
 - Find $I_o(s)$.
 - Find $i_o(t)$ for $t \geq 0$.

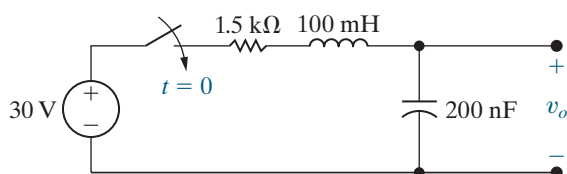
Figure P13.12



- 13.13** Find V_o and v_o in the circuit shown in Fig. P13.13 if the initial energy is zero and the switch is closed at $t = 0$.

PSPICE
MULTISIM

Figure P13.13



- 13.14** Repeat Problem 13.13 if the initial voltage on the capacitor is 20 V positive at the lower terminal.

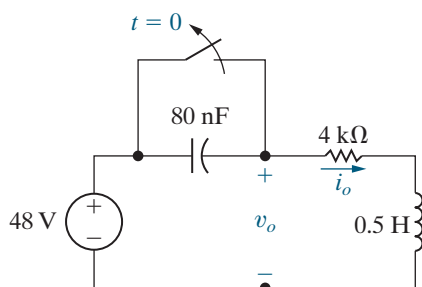
PSPICE
MULTISIM

- 13.15** The switch in the circuit in Fig. P13.15 has been closed for a long time. At $t = 0$, the switch is opened.

PSPICE
MULTISIM

- Find v_o for $t \geq 0$.
- Find i_o for $t \geq 0$.

Figure P13.15



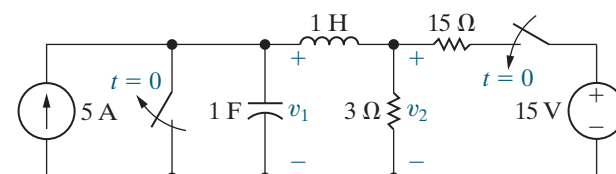
- 13.16** The dc current and dc voltage sources are applied simultaneously to the circuit in Fig. P13.16. No energy is stored in the circuit at the instant of application.

PSPICE
MULTISIM

- Derive the s -domain expressions for V_1 and V_2 .
- For $t > 0$, derive the time-domain expressions for v_1 and v_2 .

- Calculate $v_1(0^+)$ and $v_2(0^+)$.
- Compute the steady-state values of v_1 and v_2 .

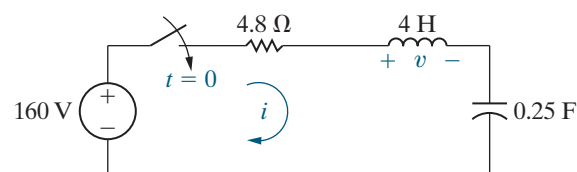
Figure P13.16



- 13.17** There is no initial energy in the circuit in Fig. P13.17 before the switch closes at $t = 0$.

- Find the s -domain expression for I .
- Find the time-domain expression for i when $t > 0$.
- Find the s -domain expression for V .
- Find the time-domain expression for v when $t > 0$.

Figure P13.17

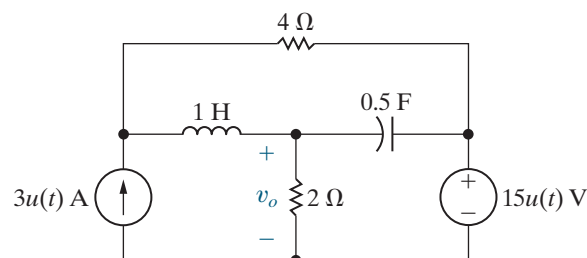


- 13.18** There is no energy stored in the circuit in Fig. P13.18 at $t = 0^-$.

PSPICE
MULTISIM

- Find V_o .
- Find v_o .
- Does your solution for v_o make sense in terms of known circuit behavior? Explain.

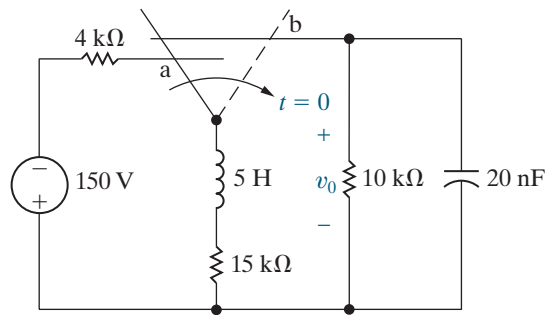
Figure P13.18



- 13.19** The make-before-break switch in the circuit in Fig. P13.19 has been in position a for a long time. At $t = 0$, it moves instantaneously to position b. Find v_o for $t \geq 0$.

PSPICE
MULTISIM

Figure P13.19

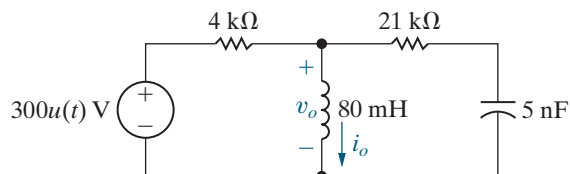


13.20 There is no energy stored in the circuit in Fig. P13.20 at $t = 0^-$.

PSPICE
MULTISIM

- Use the node voltage method to find v_o .
- Find the time domain expression for i_o .
- Do your answers in (a) and (b) make sense in terms of known circuit behavior? Explain.

Figure P13.20

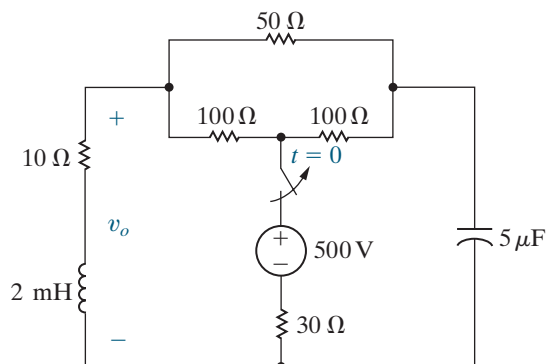


13.21 The switch in the circuit in Fig. P13.21 has been closed for a long time before opening at $t = 0$.

PSPICE
MULTISIM

- Construct the s -domain equivalent circuit for $t > 0$.
- Find V_o .
- Find v_o for $t \geq 0$.

Figure P13.21

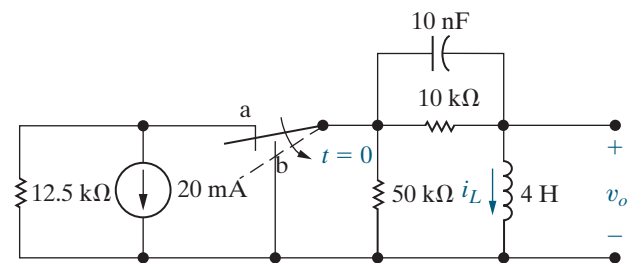


13.22 The switch in the circuit in Fig. P13.22 has been in position a for a long time. At $t = 0$, the switch moves instantaneously to position b.

PSPICE
MULTISIM

- Construct the s -domain circuit for $t > 0$.
- Find V_o .
- Find I_L .
- Find v_o for $t > 0$.
- Find i_L for $t \geq 0$.

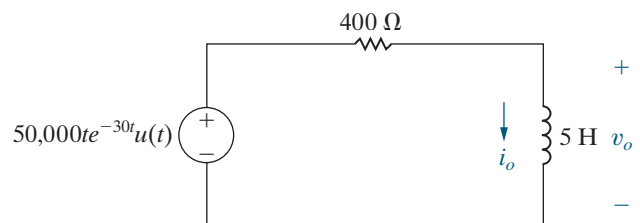
Figure P13.22



13.23 There is no energy stored in the circuit in Fig. P13.23 at the time the voltage source is energized.

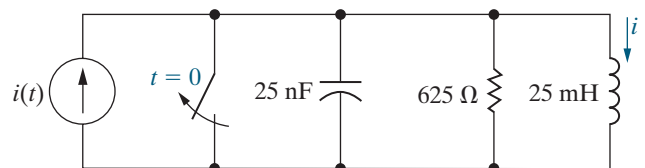
- Find V_o and I_o .
- Find v_o and i_o for $t \geq 0$.

Figure P13.23



13.24 There is no initial energy in the circuit in Fig. P13.24 before the switch opens at $t = 0$. The current source is $24 \cos 40,000t$ mA. Find $i(t)$ for $t \geq 0$.

Figure P13.24

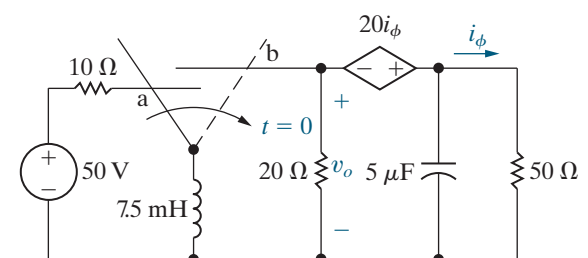


13.25 The switch in the circuit seen in Fig. P13.25 has been in position a for a long time. At $t = 0$, it moves instantaneously to position b.

PSPICE
MULTISIM

- Find V_o .
- Find v_o .

Figure P13.25

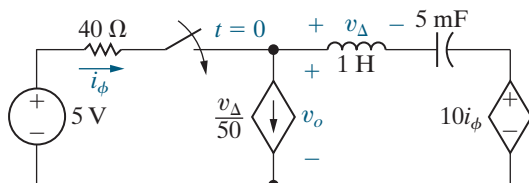


13.26 There is no energy stored in the circuit in Fig. P13.26 at the time the switch is closed.

PSPICE
MULTISIM

- Find v_o for $t \geq 0$.
- Does your solution make sense in terms of known circuit behavior? Explain.

Figure P13.26

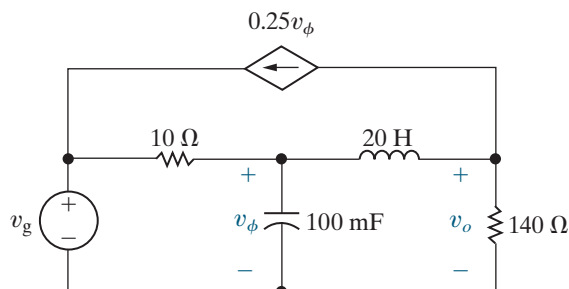


13.27 The initial energy in the circuit in Fig. P13.27 is zero. The ideal voltage source is $600u(t)$ V.

PSPICE
MULTISIM

- Find $V_o(s)$.
- Use the initial- and final-value theorems to find $v_o(0^+)$ and $v_o(\infty)$.
- Do the values obtained in (b) agree with known circuit behavior? Explain.
- Find $v_o(t)$.

Figure P13.27

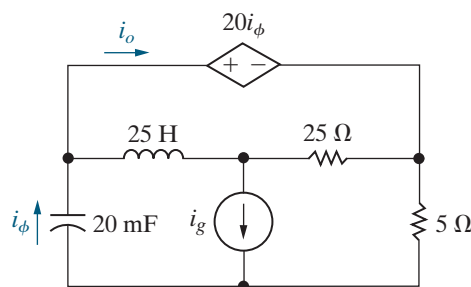


13.28 There is no energy stored in the circuit in Fig. P13.28 at the time the current source turns on. Given that $i_g = 100u(t)$ A:

PSPICE
MULTISIM

- Find $I_o(s)$.
- Use the initial- and final-value theorems to find $i_o(0^+)$ and $i_o(\infty)$.
- Determine if the results obtained in (b) agree with known circuit behavior.
- Find $i_o(t)$.

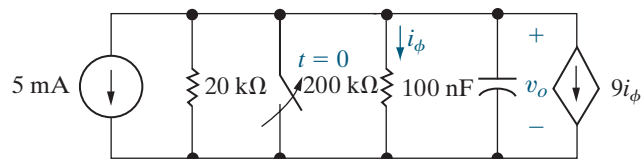
Figure P13.28



13.29 The switch in the circuit in Fig. P13.29 has been closed for a long time before opening at $t = 0$. Find v_o for $t \geq 0$.

PSPICE
MULTISIM

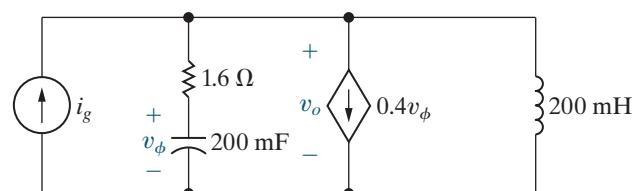
Figure P13.29



13.30 Find v_o in the circuit shown in Fig. P13.30 if $i_g = 15u(t)$ A. There is no energy stored in the circuit at $t = 0$.

PSPICE
MULTISIM

Figure P13.30

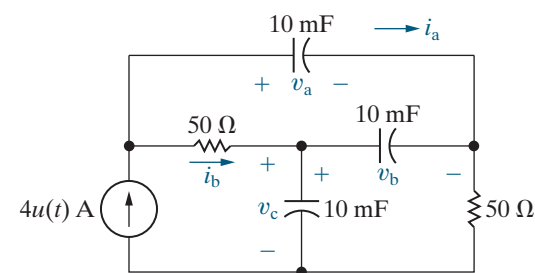


13.31 There is no energy stored in the circuit in Fig. P13.31 at the time the current source is energized.

PSPICE
MULTISIM

- Find I_a and I_b .
- Find i_a and i_b .
- Find V_a , V_b , and V_c .
- Find v_a , v_b , and v_c .
- Assume a capacitor will break down whenever its terminal voltage is 1000 V. How long after the current source turns on will one of the capacitors break down?

Figure P13.31

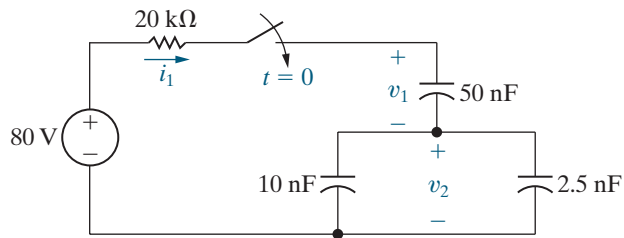


13.32 There is no energy stored in the capacitors in the circuit in Fig. P13.32 at the time the switch is closed.

PSPICE
MULTISIM

- Construct the s -domain circuit for $t > 0$.
- Find I_1 , V_1 , and V_2 .
- Find i_1 , v_1 , and v_2 .
- Do your answers for i_1 , v_1 , and v_2 make sense in terms of known circuit behavior? Explain.

Figure P13.32

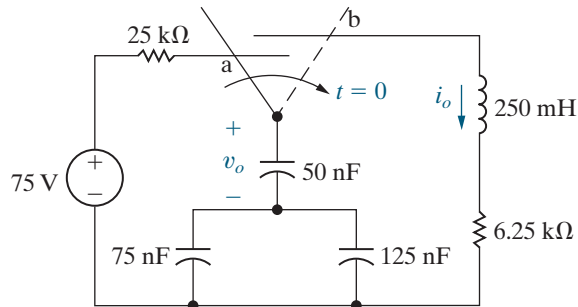


13.33 The switch in the circuit seen in Fig. P13.33 has been in position a for a long time before moving instantaneously to position b at $t = 0$.

PSPICE
MULTISIM

- Construct the s -domain equivalent circuit for $t > 0$.
- Find I_o and i_o .
- Find V_o and v_o .

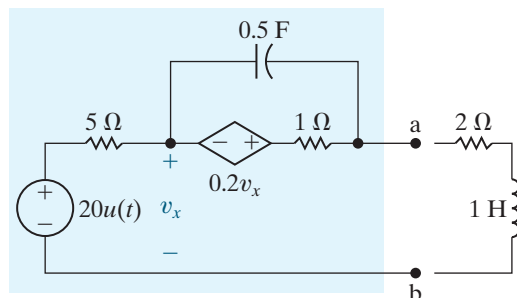
Figure P13.33



13.34 The initial charge on the capacitor in the circuit in Fig. P13.34 is zero.

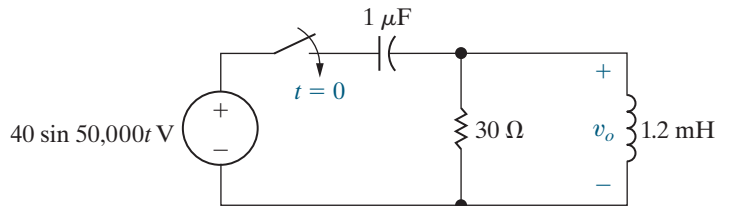
- Find the s -domain Thévenin equivalent circuit with respect to the terminals a,b.
- Find the s -domain expression for the current that the circuit delivers to a load consisting of a 1 H inductor in series with a 2Ω resistor.

Figure P13.34



13.35 There is no initial energy in the circuit in Fig. P13.35 before the switch closes at $t = 0$. Find $v_o(t)$ for $t \geq 0$.

Figure P13.35

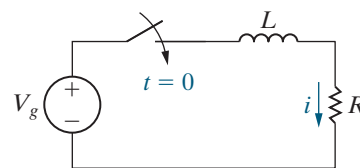


13.36 Using the expression for capacitor voltage found in Example 13.6, show that the capacitor current in Fig. 13.20 is positive for $0 < t < 200 \mu\text{s}$ and negative for $t > 200 \mu\text{s}$. Also show that at $200 \mu\text{s}$, the current is zero and that this corresponds to when dv_C/dt is zero.

13.37 The switch in the circuit shown in Fig. P13.37 has been open for a long time. The voltage of the sinusoidal source is $v_g = V_m \sin(\omega t + \phi)$. The switch closes at $t = 0$. Note that the angle ϕ in the voltage expression determines the value of the voltage at the moment when the switch closes, that is, $v_g(0) = V_m \sin \phi$.

- Use the Laplace transform method to find i for $t > 0$.
- Using the expression derived in (a), write the expression for the current after the switch has been closed for a long time.
- Using the expression derived in (a), write the expression for the transient component of i .
- Find the steady-state expression for i using the phasor method. Verify that your expression is equivalent to that obtained in (b).
- Specify the value of ϕ so that the circuit passes directly into steady-state operation when the switch is closed.

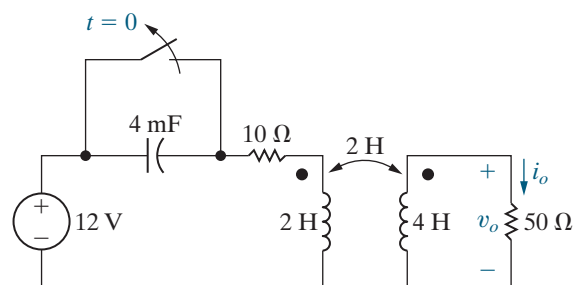
Figure P13.37



13.38 The switch in the circuit seen in Fig. P13.38 has been closed for a long time before opening at $t = 0$. Use the Laplace transform method of analysis to find i_o .

PSPICE
MULTISIM

Figure P13.38



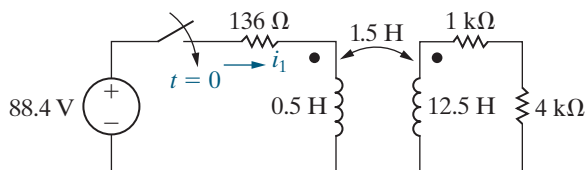
- 13.39** a) Using the results from Example 13.7 for the circuit in Fig. 13.23, verify that i_2 reaches a peak value of 481.13 mA at $t = 549.31$ ms.
 b) For the circuit shown in Fig. P13.23, find $i_1(t)$ for $t > 0$.
 c) Compute di_1/dt when i_2 is at its peak value.
 d) Express i_2 as a function of di_1/dt when i_2 is at its peak value.
 e) Use the results obtained in (c) and (d) to calculate the peak value of i_2 .

13.40 There is no energy stored in the circuit in Fig. P13.40 at the time the switch is closed.

PSPICE
MULTISIM

- a) Find I_1 .
 b) Use the initial- and final-value theorems to find $i_1(0^+)$ and $i_1(\infty)$.
 c) Find i_1 .

Figure P13.40



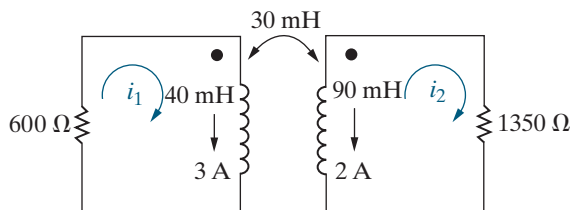
- 13.41** a) Find the current in the 4 kΩ resistor in the circuit in Fig. P13.40. The reference direction for the current is from top to bottom through the resistor.
 b) Repeat part (a) if the dot on the 12.5 H coil is reversed.

13.42 The magnetically coupled coils in the circuit seen in Fig. P13.42 carry initial currents of 3 and 2 A, as shown.

PSPICE
MULTISIM

- a) Find the initial energy stored in the circuit.
 b) Find I_1 and I_2 .
 c) Find i_1 and i_2 .
 d) Find the total energy dissipated in the 600 Ω and 1350 Ω resistors.
 e) Repeat (a)–(d), with the dot on the 90 mH inductor at the lower terminal.

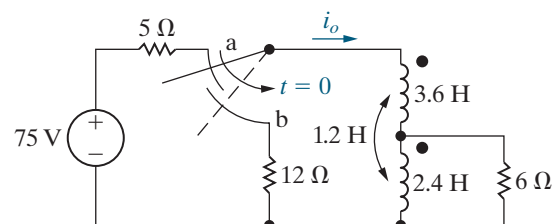
Figure P13.42



13.43 The make-before-break switch in the circuit seen in Fig. P13.43 has been in position a for a long time. At $t = 0$, it moves instantaneously to position b. Find i_o for $t \geq 0$.

PSPICE
MULTISIM

Figure P13.43

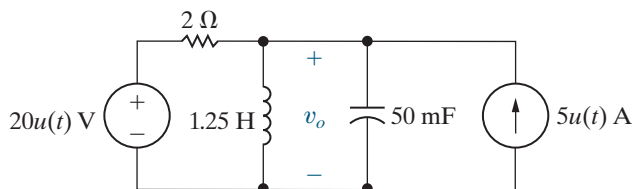


13.44 There is no energy stored in the circuit seen in Fig. P13.44 at the time the two sources are energized.

PSPICE
MULTISIM

- a) Use the principle of superposition to find V_o .
 b) Find v_o for $t > 0$.

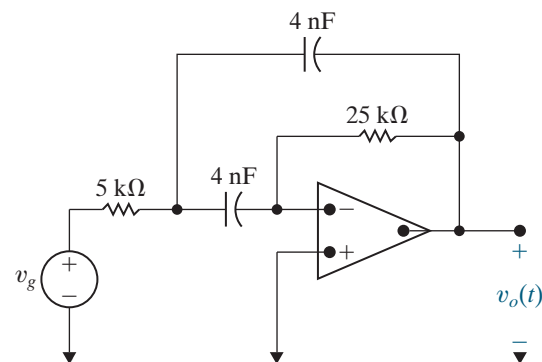
Figure P13.44



13.45 Find $v_o(t)$ in the circuit shown in Fig. P13.45 if the ideal op amp operates within its linear range and $v_g = 0.4u(t)$ V. Assume there is no energy stored in the circuit when the source is energized.

PSPICE
MULTISIM

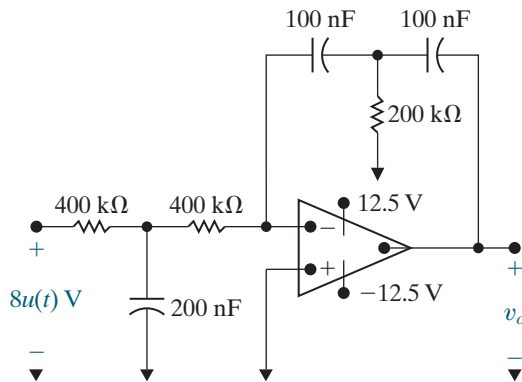
Figure P13.45



13.46 The op amp in the circuit seen in Fig. P13.46 is ideal. There is no energy stored in the capacitors at the time the circuit is energized. Determine (a) V_o , (b) v_o , and (c) how long it takes to saturate the operational amplifier.

PSPICE
MULTISIM

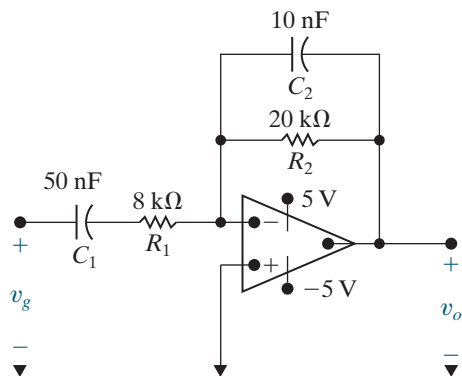
Figure P13.46



- 13.47** The op amp in the circuit shown in Fig. P13.47 is ideal. There is no energy stored in the circuit at the time it is energized. If $v_g = 20,000tu(t)$ V, find (a) V_o , (b) v_o , (c) how long it takes to saturate the operational amplifier, and (d) how small the rate of increase in v_g must be to prevent saturation.

PSPICE
MULTISIM

Figure P13.47

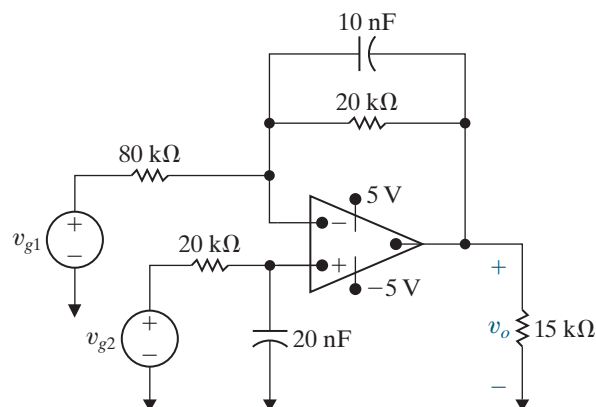


- 13.48** The op amp in the circuit shown in Fig. P13.48 is ideal. There is no energy stored in the capacitors at the instant the circuit is energized.

PSPICE
MULTISIM

- a) Find v_o if $v_{g1} = 16u(t)$ V and $v_{g2} = 8u(t)$ V.
b) How many milliseconds after the two voltage sources are turned on does the op amp saturate?

Figure P13.48

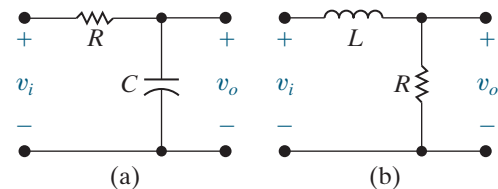


Sections 13.4–13.5

- 13.49** a) Find the transfer function $H(s) = V_o/V_i$ for the circuit shown in Fig. P13.49(a).
b) Find the transfer function $H(s) = V_o/V_i$ for the circuit shown in Fig. P13.49(b).
c) Create two different circuits that have the transfer function $H(s) = V_o/V_i = 5000/(s + 5000)$. Use components selected from Appendix H and Figs. P13.49(a) and (b).

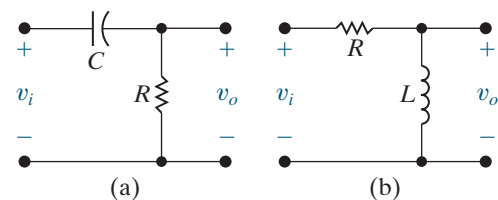
PSPICE
MULTISIM

Figure P13.49



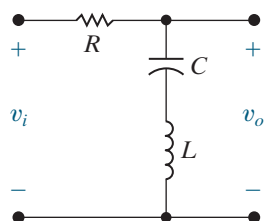
- 13.50** a) Find the transfer function $H(s) = V_o/V_i$ for the circuit shown in Fig. P13.50(a).
b) Find the transfer function $H(s) = V_o/V_i$ for the circuit shown in Fig. P13.50(b).
c) Create two different circuits that have the transfer function $H(s) = V_o/V_i = s/(s + 250)$. Use components selected from Appendix H and Figs. P13.50(a) and (b).

Figure P13.50



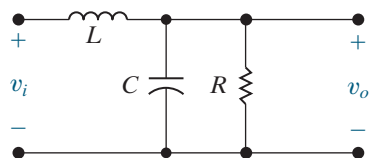
- 13.51** a) Find the transfer function $H(s) = V_o/V_i$ for the circuit shown in Fig. P13.51. Identify the poles and zeros for this transfer function.
b) Find components from Appendix H which when used in the circuit of Fig. P13.51 will result in a transfer function with two poles that are distinct real numbers. Calculate the values of the poles.
c) Find components from Appendix H which when used in the circuit of Fig. P13.51 will result in a transfer function with two poles, both with the same value. Calculate the value of the poles.
d) Find components from Appendix H which when used in the circuit of Fig. P13.51 will result in a transfer function with two poles that are complex conjugate complex numbers. Calculate the values of the poles.

Figure P13.51



- 13.52** Repeat Problem 13.51 for the circuit in Fig. P13.52. In parts (b) – (d), let $L = 1$ mH and $C = 0.1$ μ F, and use one or two resistors from Appendix H.

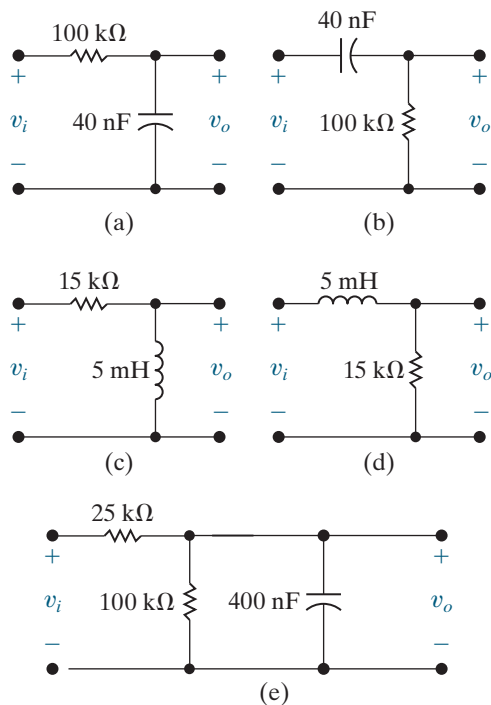
Figure P13.52



- 13.53** Find the numerical expression for the transfer function (V_o/V_i) of each circuit in Fig. P13.53 and give the numerical value of the poles and zeros of each transfer function.

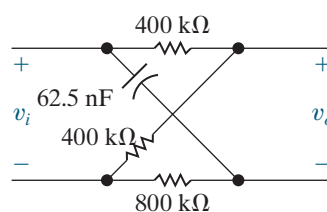
PSPICE
MULTISIM

Figure P13.53



- 13.54** a) Find the numerical expression for the transfer function $H(s) = V_o/V_i$ for the circuit in Fig. P13.54.
b) Give the numerical value of each pole and zero of $H(s)$.

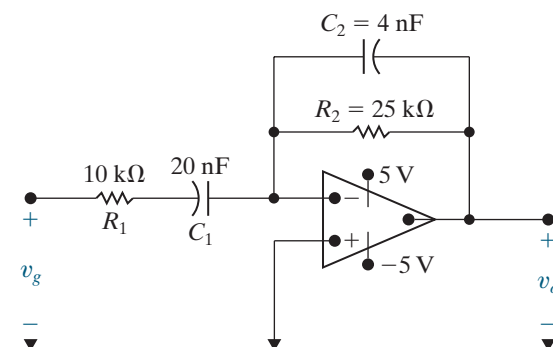
Figure P13.54



- 13.55** The operational amplifier in the circuit in Fig. P13.55 is ideal.

- a) Find the numerical expression for the transfer function $H(s) = V_o/V_g$.
b) Give the numerical value of each zero and pole of $H(s)$.
c) Use the transfer function to find the unit step response for the circuit in Fig. P13.55.

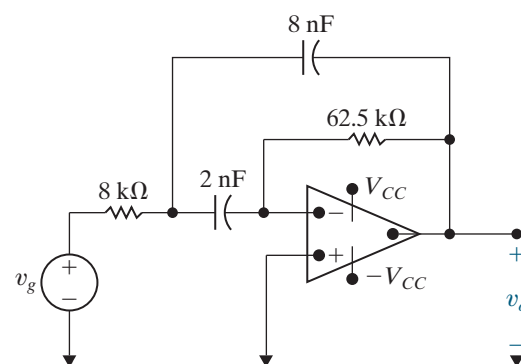
Figure P13.55



- 13.56** The operational amplifier in the circuit in Fig. P13.56 is ideal.

- a) Derive the numerical expression of the transfer function $H(s) = V_o/V_g$ for the circuit in Fig. P13.56.
b) Give the numerical value of each pole and zero of $H(s)$.
c) Use the transfer function to find the unit step response for the circuit in Fig. P13.56.

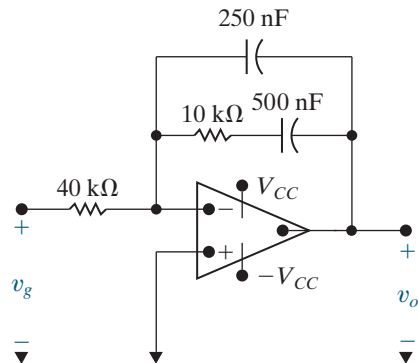
Figure P13.56



13.57 The operational amplifier in the circuit in Fig. P13.57 is ideal.

- Find the numerical expression for the transfer function $H(s) = V_o/V_g$.
- Give the numerical value of each zero and pole of $H(s)$.
- Use the transfer function to find the unit step response for the circuit in Fig. P13.57.

Figure P13.57

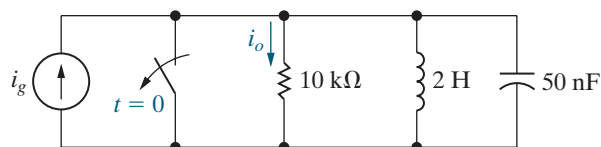


13.58 There is no energy stored in the circuit in Fig. P13.58 at the time the switch is opened. The sinusoidal current source is generating the signal $60 \cos 4000t$ mA. The response signal is the current i_o .

PSPICE
MULTISIM

- Find the transfer function I_o/I_g .
- Find $I_o(s)$.
- Describe the nature of the transient component of $i_o(t)$ without solving for $i_o(t)$.
- Describe the nature of the steady-state component of $i_o(t)$ without solving for $i_o(t)$.
- Verify the observations made in (c) and (d) by finding $i_o(t)$.

Figure P13.58

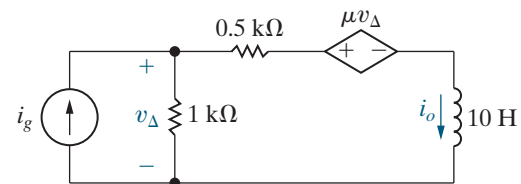


13.59 a) Find the transfer function I_o/I_g as a function of μ for the circuit seen in Fig. P13.59.

PSPICE
MULTISIM

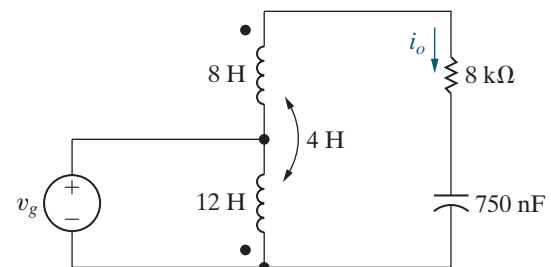
- Find the largest value of μ that will produce a bounded output signal for a bounded input signal.
- Find i_o for $\mu = -0.5, 0, 1, 1.5$, and 2 if $i_g = 10u(t)$ A.

Figure P13.59



13.60 In the circuit of Fig. P13.60 i_o is the output signal and v_g is the input signal. Find the poles and zeros of the transfer function, assuming there is no initial energy stored in the linear transformer or in the capacitor.

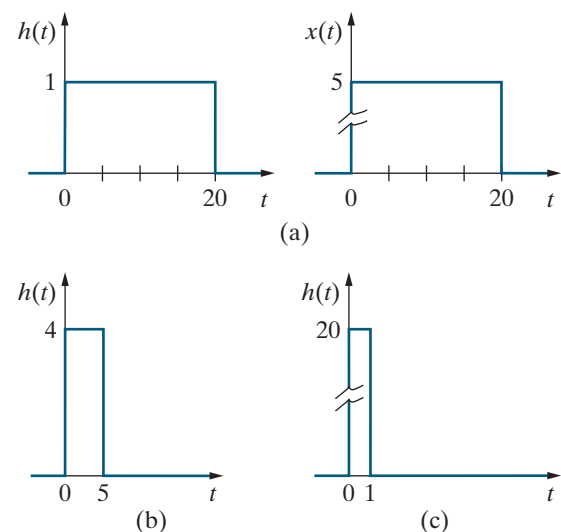
Figure P13.60



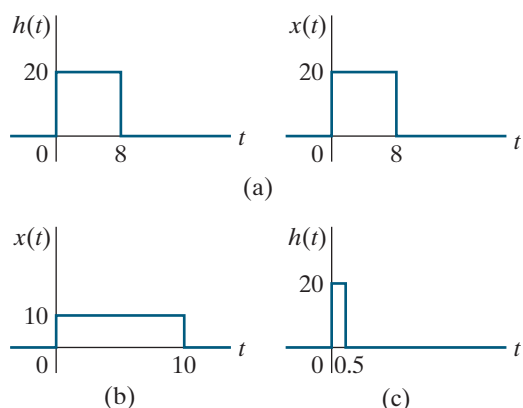
Section 13.6

- 13.61** a) Given $y(t) = x(t) * h(t)$, find $y(t)$ when $h(t)$ and $x(t)$ are the rectangular pulses shown in Fig. P13.61(a).
- Repeat (a) when $h(t)$ changes to the rectangular pulse shown in Fig. P13.61(b).
 - Repeat (a) when $h(t)$ changes to the rectangular pulse shown in Fig. P13.61(c).
 - Sketch $y(t)$ versus t for (a)–(c) on a single graph.
 - Do the sketches in (d) make sense? Explain.

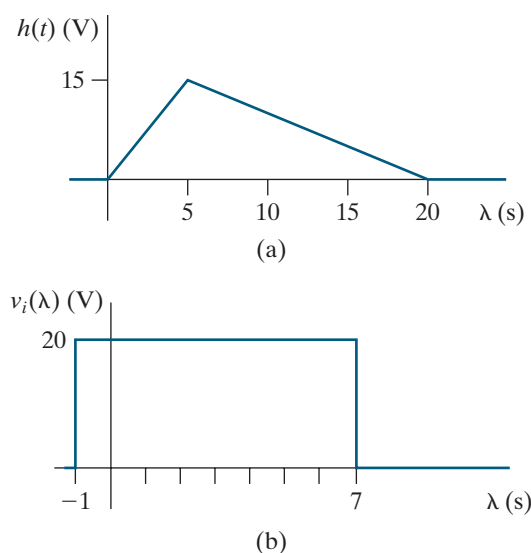
Figure P13.61



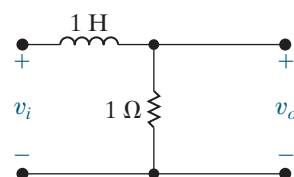
- 13.62** a) Find $x(t) * h(t)$ when $h(t)$ and $x(t)$ are the rectangular pulses shown in Fig. P13.62(a).
 b) Repeat (a) when $x(t)$ changes to the rectangular pulse shown in Fig. P13.62(b).
 c) Repeat (a) when $h(t)$ changes to the rectangular pulse shown in Fig. P13.65(c).

Figure P13.62

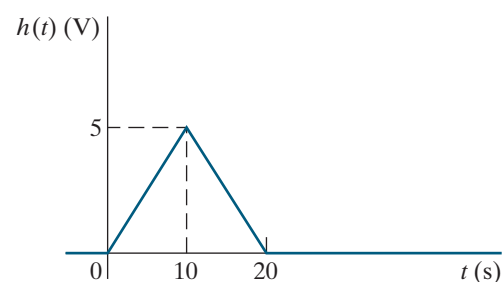
- 13.63** The voltage impulse response of a circuit is shown in Fig. P13.63(a). The input signal to the circuit is the rectangular voltage pulse shown in Fig. P13.63(b).
 a) Derive the equations for the output voltage. Note the range of time for which each equation is applicable.
 b) Sketch v_o for $0 \leq t \leq 27$ s.

Figure P13.63

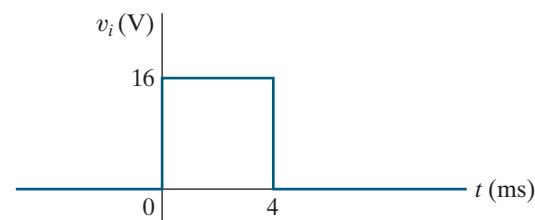
- 13.64** A rectangular voltage pulse $v_i = [u(t) - u(t-1)]$ V is applied to the circuit in Fig. P13.64. Use the convolution integral to find v_o .

Figure P13.64

- 13.65** Interchange the inductor and resistor in Problem 13.64 and again use the convolution integral to find v_o .
13.66 Assume the voltage impulse response of a circuit can be modeled by the triangular waveform shown in Fig. P13.66. The voltage input signal to this circuit is the step function $4u(t)$ V.
 a) Use the convolution integral to derive the expressions for the output voltage.
 b) Sketch the output voltage over the interval 0 to 25 s.
 c) Repeat parts (a) and (b) if the area under the voltage impulse response stays the same but the width of the impulse response narrows to 5 s.
 d) Which output waveform is closer to replicating the input waveform: (b) or (c)? Explain.

Figure P13.66

- 13.67** a) Use the convolution integral to find the output voltage of the circuit in Fig. P13.53(a) if the input voltage is the rectangular pulse shown in Fig. P13.67.
 b) Sketch $v_o(t)$ versus t for the time interval $0 \leq t \leq 10$ ms.

Figure P13.67

- 13.68** a) Repeat Problem 13.67, given that the resistor in the circuit in Fig. P13.53(a) is decreased to 10 k Ω .
 b) Does decreasing the resistor increase or decrease the memory of the circuit?
 c) Which circuit comes closer to transmitting a replica of the input voltage?

- 13.69** a) Assume the voltage impulse response of a circuit is

$$h(t) = \begin{cases} 0, & t < 0; \\ 20e^{-2t}, & t \geq 0. \end{cases}$$

Use the convolution integral to find the output voltage if the input signal is $20u(t)$ V.

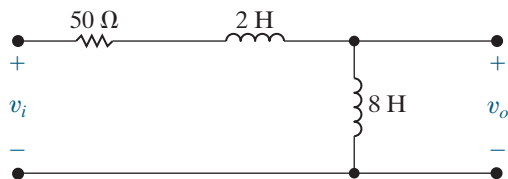
- b) Repeat (a) if the voltage impulse response is

$$h(t) = \begin{cases} 0, & t < 0; \\ 20(1 - t), & 0 \leq t \leq 1 \text{ s}; \\ 0, & t \geq 1 \text{ s}. \end{cases}$$

- c) Plot the output voltage versus time for (a) and (b) for $0 \leq t \leq 2$ s.

- 13.70** Use the convolution integral to find v_o in the circuit seen in Fig. P13.70 if $v_i = 75u(t)$ V.

Figure P13.70

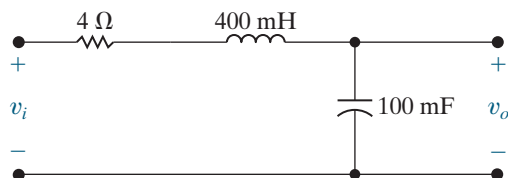


- 13.71** The input voltage in the circuit seen in Fig. P13.71 is

$$v_i = 10[u(t) - u(t - 0.1)] \text{ V}$$

- a) Use the convolution integral to find v_o .
 b) Sketch v_o for $0 \leq t \leq 1$ s.

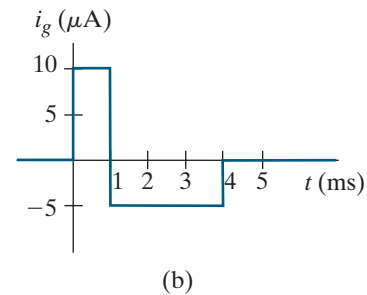
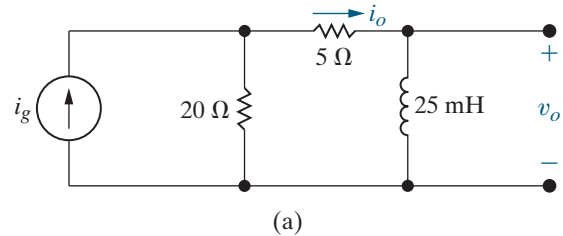
Figure P13.71



- 13.72** a) Use the convolution integral to find i_o in the circuit in Fig. P13.72(a) if i_g is the pulse shown in Fig. P13.72(b).

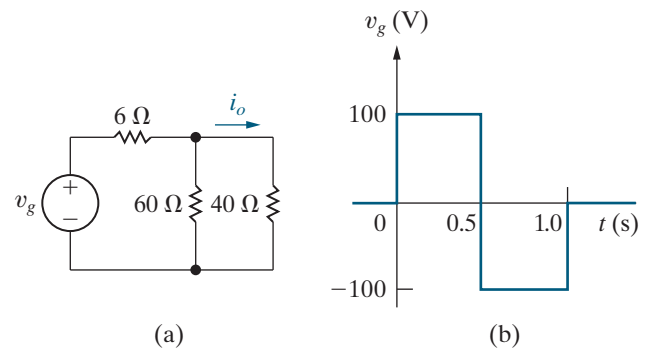
- b) Use the convolution integral to find v_o .
 c) Show that your solutions for v_o and i_o are consistent by calculating v_o and i_o at 1^- ms, 1^+ ms, 4^- ms, and 4^+ ms.

Figure P13.72



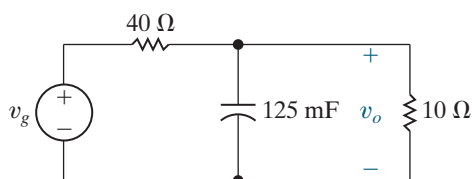
- 13.73** a) Find the impulse response of the circuit shown in Fig. P13.73(a) if v_g is the input signal and i_o is the output signal.
 b) Given that v_g has the waveform shown in Fig. P13.73(b), use the convolution integral to find i_o .
 c) Does i_o have the same waveform as v_g ? Why or why not?

Figure P13.73



- 13.74** a) Find the impulse response of the circuit seen in Fig. P13.74 if v_g is the input signal and v_o is the output signal.
 b) Assume that the voltage source has the waveform shown in Fig. P13.73(b). Use the convolution integral to find v_o .
 c) Sketch v_o for $0 \leq t \leq 1.5$ s.
 d) Does v_o have the same waveform as v_g ? Why or why not?

Figure P13.74



- 13.75 a) Show that if $y(t) = h(t) * x(t)$, then

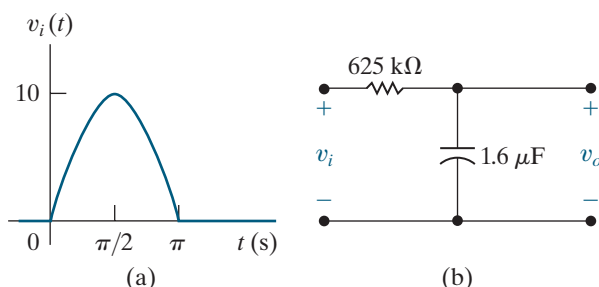
$$Y(s) = H(s)X(s).$$

- b) Use the result given in (a) to find $f(t)$ if

$$F(s) = \frac{a}{s(s+a)^2}.$$

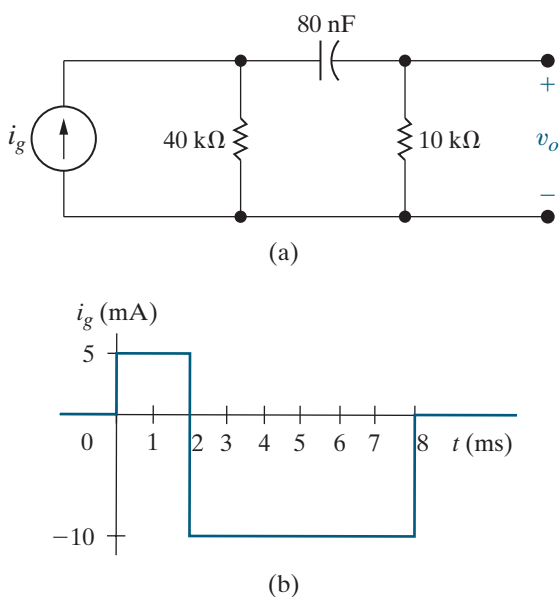
- 13.76 The sinusoidal voltage pulse shown in Fig. P13.76(a) is applied to the circuit shown in Fig. P13.76(b). Use the convolution integral to find the value of v_o at $t = 2.2$ s.

Figure P13.76



- 13.77 The current source in the circuit shown in Fig. P13.77(a) is generating the waveform shown in Fig. P13.77(b). Use the convolution integral to find v_o at $t = 7$ ms.

Figure P13.77



Section 13.7

- 13.78 When an input voltage of $240u(t)$ V is applied to a circuit, the response is known to be

$$v_o = (75 - 100e^{-800t} + 25e^{-3200t})u(t) \text{ V}.$$

What will the steady-state response be if $v_g = 40 \cos 1600t$ V?

- 13.79 The transfer function for a linear time-invariant circuit is

$$H(s) = \frac{V_o}{V_g} = \frac{10^4(s + 6000)}{(s^2 + 875s + 88 \times 10^6)}.$$

If $v_g = 12.5 \cos 8000t$ V, what is the steady-state expression for v_o ?

- 13.80 The transfer function for a linear time-invariant circuit is

$$H(s) = \frac{I_o}{I_g} = \frac{25(s + 8)}{s^2 + 60s + 150}.$$

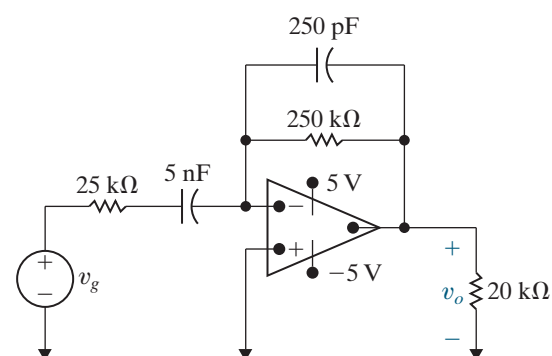
If $i_g = 10 \cos 20t$ A, what is the steady-state expression for i_o ?

- 13.81 The operational amplifier in the circuit seen in Fig. P13.81 is ideal and is operating within its linear region.

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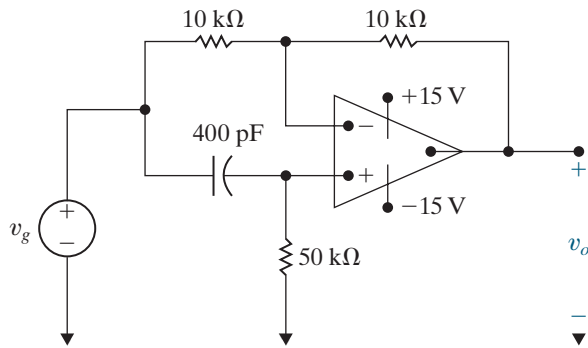
- a) Calculate the transfer function V_o/V_g .
b) If $v_g = 200\sqrt{10} \cos 8000t$ mV, what is the steady-state expression for v_o ?

Figure P13.81



- 13.82 a) For the circuit in Fig. P13.82 find the steady-state expression for v_o when $v_g = 10 \cos 50,000t$ V.
b) Replace the $50 \text{ k}\Omega$ resistor with a variable resistor and compute the value of resistance necessary to cause v_o to lead v_g by 120° .

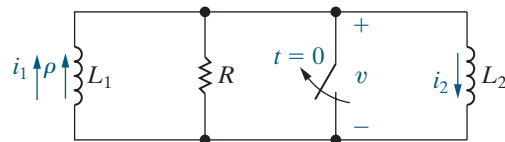
Figure P13.82



Section 13.8

13.83 The inductor L_1 in the circuit shown in Fig. P13.83 is carrying an initial current of ρ A at the instant the switch opens. Find (a) $v(t)$; (b) $i_1(t)$; (c) $i_2(t)$; and (d) $\lambda(t)$, where $\lambda(t)$ is the total flux linkage in the circuit.

Figure P13.83



13.84 a) Let $R \rightarrow \infty$ in the circuit shown in Fig. P13.83, and use the solutions derived in Problem 13.83 to find $v(t)$, $i_1(t)$, and $i_2(t)$.

b) Let $R = \infty$ in the circuit shown in Fig. P13.83 and use the Laplace transform method to find $v(t)$, $i_1(t)$, and $i_2(t)$.

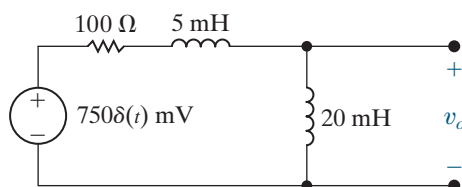
13.85 Show that after $V_0 C_e$ coulombs are transferred from C_1 to C_2 in the circuit shown in Fig. 13.47 (with $R = 0$), the voltage across each capacitor is $C_1 V_0 / (C_1 + C_2)$. (Hint: Use the conservation-of-charge principle.)

13.86 There is no energy stored in the circuit in Fig. P13.86 at the time the impulsive voltage is applied.

a) Find $v_o(t)$ for $t \geq 0$.

b) Does your solution make sense in terms of known circuit behavior? Explain.

Figure P13.86



13.87 The voltage source in the circuit in Example 13.9 is changed to a unit impulse; that is, $v_g = \delta(t)$.

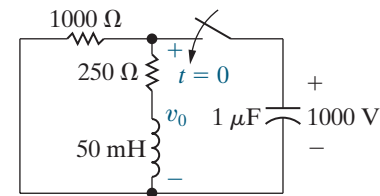
a) How much energy does the impulsive voltage source store in the capacitor?

b) How much energy does it store in the inductor?

c) Use the transfer function to find $v_o(t)$.

d) Show that the response found in (c) is identical to the response generated by first charging the capacitor to 1000 V and then releasing the charge to the circuit, as shown in Fig. P13.87.

Figure P13.87

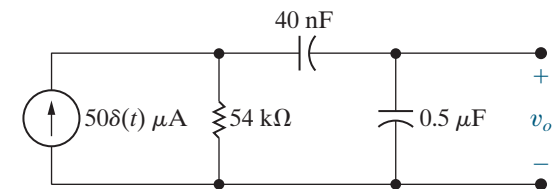


13.88 There is no energy stored in the circuit in Fig. P13.88 at the time the impulsive current is applied.

a) Find v_o for $t \geq 0^+$.

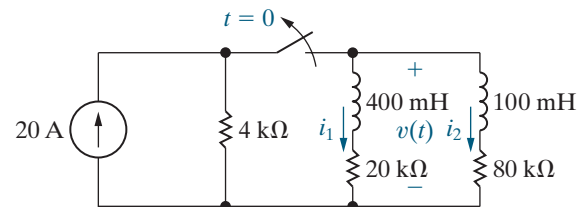
b) Does your solution make sense in terms of known circuit behavior? Explain.

Figure P13.88



13.89 The switch in the circuit in Fig. P13.89 has been closed for a long time. The switch opens at $t = 0$. Compute (a) $i_1(0^-)$; (b) $i_1(0^+)$; (c) $i_2(0^-)$; (d) $i_2(0^+)$; (e) $i_1(t)$; (f) $i_2(t)$; and (g) $v(t)$.

Figure P13.89



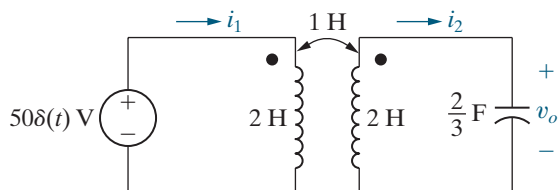
13.90 There is no energy stored in the circuit in Fig. P13.90 at the time the impulsive voltage is applied.

a) Find i_1 for $t \geq 0^+$.

b) Find i_2 for $t \geq 0^+$.

- c) Find v_o for $t \geq 0^+$.
 d) Do your solutions for i_1 , i_2 , and v_o make sense in terms of known circuit behavior? Explain.

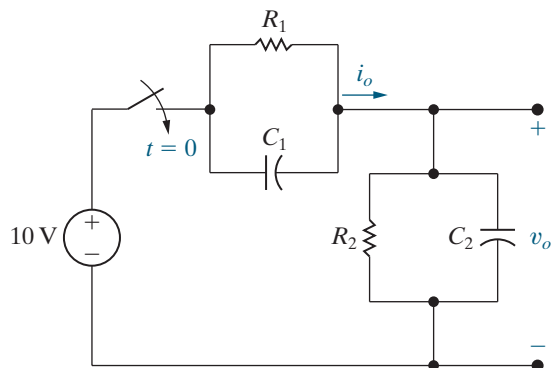
Figure P13.90



- 13.91** The parallel combination of R_2 and C_2 in the circuit shown in Fig. P13.91 represents the input circuit to a cathode-ray oscilloscope (CRO). The parallel combination of R_1 and C_1 is a circuit model of a compensating lead that is used to connect the CRO to the source. There is no energy stored in C_1 or C_2 at the time when the 10 V source is connected to the CRO via the compensating lead. The circuit values are $C_1 = 5$ pF, $C_2 = 20$ pF, $R_1 = 1$ MΩ, and $R_2 = 4$ MΩ.

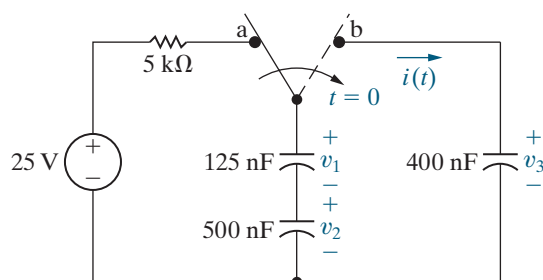
- a) Find v_o .
 b) Find i_o .
 c) Repeat (a) and (b) given C_1 is changed to 80 pF.

Figure P13.91



- 13.92** Show that if $R_1 C_1 = R_2 C_2$ in the circuit shown in Fig. P13.91, v_o will be a scaled replica of the source voltage.
13.93 The switch in the circuit in Fig. P13.93 has been in position a for a long time. At $t = 0$, the switch moves to position b. Compute (a) $v_1(0^-)$; (b) $v_2(0^-)$; (c) $v_3(0^-)$; (d) $i(t)$; (e) $v_1(0^+)$; (f) $v_2(0^+)$; and (g) $v_3(0^+)$.

Figure P13.93



Sections 13.1–13.8

13.94 Assume the line-to-neutral voltage V_o in the 60 Hz circuit of Fig. 13.59 is $120\angle 0^\circ$ V (rms). Load R_a is absorbing 720 W; load R_b is absorbing 450 W; and the inductive load is absorbing 360 magnetizing VAR. The inductive reactance of the line is $2\ \Omega$. Assume V_g does not change after the switch opens.

PRACTICAL PERSPECTIVE

- a) Calculate the initial value of $i_2(t)$ and $i_o(t)$.
 b) Find V_o , $v_o(t)$, and $v_o(0^+)$ using the s -domain circuit of Fig. 13.60.
 c) Test the steady-state component of v_o using phasor domain analysis.
 d) Using a computer program of your choice, plot v_o vs. t for $0 \leq t \leq 20$ ms.

13.95 Assume the switch in the circuit described in Problem 13.94 opens at the instant the sinusoidal steady-state voltage v_o is zero and going positive, i.e., $v_o = 120\sqrt{2} \sin 120\pi t$ V.

PRACTICAL PERSPECTIVE

- a) Find $v_o(t)$ for $t \geq 0$.
 b) Using a computer program of your choice, plot $v_o(t)$ vs. t for $0 \leq t \leq 20$ ms.
 c) Compare the disturbance in the voltage in part (a) with that obtained in part (c) of Problem 13.94.

13.96 The purpose of this problem is to show that the line-to-neutral voltage in the circuit described in Problem 13.94 can go directly into steady state if the load R_b is disconnected from the circuit at precisely the right time. Let $v_o = V_m \cos(120\pi t + \theta)$ V, where $V_m = 120\sqrt{2}$. Assume v_g does not change after R_b is disconnected.

PRACTICAL PERSPECTIVE

- a) Find the value of θ (in degrees) so that v_o goes directly into steady-state operation when the load R_b is disconnected.
 b) For the value of θ found in part (a), find $v_o(t)$ for $t \geq 0$.
 c) Using a computer program of your choice, plot on a single graph, for -10 ms $\leq t \leq 10$ ms, $v_o(t)$ before and after load R_b is disconnected.